

# ABCD ADHD menarche analysis: Output for longitudinal supplementary/sensitivity analyses

## Model 2 longitudinal ADHD present output

```
      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8266999 0.004585641 0.8172962 0.8353203
```

```
Family: poisson
Links: mu = log
Formula: internalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_pres + age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family\_id (Number of levels: 3915)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.844    | 0.015     | 0.815    | 0.873    | 1.000 | 1907     | 3757     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.109    | 0.033     | 0.052    | 0.181    | 1.004 | 994      | 1511     |

Regression Coefficients:

|                           | Estimate | Est.Error | 1-95% CI |
|---------------------------|----------|-----------|----------|
| Intercept                 | 1.219    | 0.037     | 1.145    |
| menarche_status_p_wave_2Y | 0.170    | 0.033     | 0.106    |
| adhd_status_presY         | 0.208    | 0.055     | 0.102    |
| age_wave_2_sc             | -0.017   | 0.030     | -0.078   |
| ethnraceblack_nh          | -0.404   | 0.055     | -0.512   |
| ethnracehispanic          | -0.103   | 0.048     | -0.198   |

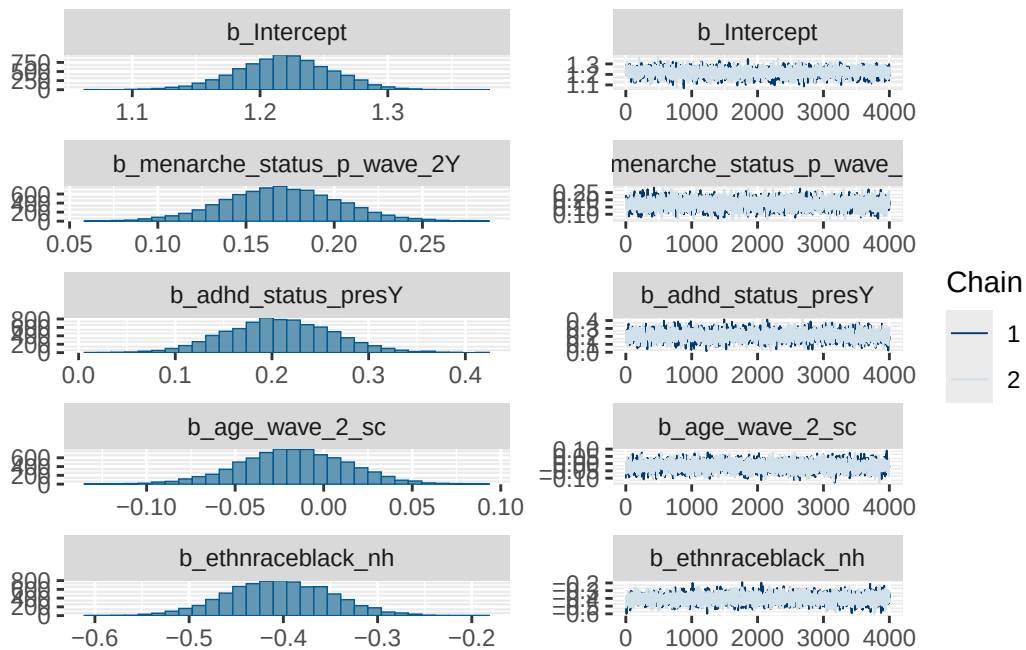
|                                             |          |       |          |          |
|---------------------------------------------|----------|-------|----------|----------|
| ethnraceother                               | 0.001    | 0.051 | -0.098   |          |
| inr_wave_0_sc                               | 0.079    | 0.035 | 0.010    |          |
| internalising_wave_0_sc                     | 0.850    | 0.027 | 0.797    |          |
| menarche_status_p_wave_2Y:adhd_status_presY | 0.058    | 0.087 | -0.113   |          |
|                                             | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
| Intercept                                   | 1.290    | 1.000 | 2055     | 3567     |
| menarche_status_p_wave_2Y                   | 0.236    | 1.001 | 1993     | 3577     |
| adhd_status_presY                           | 0.318    | 1.000 | 1975     | 3949     |
| age_wave_2_sc                               | 0.041    | 1.000 | 2401     | 4049     |
| ethnraceblack_nh                            | -0.296   | 1.000 | 2056     | 3413     |
| ethnracehispanic                            | -0.009   | 1.000 | 1693     | 3494     |
| ethnraceother                               | 0.102    | 1.001 | 1711     | 2988     |
| inr_wave_0_sc                               | 0.148    | 1.000 | 1899     | 3569     |
| internalising_wave_0_sc                     | 0.902    | 1.001 | 1499     | 2913     |
| menarche_status_p_wave_2Y:adhd_status_presY | 0.227    | 1.001 | 1589     | 3268     |

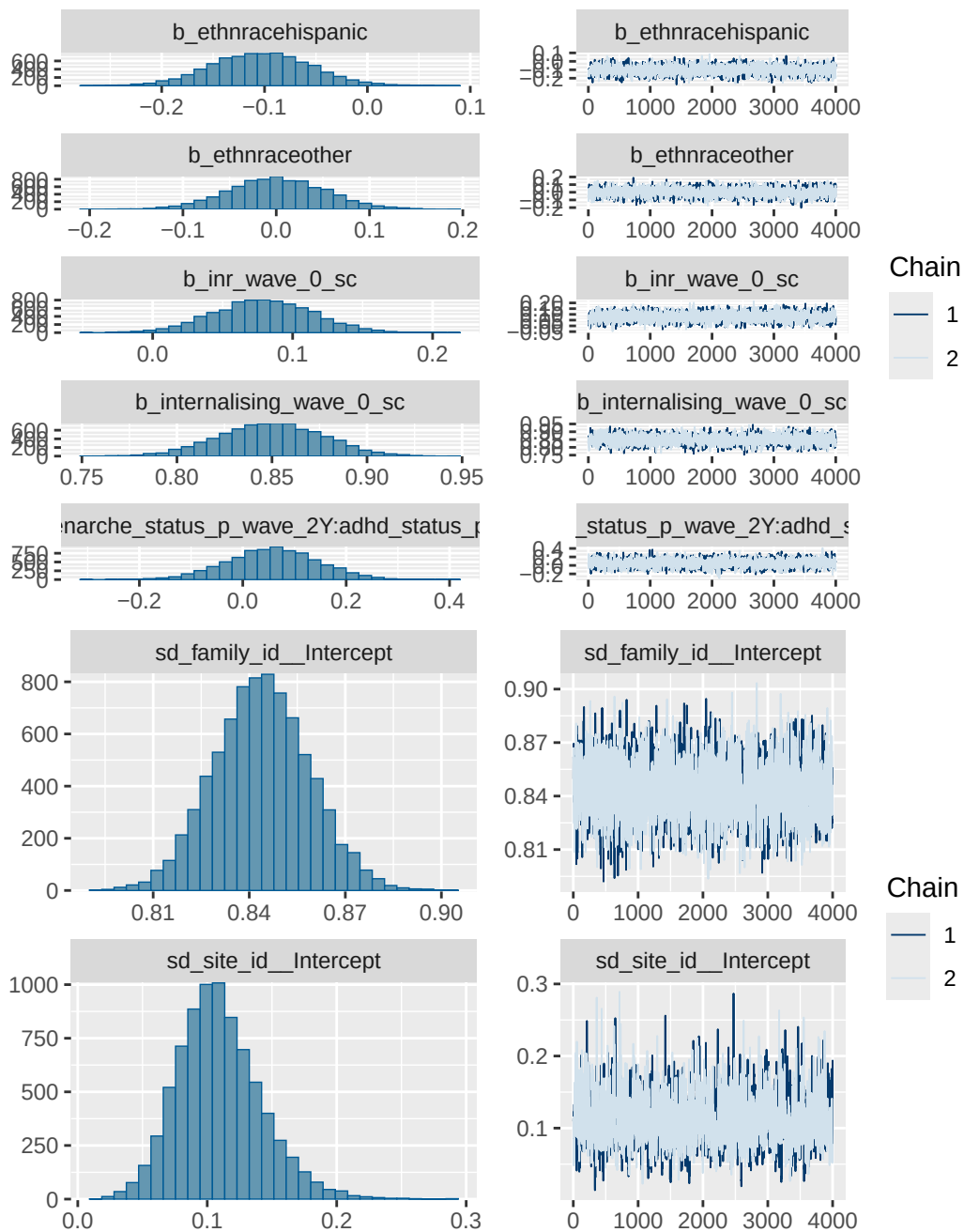
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

## Model 2 longitudinal ADHD present diagnostics

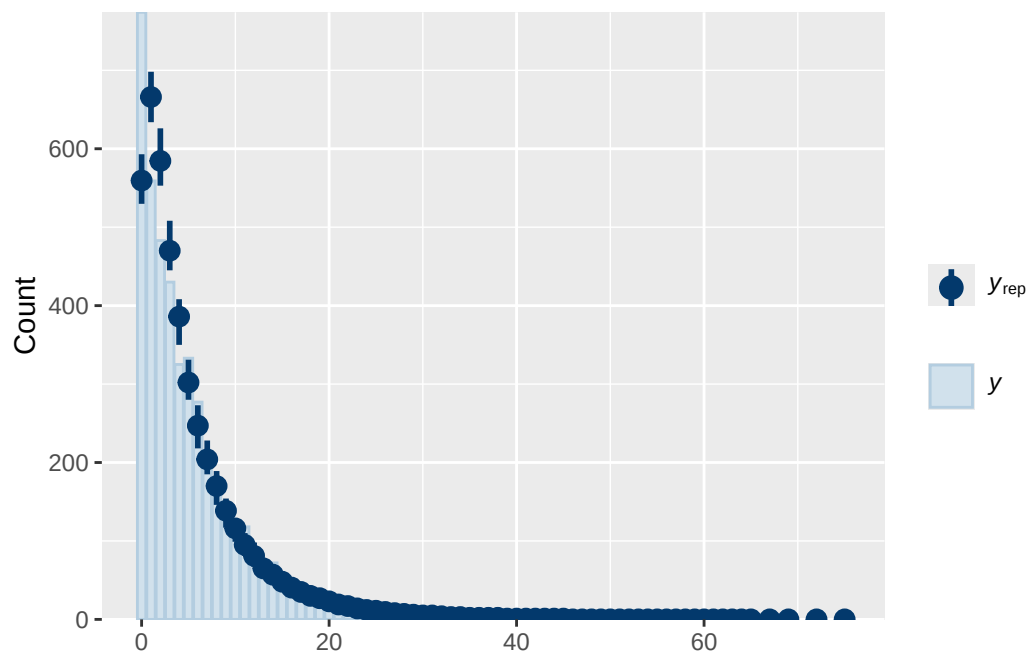
| prior                           | class        | coef                                        |
|---------------------------------|--------------|---------------------------------------------|
| normal(0, 1)                    | b            |                                             |
| normal(0, 1)                    | b            | adhd_status_presY                           |
| normal(0, 1)                    | b            | age_wave_2_sc                               |
| normal(0, 1)                    | b            | ethnraceblack_nh                            |
| normal(0, 1)                    | b            | ethnracehispanic                            |
| normal(0, 1)                    | b            | ethnraceother                               |
| normal(0, 1)                    | b            | inr_wave_0_sc                               |
| normal(0, 1)                    | b            | internalising_wave_0_sc                     |
| normal(0, 1)                    | b            | menarche_status_p_wave_2Y                   |
| normal(0, 1)                    | b            | menarche_status_p_wave_2Y:adhd_status_presY |
| student_t(3, 1.4, 2.5)          | Intercept    |                                             |
| student_t(3, 0, 2.5)            | sd           |                                             |
| student_t(3, 0, 2.5)            | sd           |                                             |
| student_t(3, 0, 2.5)            | sd           | Intercept                                   |
| student_t(3, 0, 2.5)            | sd           |                                             |
| student_t(3, 0, 2.5)            | sd           | Intercept                                   |
| group resp dpar nlpar lb ub tag | source       |                                             |
|                                 | user         |                                             |
|                                 | (vectorized) |                                             |

|           |         |              |
|-----------|---------|--------------|
|           |         | (vectorized) |
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|           |         | (vectorized) |
|           |         | (vectorized) |
|           |         | (vectorized) |
|           | default |              |
|           | default |              |
| family_id | 0       | (vectorized) |
| family_id | 0       | (vectorized) |
| site_id   | 0       | (vectorized) |
| site_id   | 0       | (vectorized) |

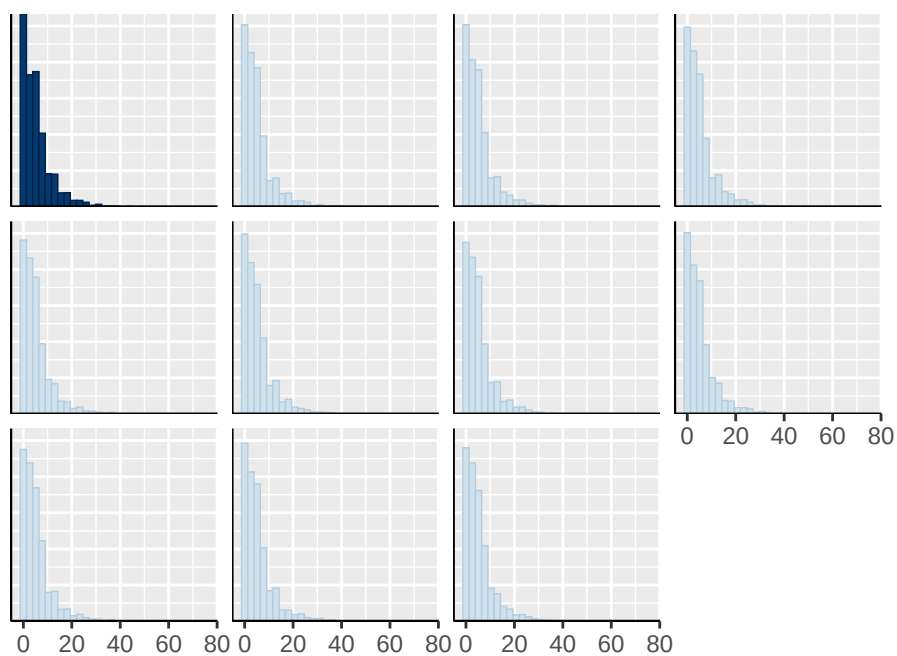




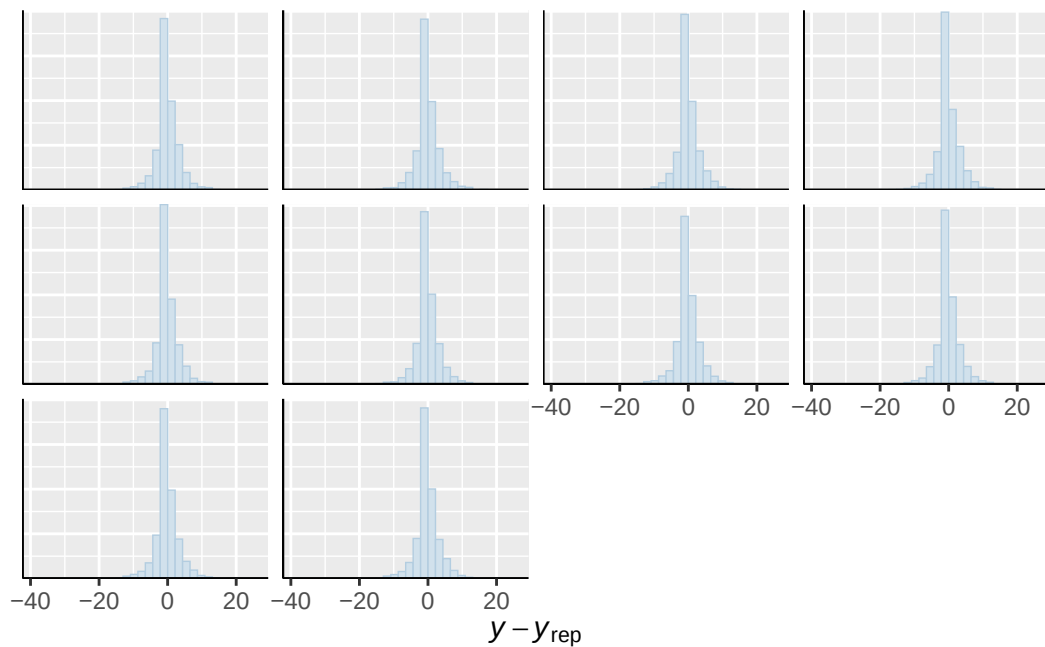




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

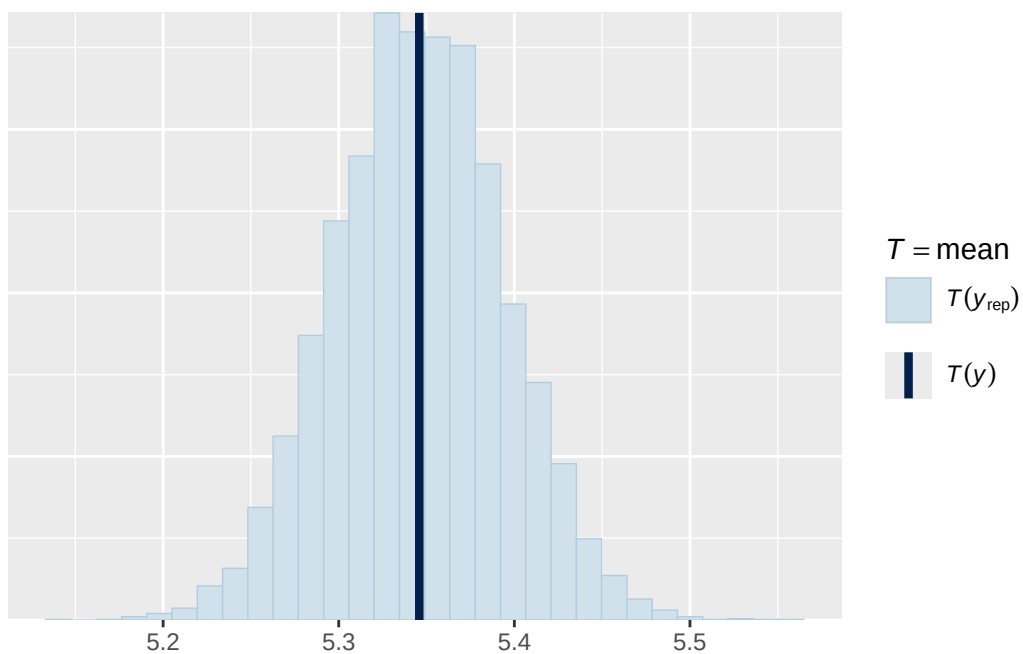


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

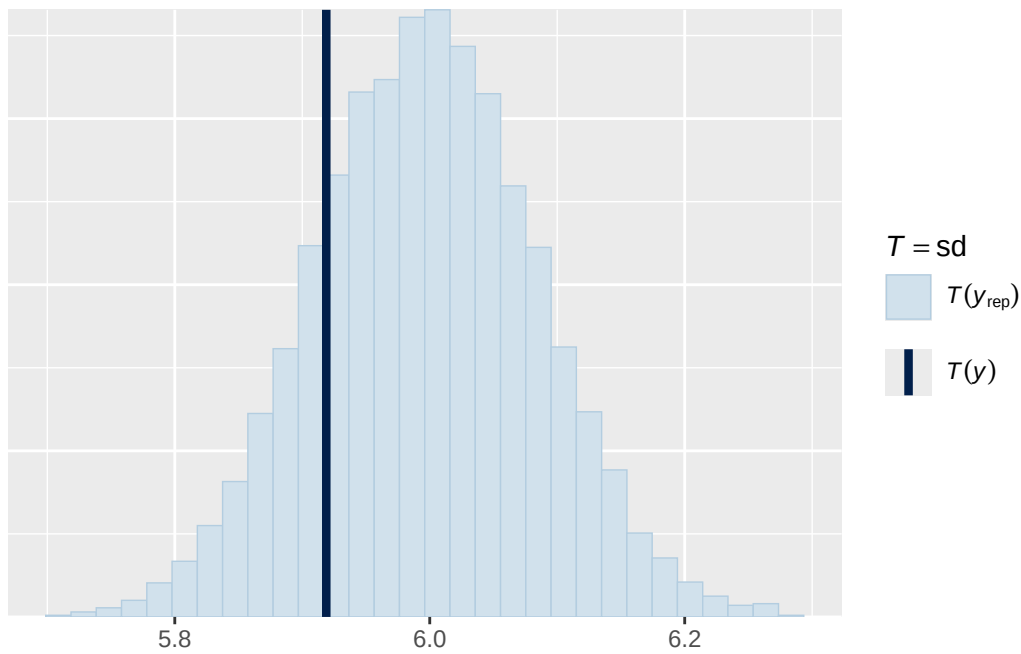


Using all posterior draws for ppc type 'stat' by default.

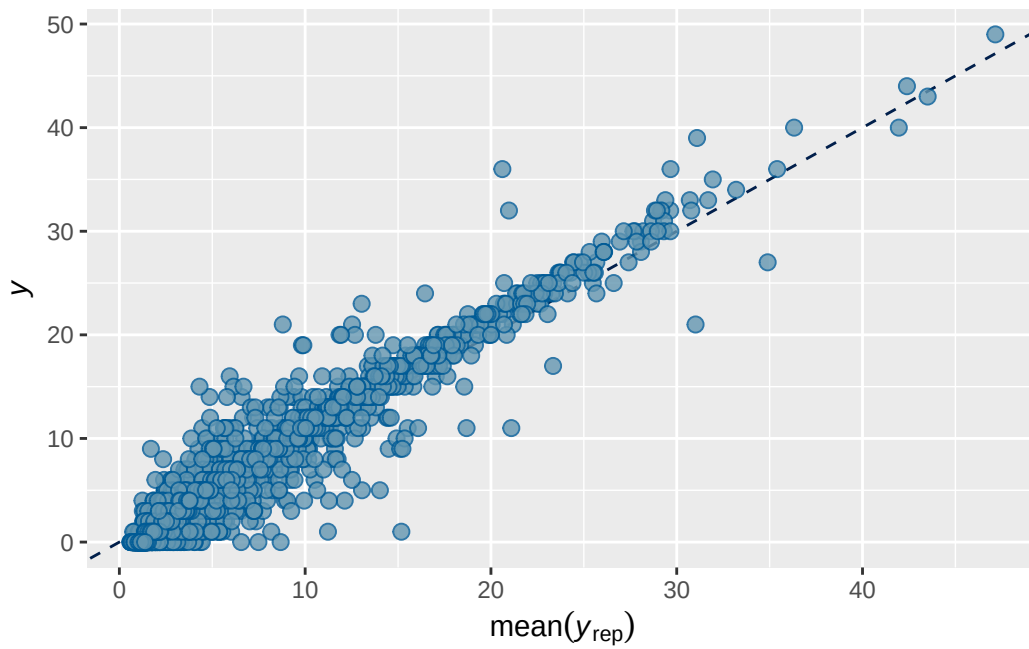
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 2B longitudinal ADHD present output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8274823 0.004586237 0.8183964 0.8362702

```

```

Family: poisson
Links: mu = log
Formula: internalising_wave_3 ~ breast_development_p_wave_2 * adhd_status_pres + age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.849    0.015   0.821   0.878 1.001    1081    2057

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.114    0.033   0.057   0.186 1.003     431     364

```

### Regression Coefficients:

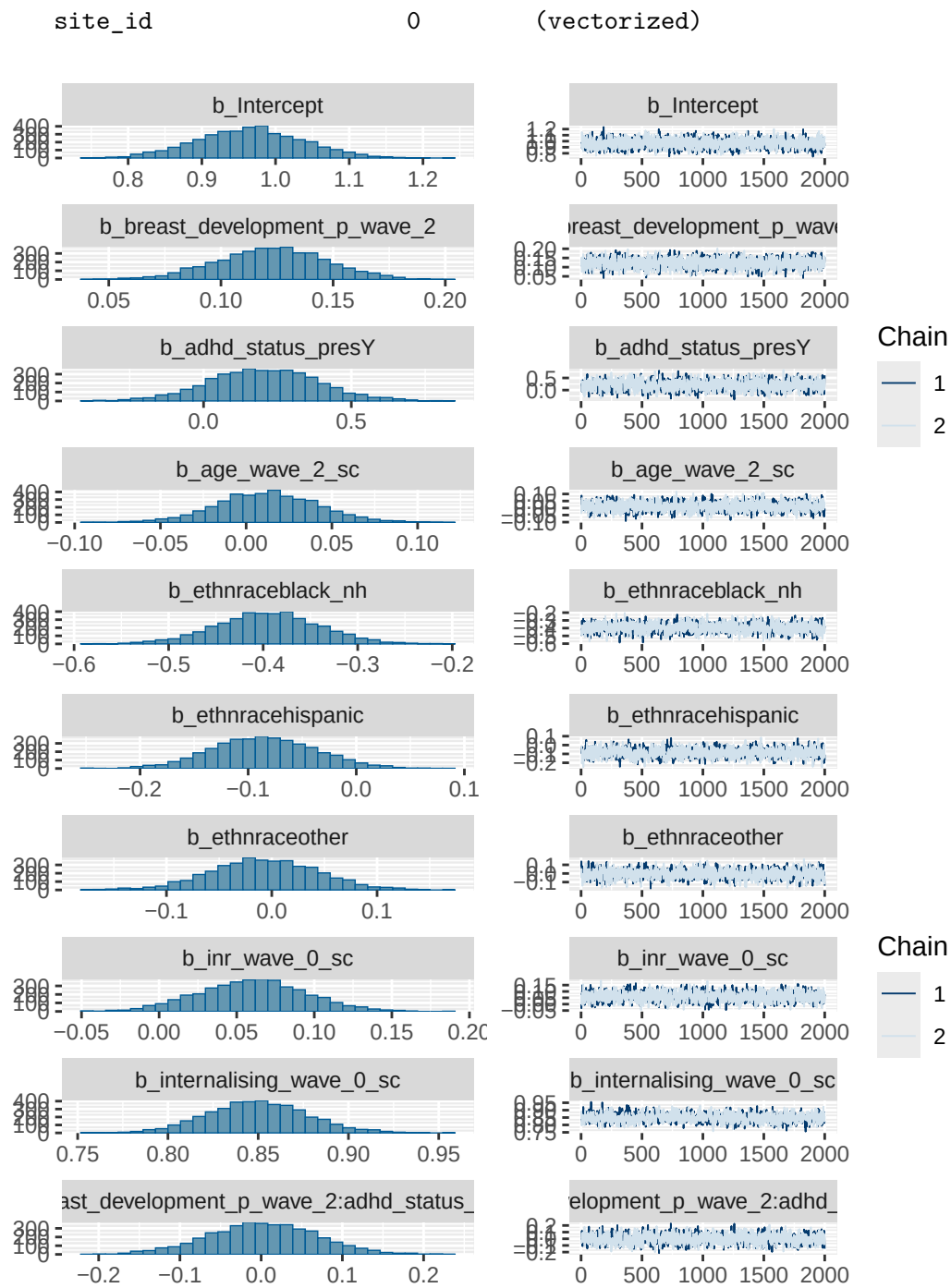
|                                               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------------------------------------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept                                     | 0.965    | 0.072     | 0.825    |          |      |          |          |
| breast_development_p_wave_2                   | 0.123    | 0.024     | 0.074    |          |      |          |          |
| adhd_status_presY                             | 0.204    | 0.183     | -0.156   |          |      |          |          |
| age_wave_2_sc                                 | 0.013    | 0.029     | -0.046   |          |      |          |          |
| ethnraceblack_nh                              | -0.395   | 0.055     | -0.505   |          |      |          |          |
| ethnracehispanic                              | -0.085   | 0.048     | -0.179   |          |      |          |          |
| ethnraceother                                 | -0.007   | 0.051     | -0.106   |          |      |          |          |
| inr_wave_0_sc                                 | 0.060    | 0.035     | -0.007   |          |      |          |          |
| internalising_wave_0_sc                       | 0.850    | 0.027     | 0.797    |          |      |          |          |
| breast_development_p_wave_2:adhd_status_presY | 0.002    | 0.067     | -0.131   |          |      |          |          |
|                                               | u-95% CI | Rhat      | Bulk_ESS | Tail_ESS |      |          |          |
| Intercept                                     | 1.106    | 1.001     | 1261     | 1930     |      |          |          |
| breast_development_p_wave_2                   | 0.169    | 1.000     | 1223     | 2060     |      |          |          |
| adhd_status_presY                             | 0.571    | 1.002     | 1089     | 1992     |      |          |          |
| age_wave_2_sc                                 | 0.071    | 1.001     | 1292     | 2223     |      |          |          |
| ethnraceblack_nh                              | -0.288   | 1.001     | 1174     | 2007     |      |          |          |
| ethnracehispanic                              | 0.010    | 1.002     | 1073     | 1838     |      |          |          |
| ethnraceother                                 | 0.092    | 1.001     | 1146     | 1819     |      |          |          |
| inr_wave_0_sc                                 | 0.129    | 1.001     | 1134     | 2084     |      |          |          |

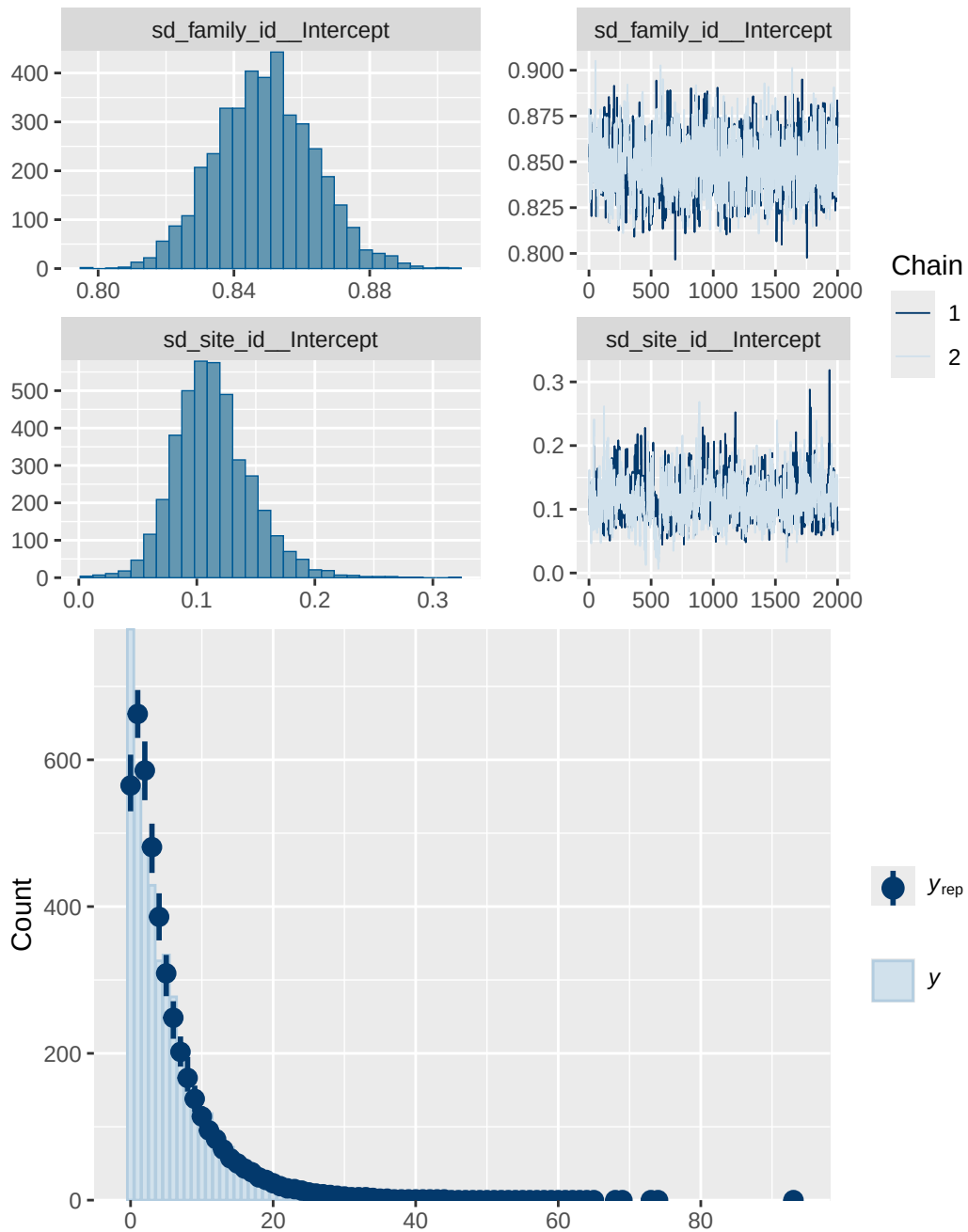
|                                               |       |       |      |      |
|-----------------------------------------------|-------|-------|------|------|
| internalising_wave_0_sc                       | 0.905 | 1.015 | 880  | 1394 |
| breast_development_p_wave_2:adhd_status_presY | 0.133 | 1.001 | 1084 | 1958 |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

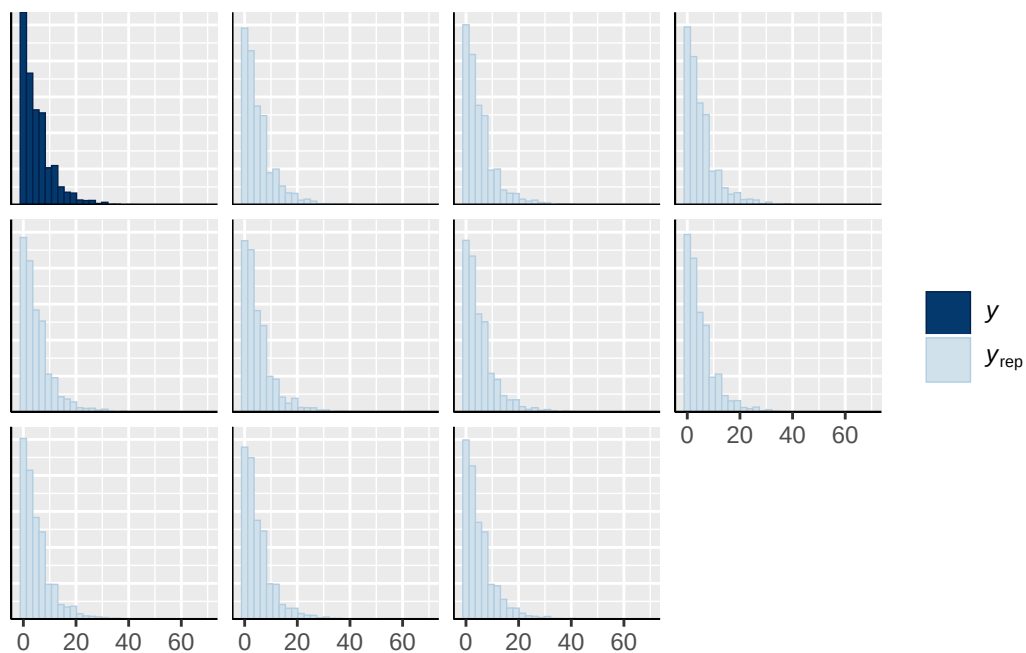
## Model 2B longitudinal ADHD present diagnostics

|                                 | prior        | class        | coef                                          |
|---------------------------------|--------------|--------------|-----------------------------------------------|
|                                 | normal(0, 1) | b            |                                               |
|                                 | normal(0, 1) | b            | adhd_status_presY                             |
|                                 | normal(0, 1) | b            | age_wave_2_sc                                 |
|                                 | normal(0, 1) | b            | breast_development_p_wave_2                   |
|                                 | normal(0, 1) | b            | breast_development_p_wave_2:adhd_status_presY |
|                                 | normal(0, 1) | b            | ethnraceblack_nh                              |
|                                 | normal(0, 1) | b            | ethnracehispanic                              |
|                                 | normal(0, 1) | b            | ethnraceother                                 |
|                                 | normal(0, 1) | b            | inr_wave_0_sc                                 |
|                                 | normal(0, 1) | b            | internalising_wave_0_sc                       |
| student_t(3, 1.4, 2.5)          | Intercept    |              |                                               |
| student_t(3, 0, 2.5)            | sd           |              |                                               |
| student_t(3, 0, 2.5)            | sd           |              |                                               |
| student_t(3, 0, 2.5)            | sd           |              | Intercept                                     |
| student_t(3, 0, 2.5)            | sd           |              |                                               |
| student_t(3, 0, 2.5)            | sd           |              | Intercept                                     |
| group resp dpar nlpar lb ub tag |              | source       |                                               |
|                                 |              | user         |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | default      |                                               |
|                                 | 0            | default      |                                               |
| family_id                       | 0            | (vectorized) |                                               |
| family_id                       | 0            | (vectorized) |                                               |
| site_id                         | 0            | (vectorized) |                                               |

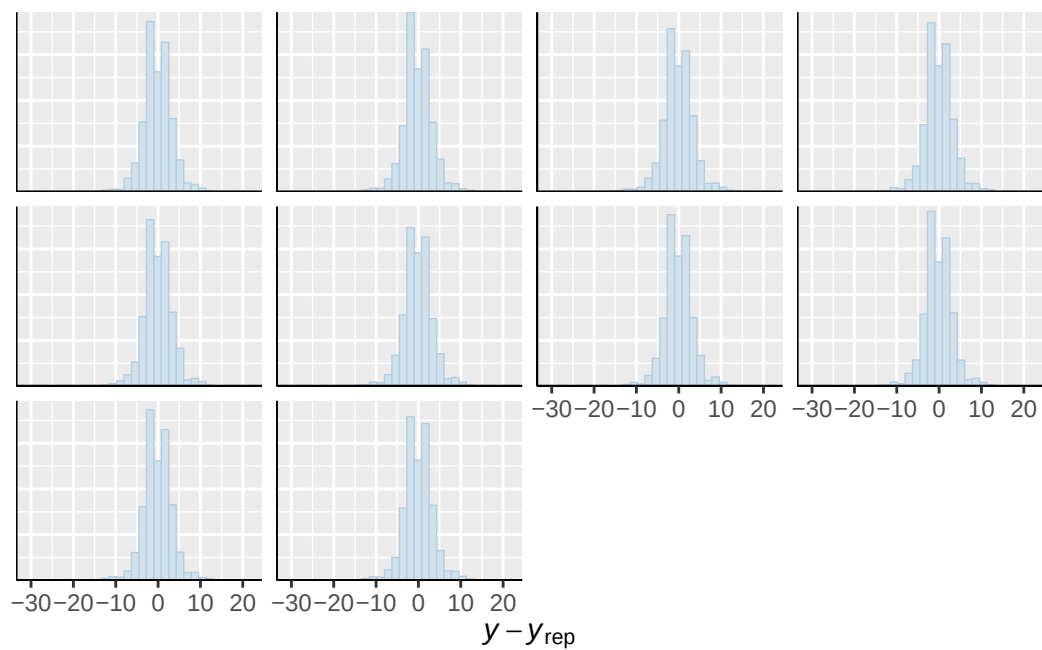




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



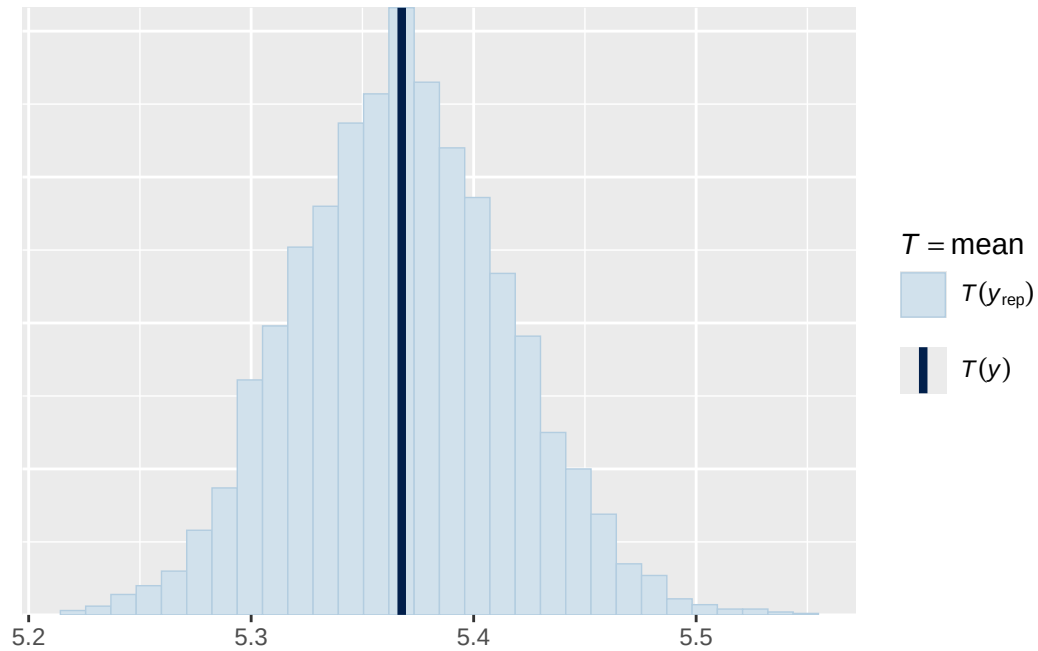
Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



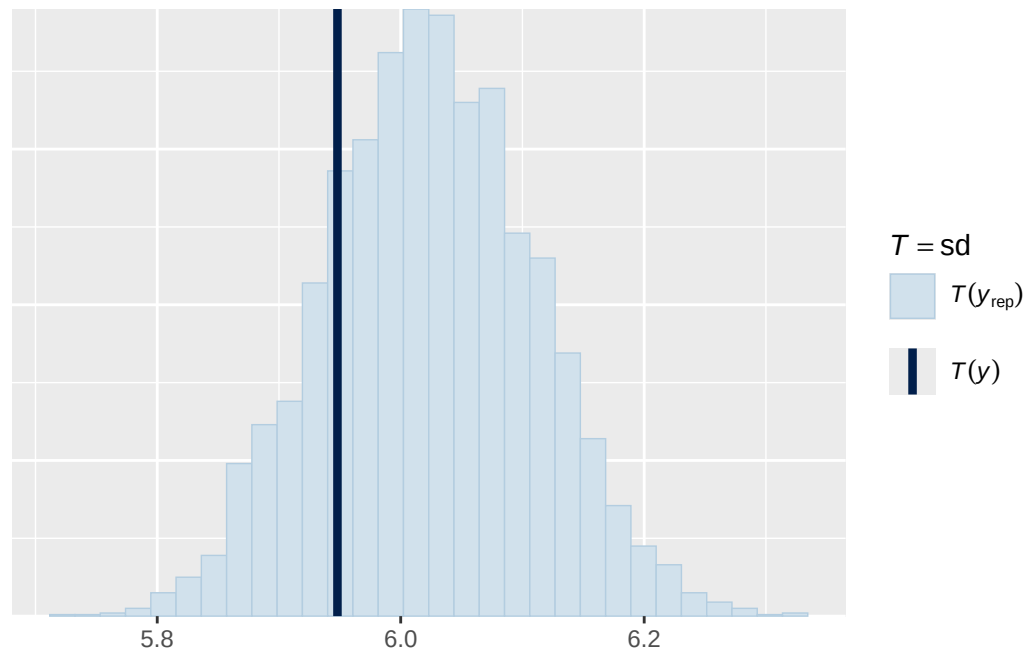
Using all posterior draws for ppc type 'stat' by default.



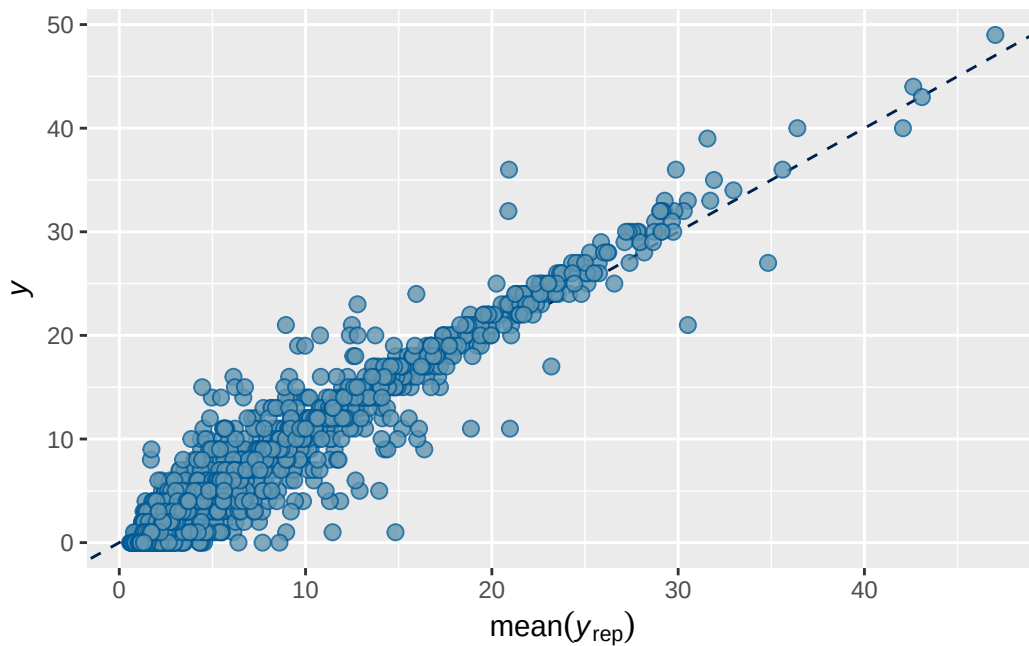
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 3 longitudinal ADHD present output

|    | Estimate  | Est.Error  | Q2.5      | Q97.5     |
|----|-----------|------------|-----------|-----------|
| R2 | 0.8212615 | 0.00533859 | 0.8102805 | 0.8313783 |

```
Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_pres + +age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family\_id (Number of levels: 3915)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 1.026    | 0.021     | 0.986    | 1.067    | 1.002 | 1495     | 3351     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.070    | 0.037     | 0.006    | 0.151    | 1.001 | 602      | 1363     |

# Regression Coefficients:

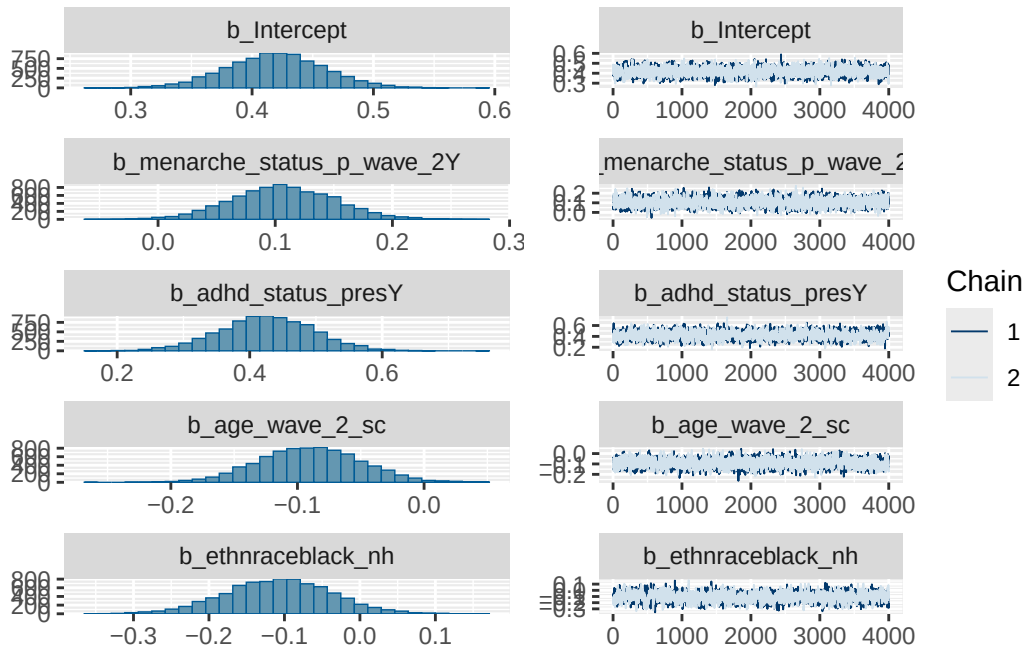
|                                             | Estimate | Est.Error | 1-95% CI |          |  |
|---------------------------------------------|----------|-----------|----------|----------|--|
| Intercept                                   | 0.419    | 0.040     | 0.339    |          |  |
| menarche_status_p_wave_2Y                   | 0.107    | 0.044     | 0.023    |          |  |
| adhd_status_presY                           | 0.425    | 0.071     | 0.285    |          |  |
| age_wave_2_sc                               | -0.090   | 0.040     | -0.169   |          |  |
| ethnraceblack_nh                            | -0.108   | 0.071     | -0.248   |          |  |
| ethnracehispanic                            | -0.113   | 0.059     | -0.231   |          |  |
| ethnraceother                               | -0.038   | 0.065     | -0.167   |          |  |
| inr_wave_0_sc                               | -0.006   | 0.045     | -0.091   |          |  |
| externalising_wave_0_sc                     | 1.029    | 0.035     | 0.961    |          |  |
| menarche_status_p_wave_2Y:adhd_status_presY | 0.179    | 0.108     | -0.037   |          |  |
|                                             | u-95% CI | Rhat      | Bulk_ESS | Tail_ESS |  |
| Intercept                                   | 0.499    | 1.000     | 2211     | 3678     |  |
| menarche_status_p_wave_2Y                   | 0.194    | 1.001     | 2205     | 3690     |  |
| adhd_status_presY                           | 0.564    | 1.001     | 1595     | 2521     |  |
| age_wave_2_sc                               | -0.012   | 1.001     | 2276     | 3214     |  |
| ethnraceblack_nh                            | 0.030    | 1.003     | 1493     | 2467     |  |
| ethnracehispanic                            | -0.001   | 1.000     | 2006     | 3547     |  |
| ethnraceother                               | 0.089    | 1.001     | 1677     | 3299     |  |
| inr_wave_0_sc                               | 0.085    | 1.000     | 1889     | 3855     |  |
| externalising_wave_0_sc                     | 1.097    | 1.003     | 1583     | 2501     |  |
| menarche_status_p_wave_2Y:adhd_status_presY | 0.384    | 1.002     | 1491     | 2666     |  |

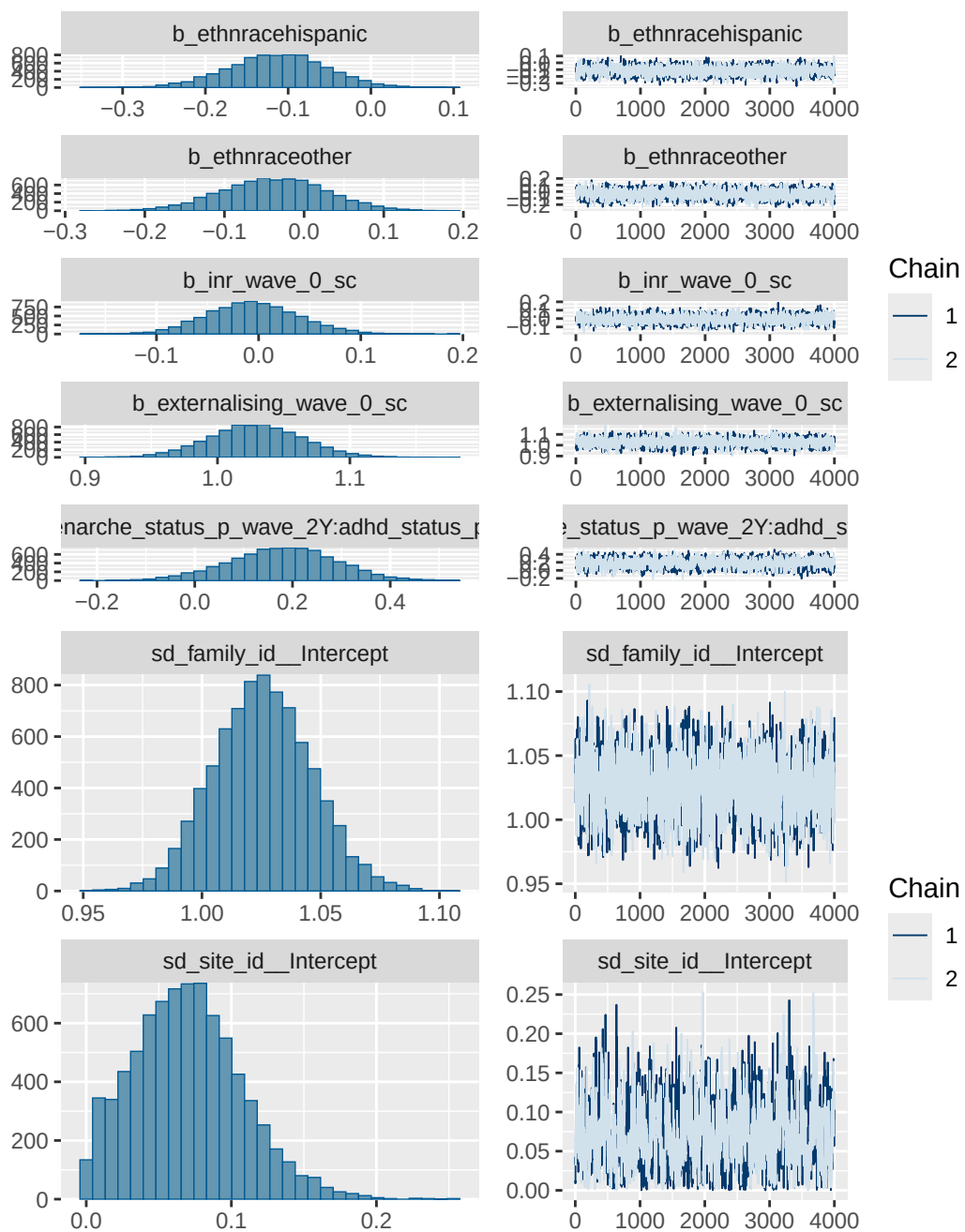
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

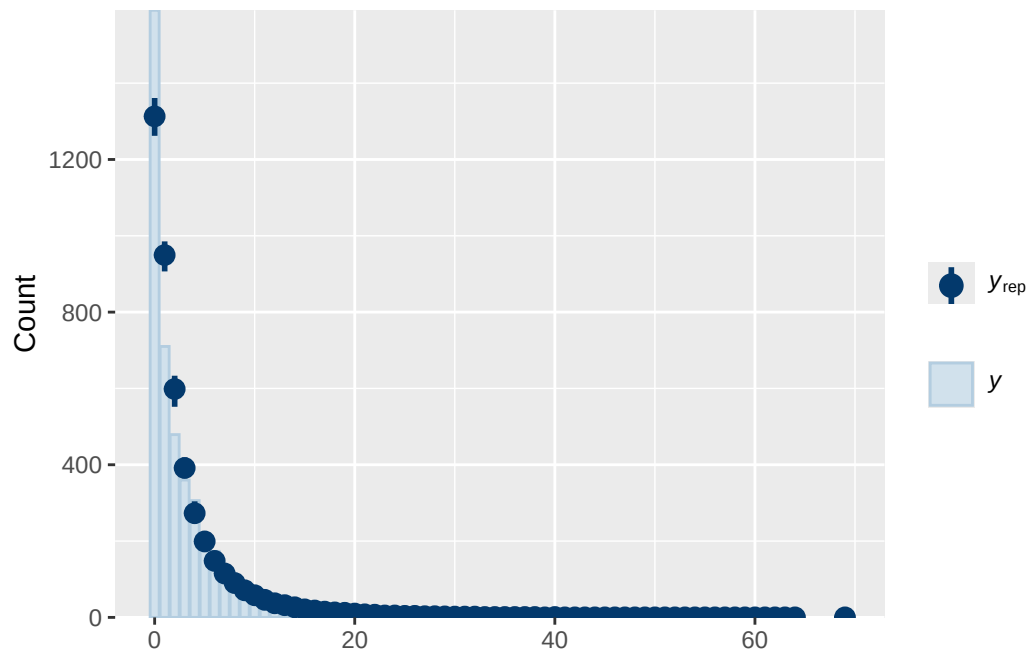
## Model 3 longitudinal ADHD present diagnostics

| prior        | class | coef                                        |
|--------------|-------|---------------------------------------------|
| normal(0, 1) | b     |                                             |
| normal(0, 1) | b     | adhd_status_presY                           |
| normal(0, 1) | b     | age_wave_2_sc                               |
| normal(0, 1) | b     | ethnraceblack_nh                            |
| normal(0, 1) | b     | ethnracehispanic                            |
| normal(0, 1) | b     | ethnraceother                               |
| normal(0, 1) | b     | externalising_wave_0_sc                     |
| normal(0, 1) | b     | inr_wave_0_sc                               |
| normal(0, 1) | b     | menarche_status_p_wave_2Y                   |
| normal(0, 1) | b     | menarche_status_p_wave_2Y:adhd_status_presY |

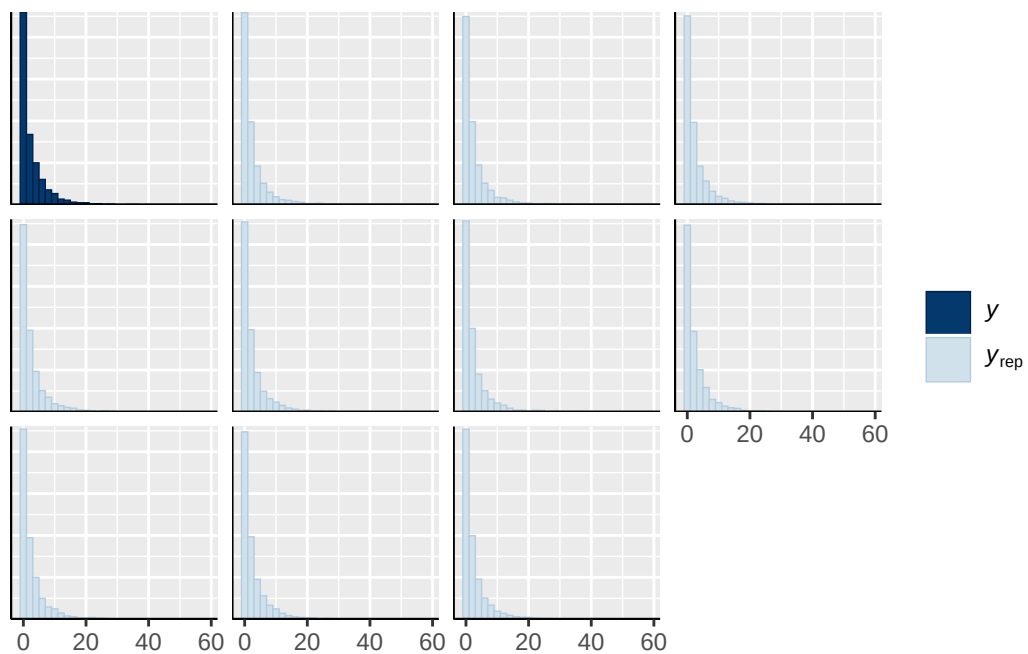
|                                 |              |              |
|---------------------------------|--------------|--------------|
| student_t(3, 0, 2.9)            | Intercept    |              |
| student_t(3, 0, 2.9)            | sd           |              |
| student_t(3, 0, 2.9)            | sd           |              |
| student_t(3, 0, 2.9)            | sd           | Intercept    |
| student_t(3, 0, 2.9)            | sd           |              |
| student_t(3, 0, 2.9)            | sd           | Intercept    |
| group resp dpar nlpar lb ub tag | source       |              |
|                                 | user         |              |
|                                 | (vectorized) |              |
|                                 | (vectorized) |              |
|                                 | (vectorized) |              |
|                                 | (vectorized) |              |
|                                 | (vectorized) |              |
|                                 | (vectorized) |              |
|                                 | (vectorized) |              |
|                                 | (vectorized) |              |
|                                 | default      |              |
|                                 | default      |              |
| family_id                       | 0            | (vectorized) |
| family_id                       | 0            | (vectorized) |
| site_id                         | 0            | (vectorized) |
| site_id                         | 0            | (vectorized) |



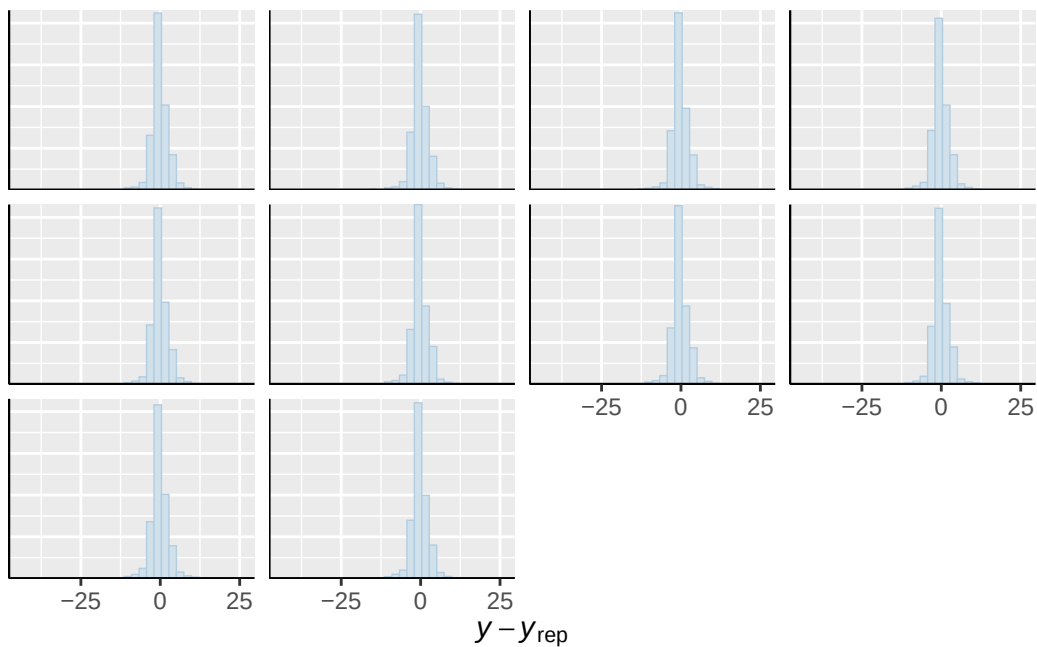




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

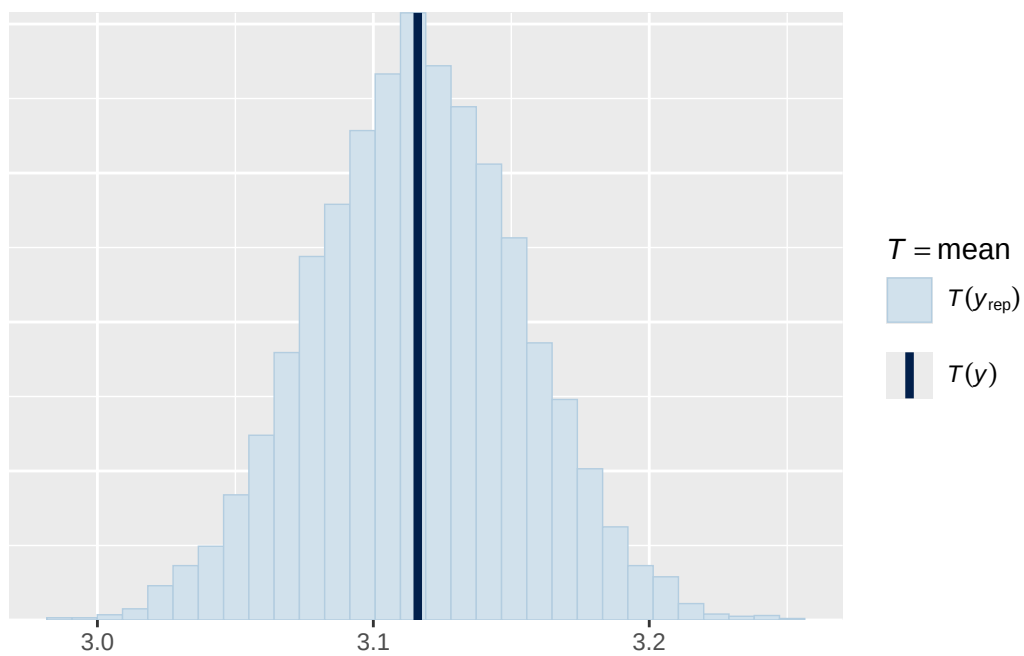


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

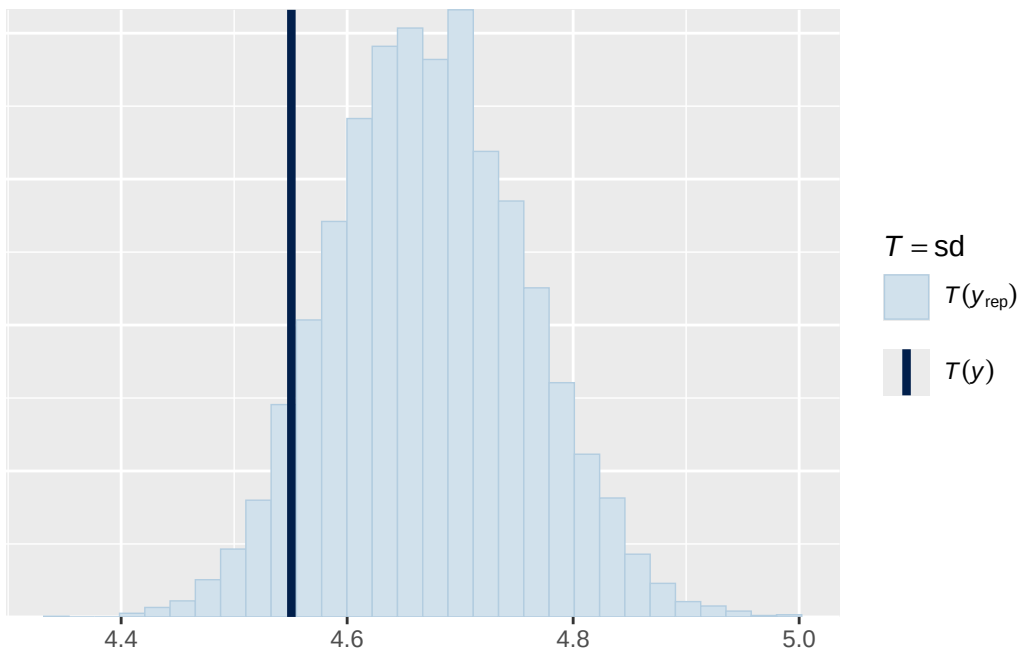


Using all posterior draws for ppc type 'stat' by default.

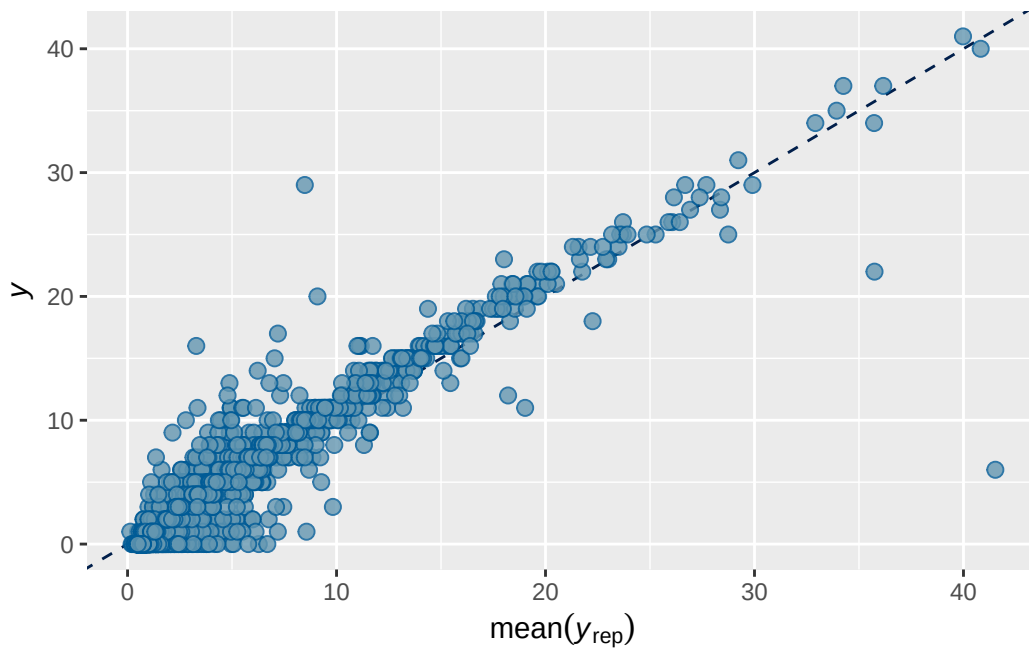
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter\_avg' by default.





## Model 3B longitudinal ADHD present output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8238293 0.005235983 0.8129815 0.8334161

```

```

Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2 * adhd_status_pres + age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.034    0.021   0.994   1.074 1.003    1919    3560

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.070    0.038   0.006   0.149 1.003     603    1245

```

### Regression Coefficients:

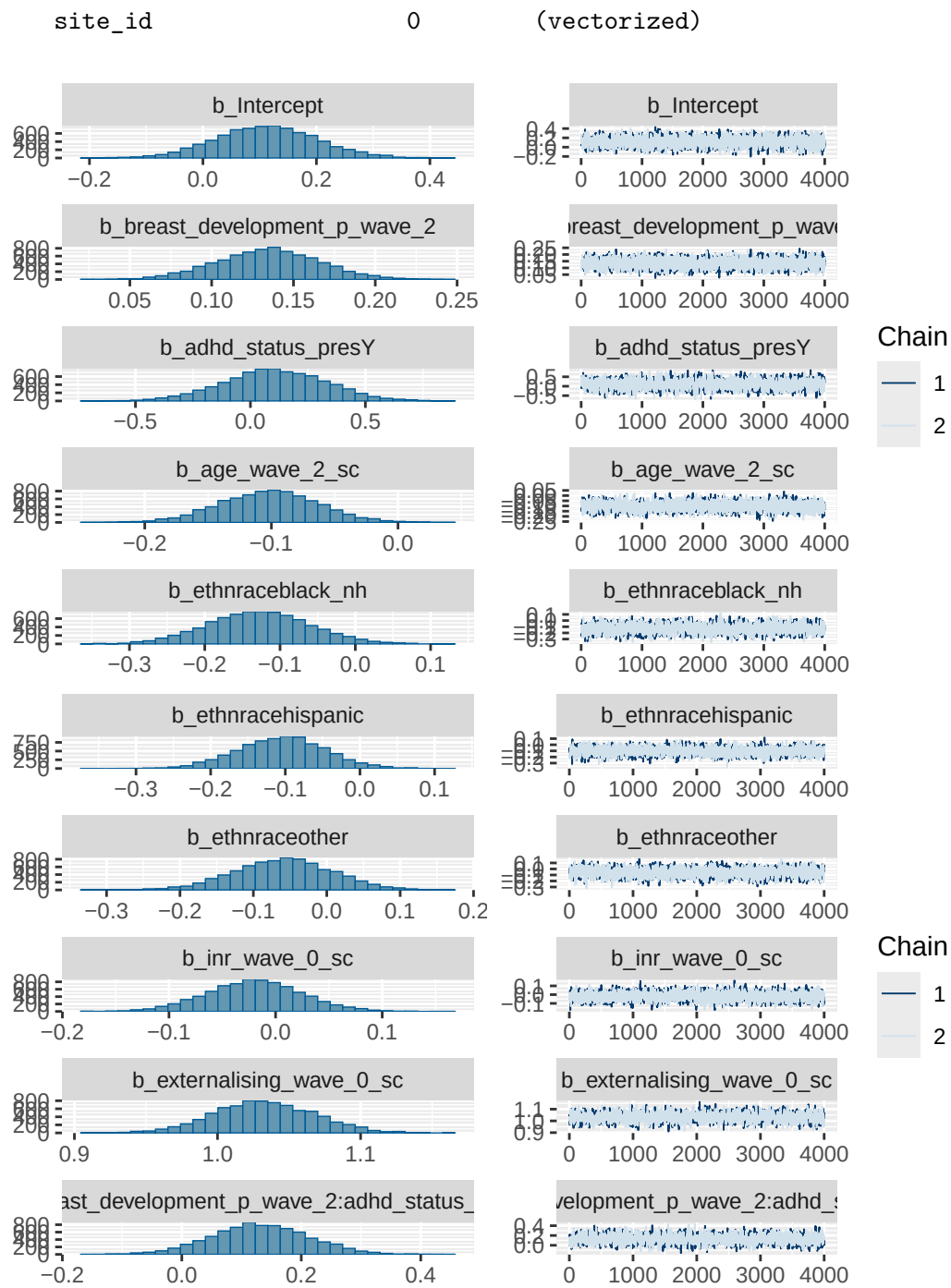
|                                               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------------------------------------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept                                     | 0.112    | 0.089     | -0.060   |          |      |          |          |
| breast_development_p_wave_2                   | 0.134    | 0.031     | 0.073    |          |      |          |          |
| adhd_status_presY                             | 0.107    | 0.222     | -0.331   |          |      |          |          |
| age_wave_2_sc                                 | -0.102   | 0.039     | -0.178   |          |      |          |          |
| ethnraceblack_nh                              | -0.128   | 0.068     | -0.260   |          |      |          |          |
| ethnracehispanic                              | -0.105   | 0.058     | -0.219   |          |      |          |          |
| ethnraceother                                 | -0.059   | 0.067     | -0.191   |          |      |          |          |
| inr_wave_0_sc                                 | -0.020   | 0.044     | -0.107   |          |      |          |          |
| externalising_wave_0_sc                       | 1.033    | 0.035     | 0.965    |          |      |          |          |
| breast_development_p_wave_2:adhd_status_presY | 0.131    | 0.081     | -0.028   |          |      |          |          |
|                                               |          |           |          | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
| Intercept                                     | 0.289    | 1.000     | 2241     | 3971     |      |          |          |
| breast_development_p_wave_2                   | 0.196    | 1.001     | 2257     | 4052     |      |          |          |
| adhd_status_presY                             | 0.537    | 1.001     | 1455     | 2891     |      |          |          |
| age_wave_2_sc                                 | -0.025   | 1.001     | 1864     | 3311     |      |          |          |
| ethnraceblack_nh                              | 0.005    | 1.001     | 1444     | 2701     |      |          |          |
| ethnracehispanic                              | 0.010    | 1.002     | 1820     | 3539     |      |          |          |
| ethnraceother                                 | 0.069    | 1.001     | 1609     | 3337     |      |          |          |
| inr_wave_0_sc                                 | 0.068    | 1.001     | 1776     | 3324     |      |          |          |

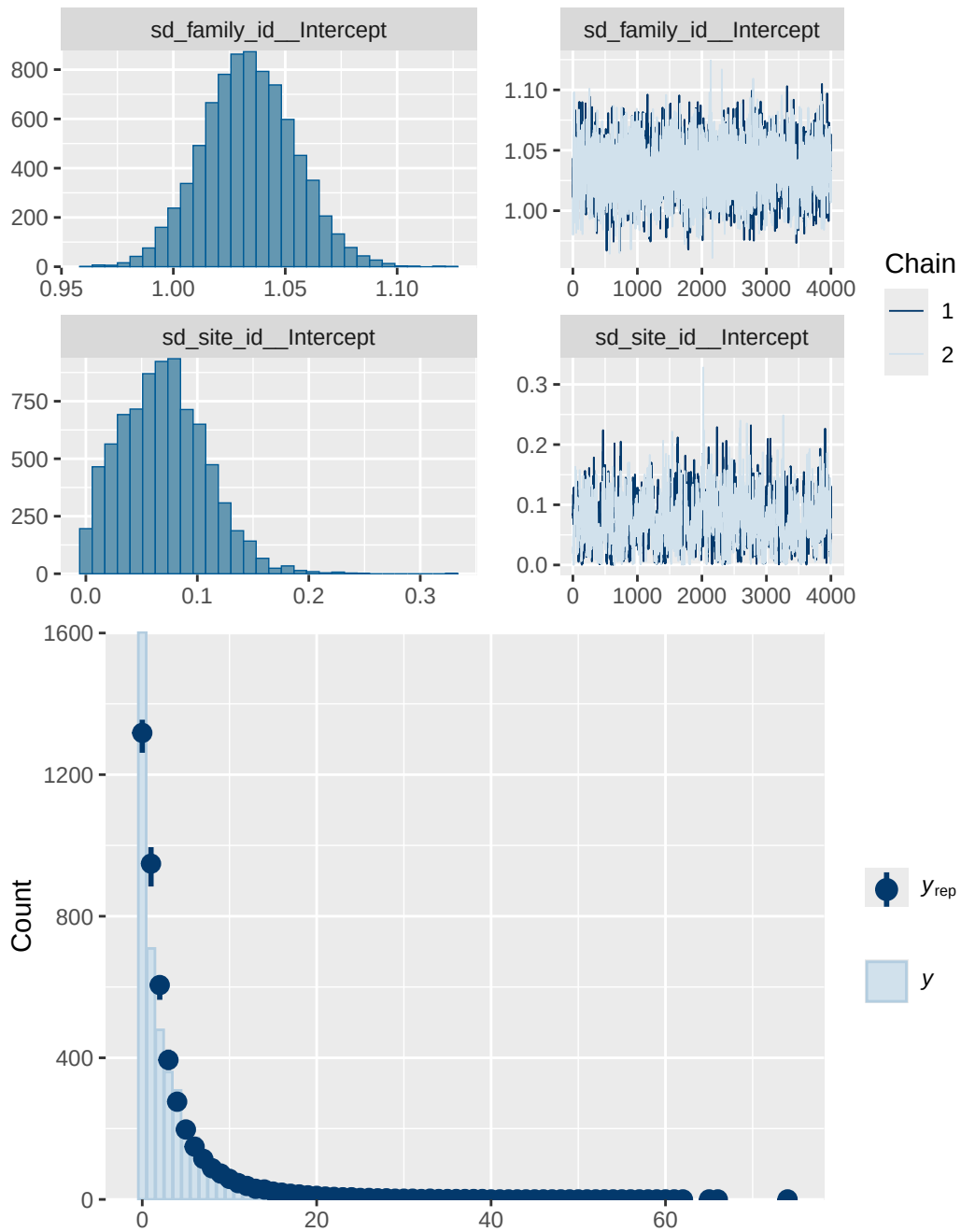
|                                               |       |       |      |      |
|-----------------------------------------------|-------|-------|------|------|
| externalising_wave_0_sc                       | 1.100 | 1.003 | 1240 | 2642 |
| breast_development_p_wave_2:adhd_status_presY | 0.289 | 1.001 | 1403 | 2937 |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

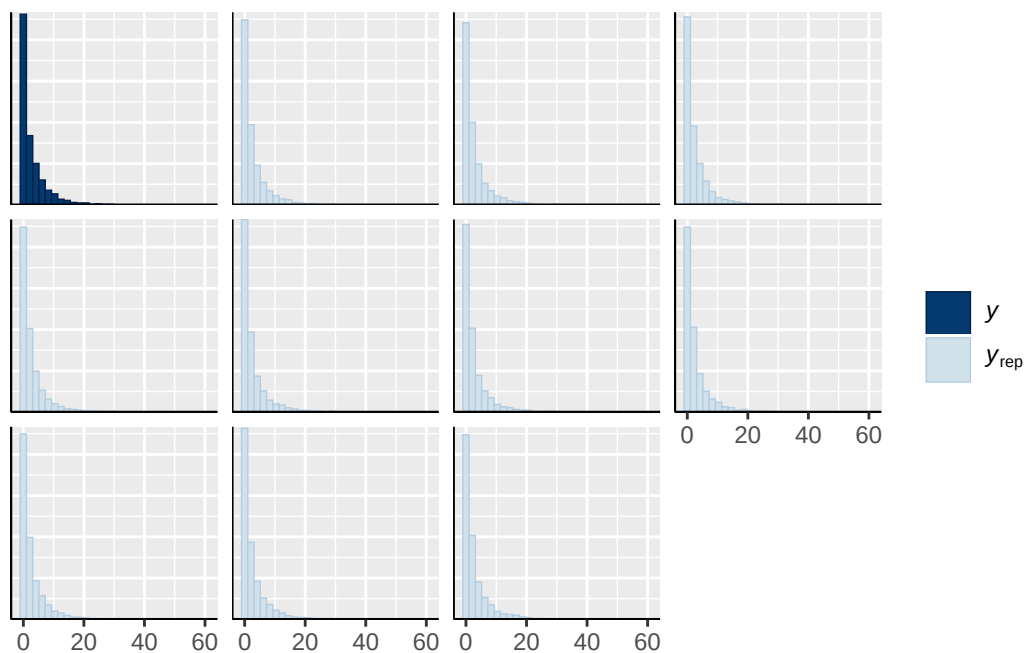
### Model 3B longitudinal ADHD present diagnostics

|                                 | prior        | class        | coef                                          |
|---------------------------------|--------------|--------------|-----------------------------------------------|
|                                 | normal(0, 1) | b            |                                               |
|                                 | normal(0, 1) | b            | adhd_status_presY                             |
|                                 | normal(0, 1) | b            | age_wave_2_sc                                 |
|                                 | normal(0, 1) | b            | breast_development_p_wave_2                   |
|                                 | normal(0, 1) | b            | breast_development_p_wave_2:adhd_status_presY |
|                                 | normal(0, 1) | b            | ethnraceblack_nh                              |
|                                 | normal(0, 1) | b            | ethnracehispanic                              |
|                                 | normal(0, 1) | b            | ethnraceother                                 |
|                                 | normal(0, 1) | b            | externalising_wave_0_sc                       |
|                                 | normal(0, 1) | b            | inr_wave_0_sc                                 |
| student_t(3, 0, 2.9)            | Intercept    |              |                                               |
| student_t(3, 0, 2.9)            | sd           |              |                                               |
| student_t(3, 0, 2.9)            | sd           |              |                                               |
| student_t(3, 0, 2.9)            | sd           |              | Intercept                                     |
| student_t(3, 0, 2.9)            | sd           |              |                                               |
| student_t(3, 0, 2.9)            | sd           |              | Intercept                                     |
| group resp dpar nlpar lb ub tag |              | source       |                                               |
|                                 |              | user         |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | (vectorized) |                                               |
|                                 |              | default      |                                               |
|                                 | 0            | default      |                                               |
| family_id                       | 0            | (vectorized) |                                               |
| family_id                       | 0            | (vectorized) |                                               |
| site_id                         | 0            | (vectorized) |                                               |

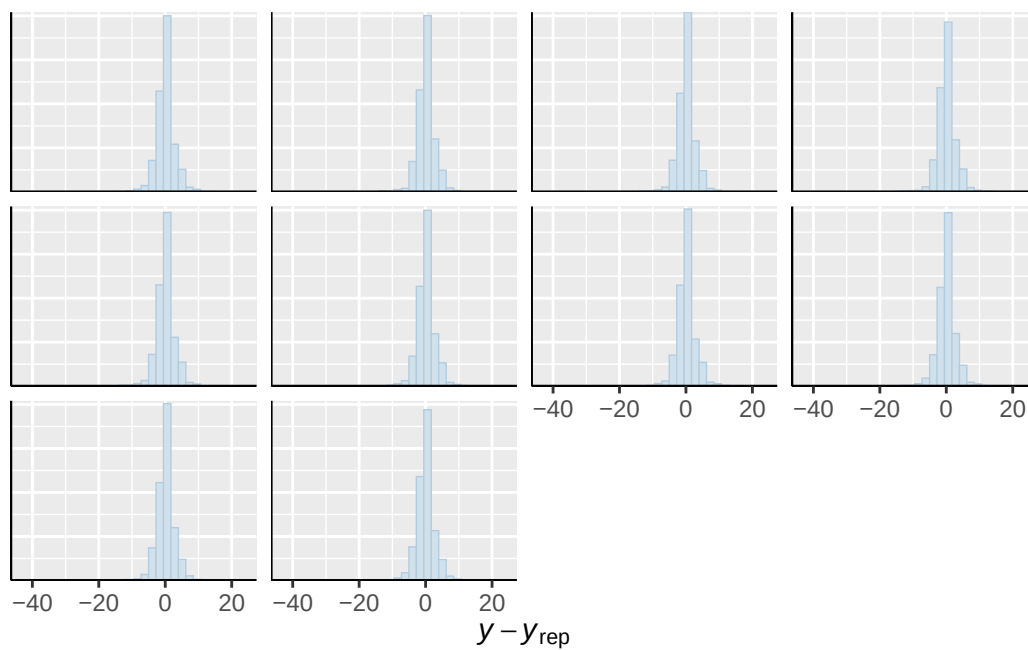




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

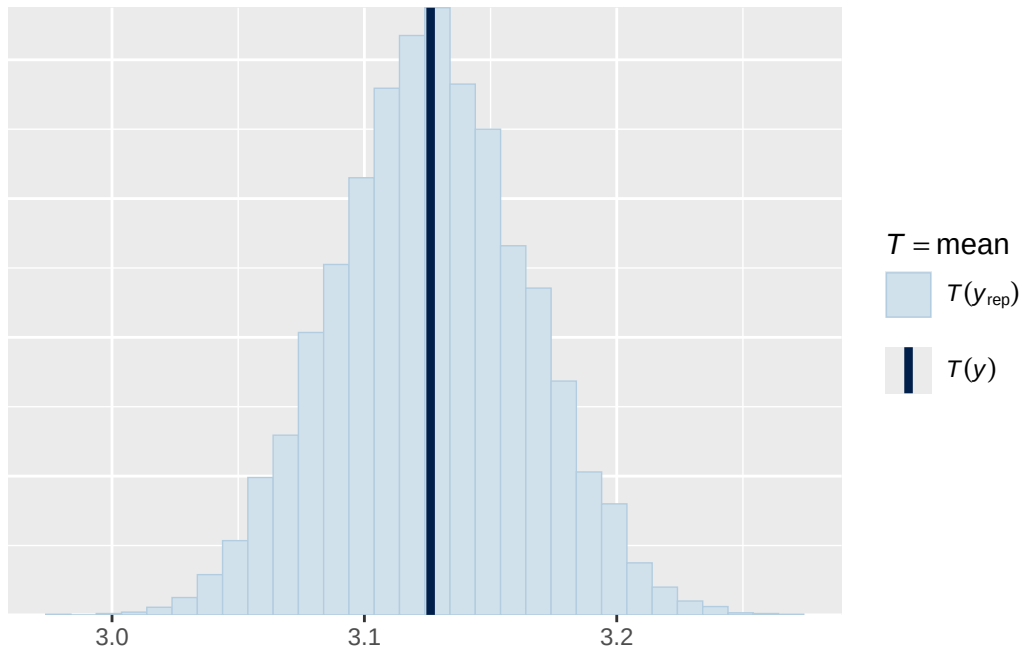


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

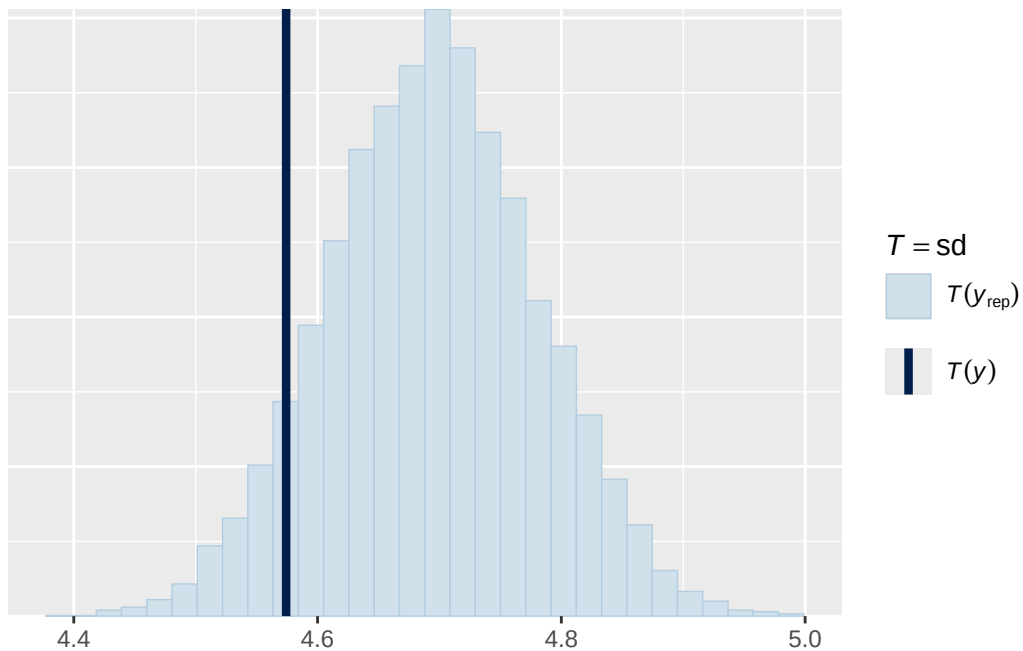


Using all posterior draws for ppc type 'stat' by default.

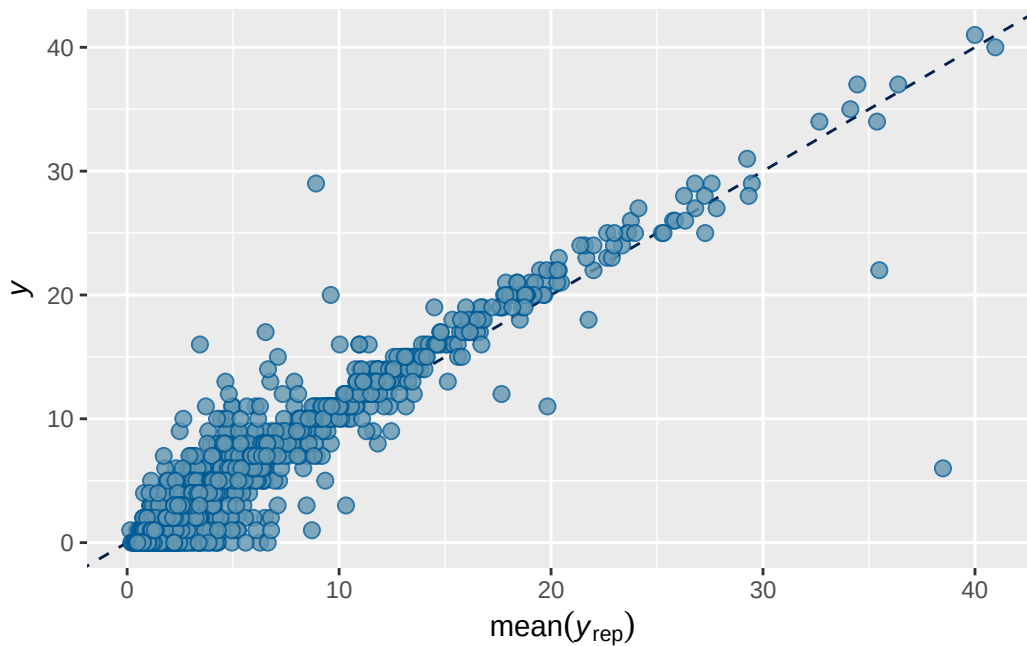
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1 longitudinal observed output

|    | Estimate | Est.Error   | Q2.5      | Q97.5     |
|----|----------|-------------|-----------|-----------|
| R2 | 0.739687 | 0.007910787 | 0.7238315 | 0.7547301 |

```
Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: obs_df_wide (Number of observations: 4043)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family\_id (Number of levels: 3513)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.878    | 0.022     | 0.836    | 0.922    | 1.000 | 2879     | 4782     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.078    | 0.038     | 0.008    | 0.157    | 1.002 | 1060     | 2028     |

# Regression Coefficients:

|                           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS |
|---------------------------|----------|-----------|----------|----------|-------|----------|
| Intercept                 | 0.151    | 0.040     | 0.072    | 0.229    | 1.000 | 6594     |
| menarche_status_p_wave_2Y | 0.114    | 0.044     | 0.029    | 0.201    | 1.000 | 6832     |
| age_wave_2_sc             | -0.121   | 0.043     | -0.207   | -0.037   | 1.000 | 7110     |
| ethnraceblack_nh          | -0.074   | 0.070     | -0.214   | 0.063    | 1.000 | 6034     |
| ethnracehispanic          | -0.040   | 0.062     | -0.161   | 0.082    | 1.001 | 5041     |
| ethnraceother             | 0.001    | 0.064     | -0.123   | 0.130    | 1.001 | 6102     |
| inr_wave_0_sc             | 0.089    | 0.045     | 0.001    | 0.175    | 1.000 | 5654     |
| adhd_traits_wave_0_sc     | 1.349    | 0.035     | 1.282    | 1.418    | 1.001 | 4610     |
| Tail_ESS                  |          |           |          |          |       |          |
| Intercept                 | 5853     |           |          |          |       |          |
| menarche_status_p_wave_2Y | 6557     |           |          |          |       |          |
| age_wave_2_sc             | 6797     |           |          |          |       |          |
| ethnraceblack_nh          | 6380     |           |          |          |       |          |
| ethnracehispanic          | 5955     |           |          |          |       |          |
| ethnraceother             | 6242     |           |          |          |       |          |
| inr_wave_0_sc             | 5872     |           |          |          |       |          |
| adhd_traits_wave_0_sc     | 5780     |           |          |          |       |          |

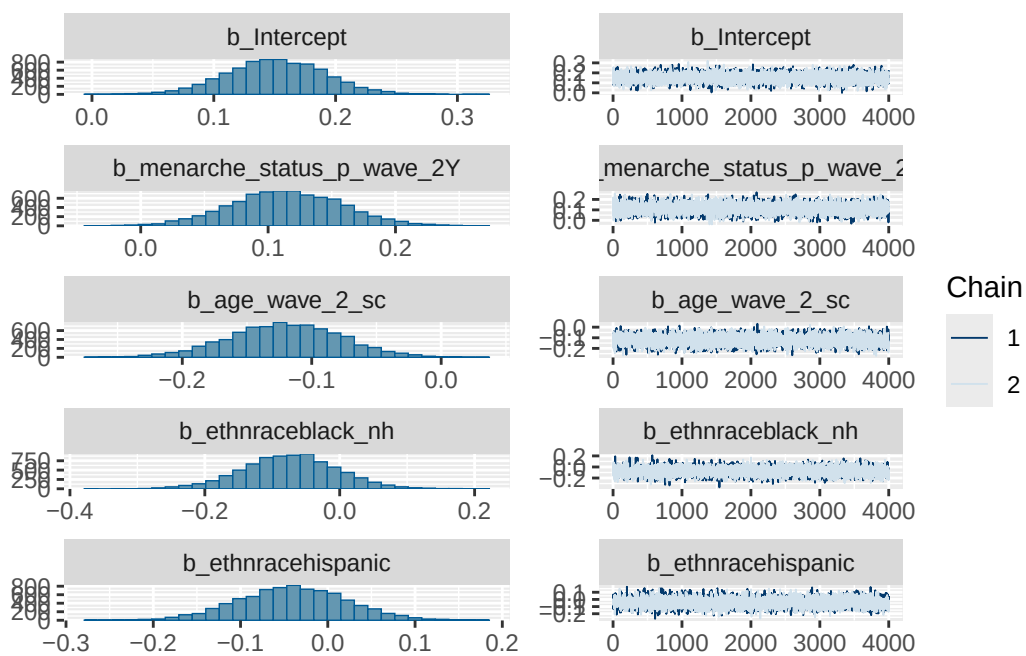
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

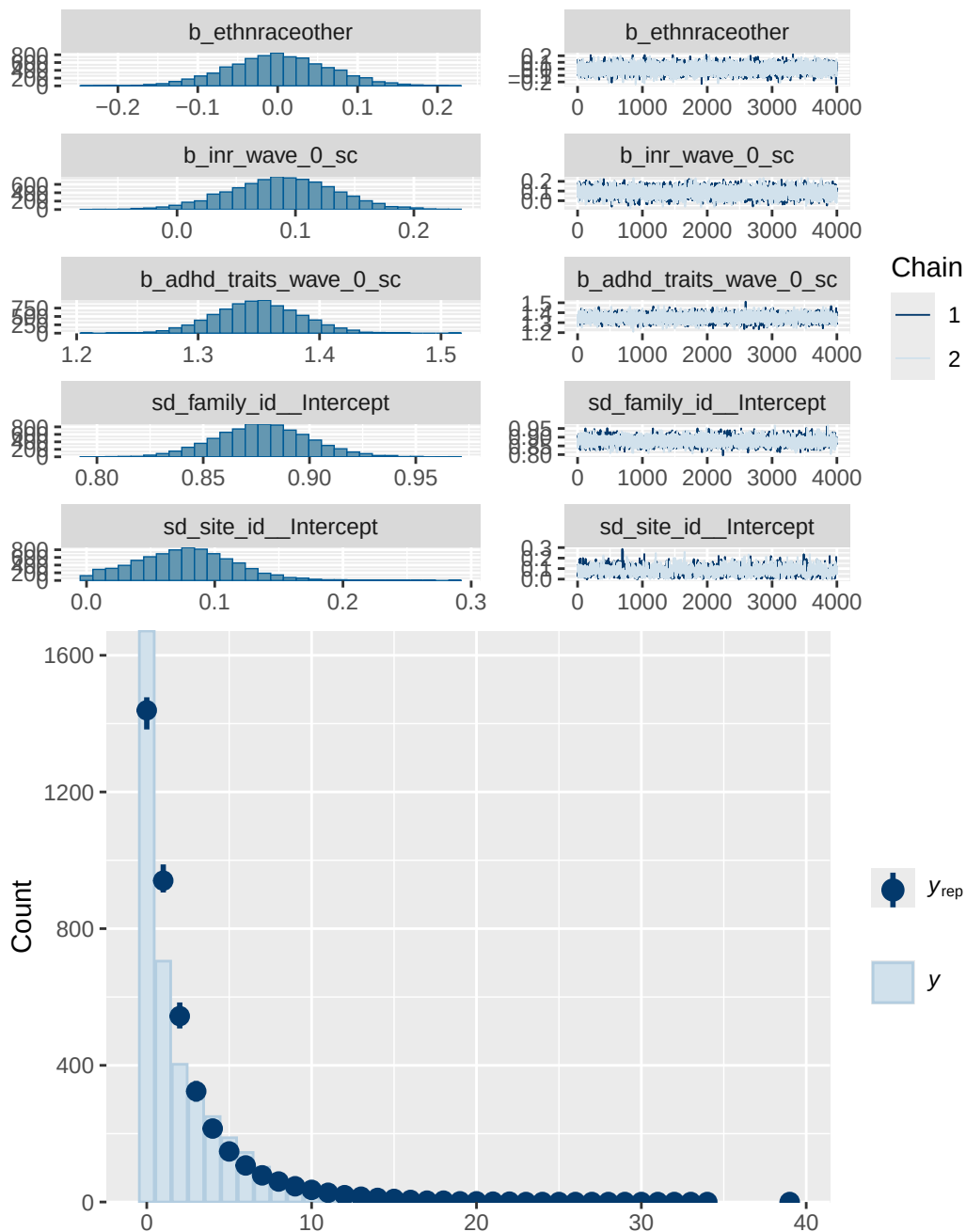
## Model 1 longitudinal observed diagnostics

| prior                | class     | coef                      | group     | resp | dpar |
|----------------------|-----------|---------------------------|-----------|------|------|
| normal(0, 1)         | b         |                           |           |      |      |
| normal(0, 1)         | b         | adhd_traits_wave_0_sc     |           |      |      |
| normal(0, 1)         | b         | age_wave_2_sc             |           |      |      |
| normal(0, 1)         | b         | ethnraceblack_nh          |           |      |      |
| normal(0, 1)         | b         | ethnracehispanic          |           |      |      |
| normal(0, 1)         | b         | ethnraceother             |           |      |      |
| normal(0, 1)         | b         | inr_wave_0_sc             |           |      |      |
| normal(0, 1)         | b         | menarche_status_p_wave_2Y |           |      |      |
| student_t(3, 0, 2.9) | Intercept |                           |           |      |      |
| student_t(3, 0, 2.9) | sd        |                           |           |      |      |
| student_t(3, 0, 2.9) | sd        |                           | family_id |      |      |
| student_t(3, 0, 2.9) | sd        | Intercept                 | family_id |      |      |
| student_t(3, 0, 2.9) | sd        |                           | site_id   |      |      |
| student_t(3, 0, 2.9) | sd        | Intercept                 | site_id   |      |      |

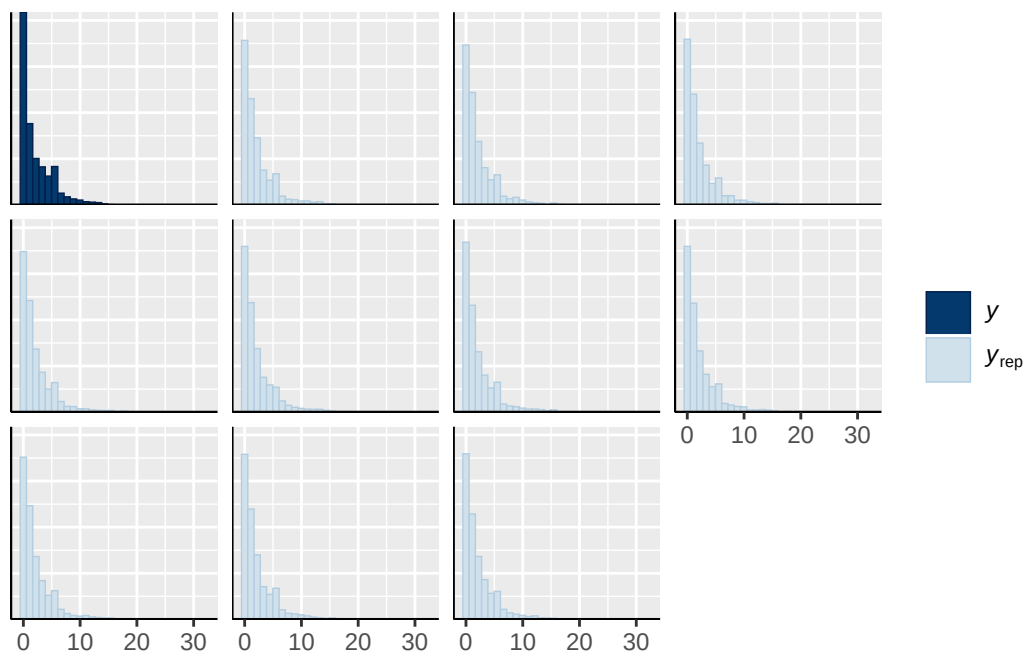


| nlpar | lb | ub | tag | source       | user |
|-------|----|----|-----|--------------|------|
|       |    |    |     | (vectorized) |      |
|       |    |    |     | (vectorized) |      |
|       |    |    |     | (vectorized) |      |
|       |    |    |     | (vectorized) |      |
|       |    |    |     | (vectorized) |      |
|       |    |    |     | (vectorized) |      |
|       |    |    |     | (vectorized) |      |
|       |    |    |     | default      |      |
| 0     |    |    |     | default      |      |
| 0     |    |    |     | (vectorized) |      |
| 0     |    |    |     | (vectorized) |      |
| 0     |    |    |     | (vectorized) |      |
| 0     |    |    |     | (vectorized) |      |

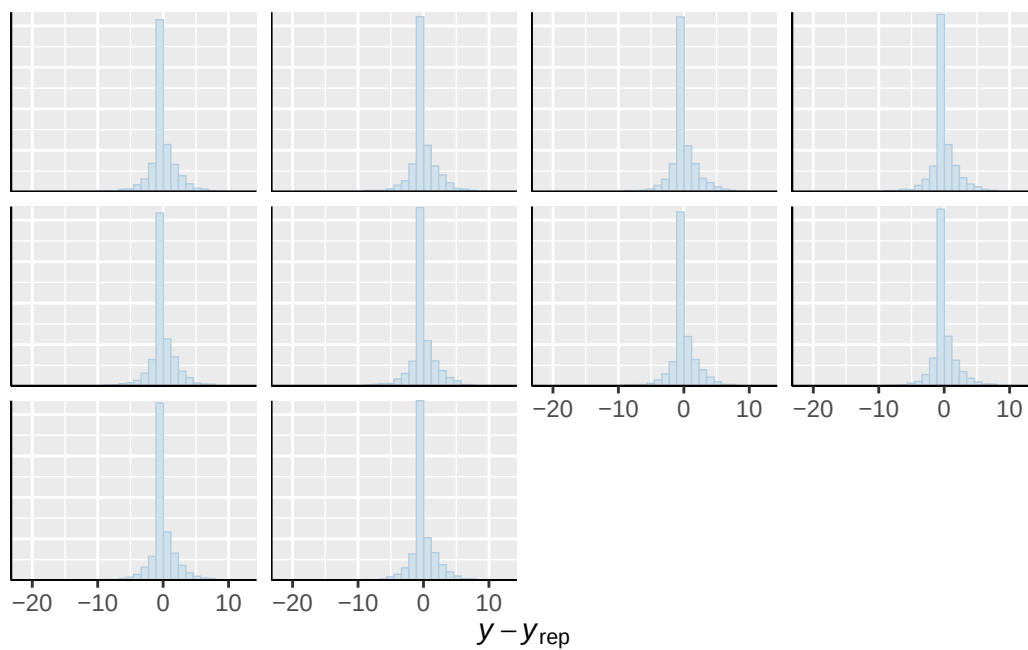




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

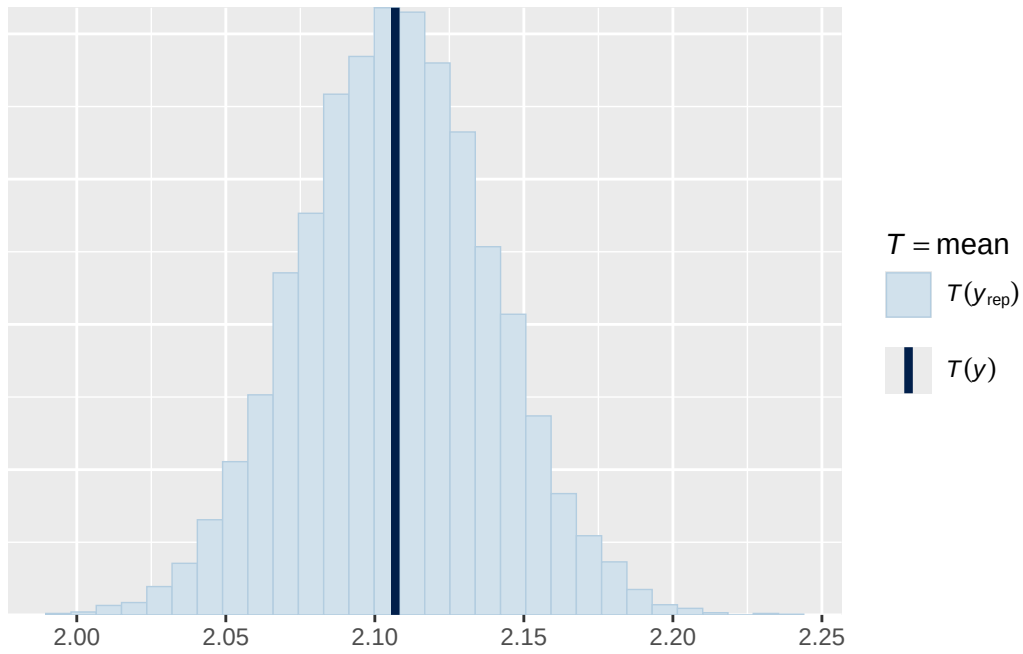


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

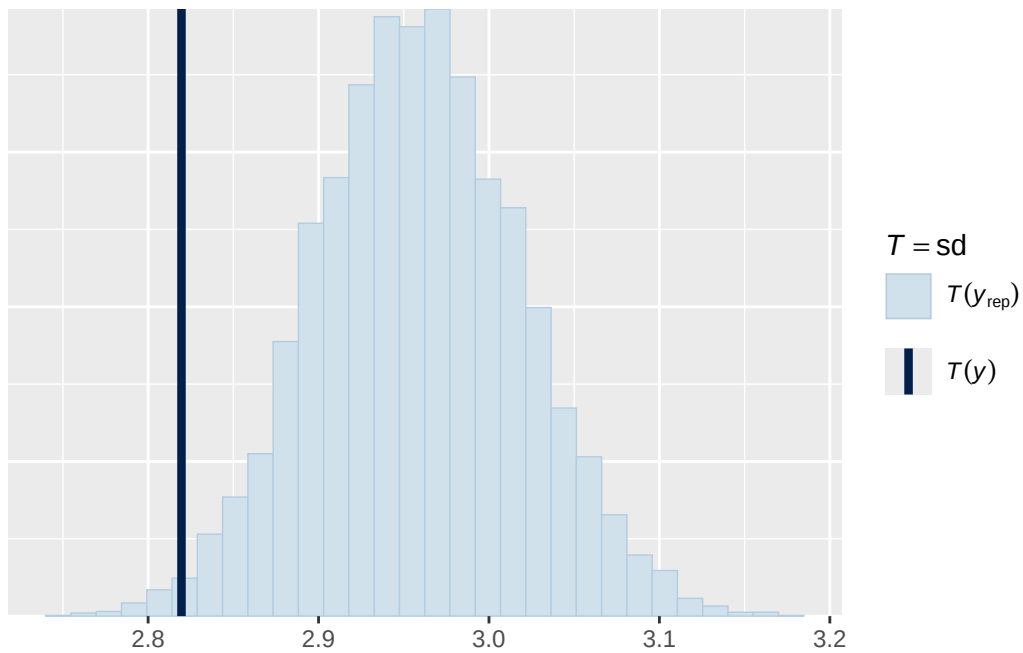


Using all posterior draws for ppc type 'stat' by default.

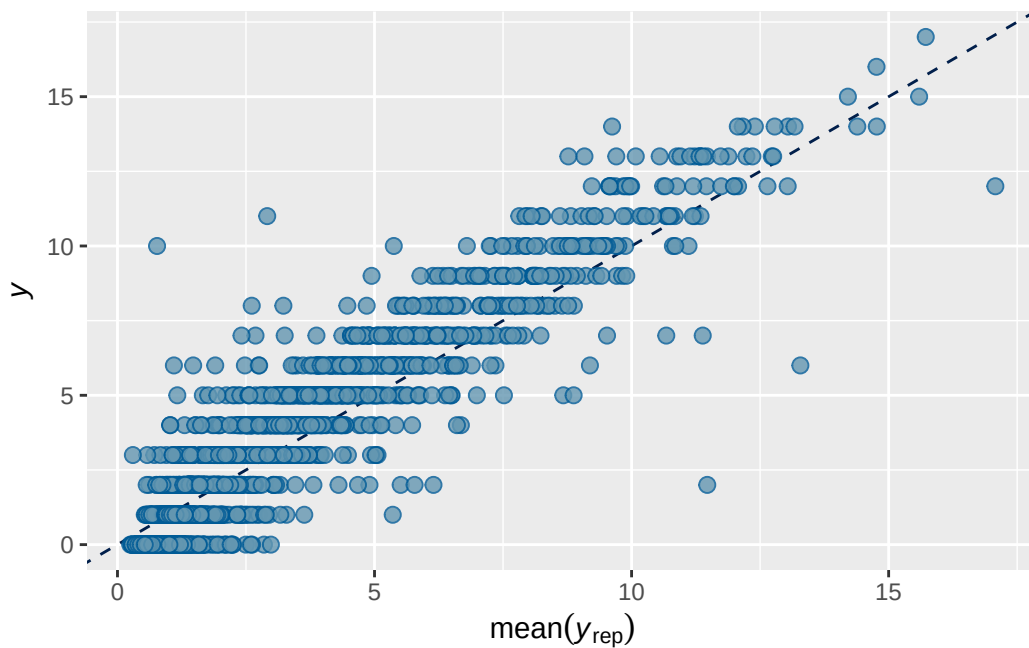
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1B longitudinal observed output

|    | Estimate  | Est.Error   | Q2.5     | Q97.5     |
|----|-----------|-------------|----------|-----------|
| R2 | 0.7391037 | 0.007829595 | 0.723389 | 0.7539044 |

Family: poisson  
 Links: mu = log  
 Formula: adhd\_traits\_wave\_3 ~ breast\_development\_p\_wave\_2\_sc + age\_wave\_2\_sc + ethnrace + in  
 Data: obs\_df\_wide (Number of observations: 4065)  
 Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;  
 total post-warmup draws = 8000

Multilevel Hyperparameters:

~family\_id (Number of levels: 3531)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.877    | 0.022     | 0.836    | 0.920    | 1.000 | 2348     | 4033     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.082    | 0.039     | 0.008    | 0.163    | 1.001 | 884      | 1020     |

# Regression Coefficients:

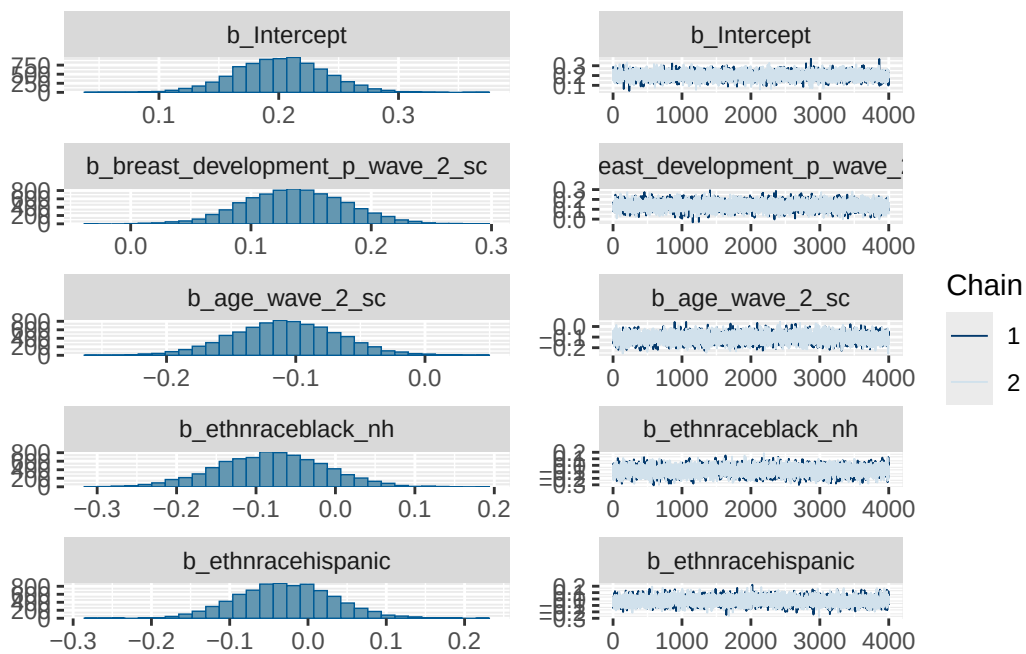
|                                | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  |
|--------------------------------|----------|-----------|----------|----------|-------|
| Intercept                      | 0.201    | 0.037     | 0.127    | 0.273    | 1.001 |
| breast_development_p_wave_2_sc | 0.137    | 0.042     | 0.056    | 0.219    | 1.001 |
| age_wave_2_sc                  | -0.108   | 0.041     | -0.189   | -0.028   | 1.001 |
| ethnraceblack_nh               | -0.081   | 0.067     | -0.213   | 0.049    | 1.001 |
| ethnracehispanic               | -0.028   | 0.061     | -0.144   | 0.092    | 1.001 |
| ethnraceother                  | -0.005   | 0.064     | -0.133   | 0.119    | 1.001 |
| inr_wave_0_sc                  | 0.077    | 0.045     | -0.010   | 0.163    | 1.001 |
| adhd_traits_wave_0_sc          | 1.344    | 0.035     | 1.274    | 1.413    | 1.001 |
|                                | Bulk_ESS | Tail_ESS  |          |          |       |
| Intercept                      | 3308     | 4150      |          |          |       |
| breast_development_p_wave_2_sc | 3280     | 4597      |          |          |       |
| age_wave_2_sc                  | 3034     | 4586      |          |          |       |
| ethnraceblack_nh               | 2708     | 4539      |          |          |       |
| ethnracehispanic               | 2348     | 4339      |          |          |       |
| ethnraceother                  | 3196     | 4514      |          |          |       |
| inr_wave_0_sc                  | 2921     | 4731      |          |          |       |
| adhd_traits_wave_0_sc          | 2155     | 3988      |          |          |       |

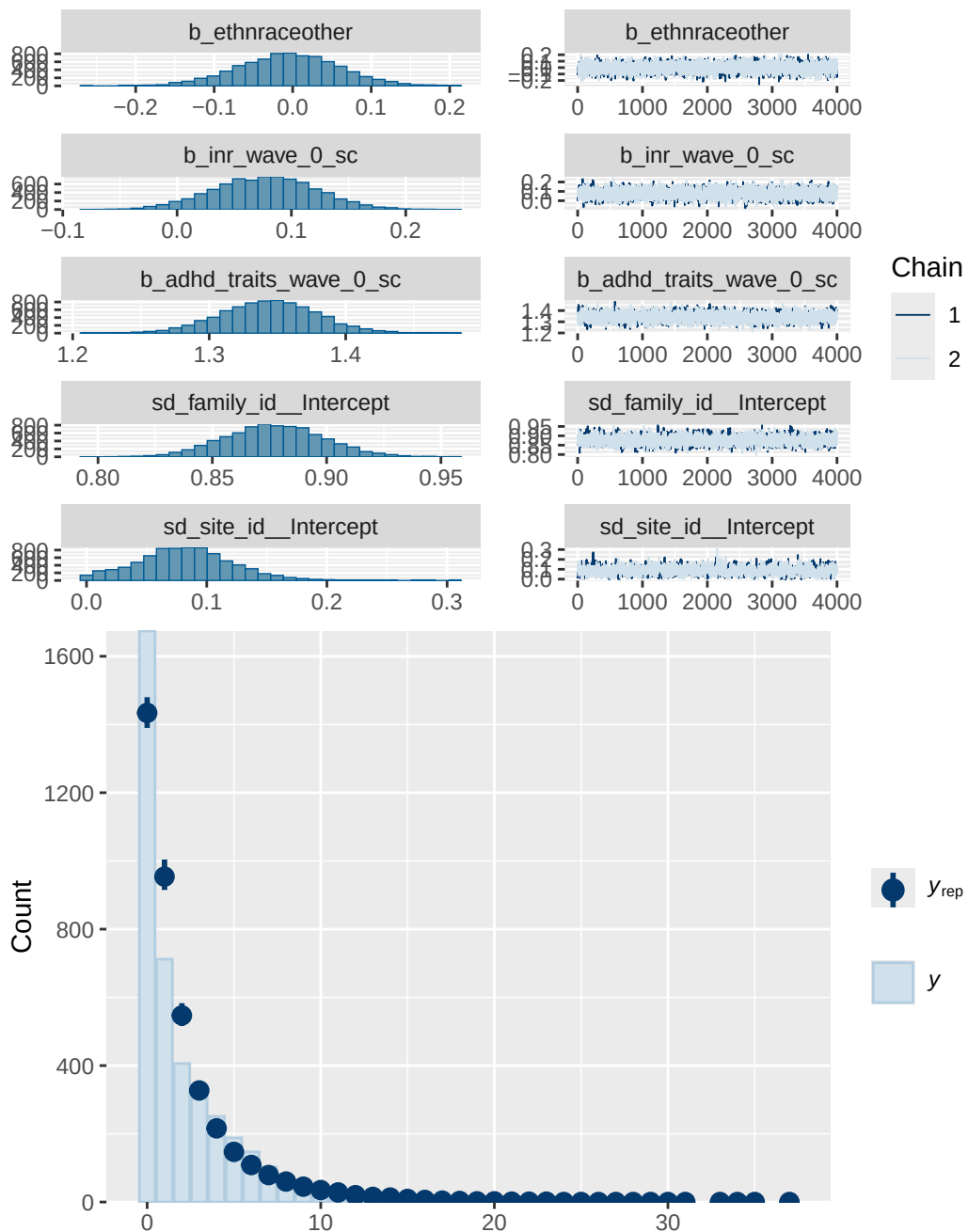
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

## Model 1B longitudinal observed diagnostics

| prior                | class     | coef                           | group     | resp |
|----------------------|-----------|--------------------------------|-----------|------|
| normal(0, 1)         | b         |                                |           |      |
| normal(0, 1)         | b         | adhd_traits_wave_0_sc          |           |      |
| normal(0, 1)         | b         | age_wave_2_sc                  |           |      |
| normal(0, 1)         | b         | breast_development_p_wave_2_sc |           |      |
| normal(0, 1)         | b         | ethnraceblack_nh               |           |      |
| normal(0, 1)         | b         | ethnracehispanic               |           |      |
| normal(0, 1)         | b         | ethnraceother                  |           |      |
| normal(0, 1)         | b         | inr_wave_0_sc                  |           |      |
| student_t(3, 0, 2.7) | Intercept |                                |           |      |
| student_t(3, 0, 2.7) | sd        |                                |           |      |
| student_t(3, 0, 2.7) | sd        |                                | family_id |      |
| student_t(3, 0, 2.7) | sd        | Intercept                      | family_id |      |
| student_t(3, 0, 2.7) | sd        |                                | site_id   |      |
| student_t(3, 0, 2.7) | sd        | Intercept                      | site_id   |      |

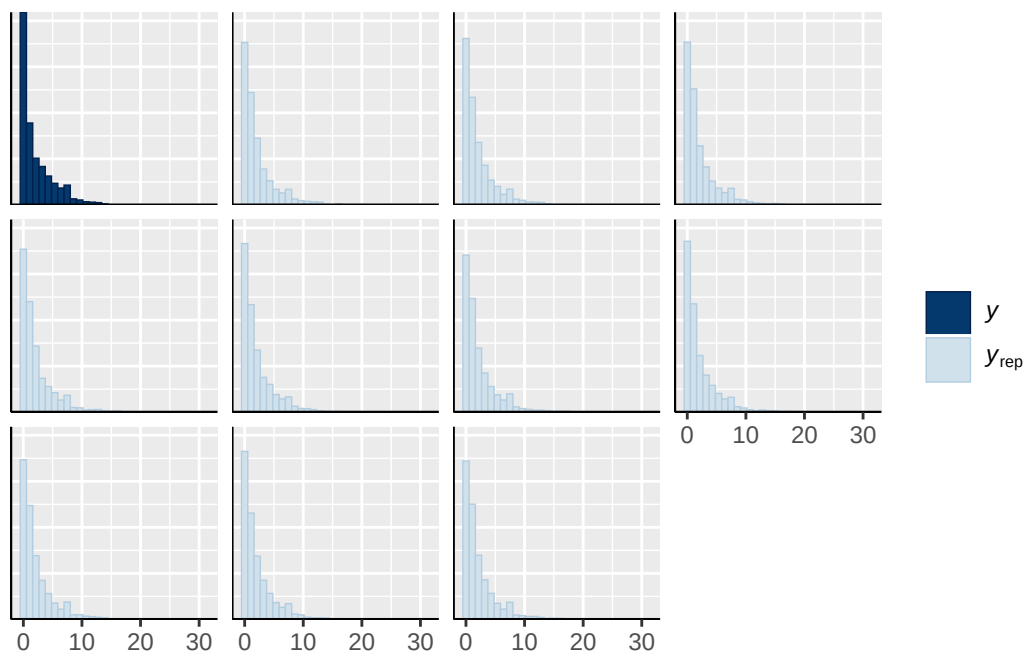
| dpar | nlpar | lb | ub | tag | source       | user |
|------|-------|----|----|-----|--------------|------|
|      |       |    |    |     | (vectorized) |      |
|      |       |    |    |     | (vectorized) |      |
|      |       |    |    |     | (vectorized) |      |
|      |       |    |    |     | (vectorized) |      |
|      |       |    |    |     | (vectorized) |      |
|      |       |    |    |     | (vectorized) |      |
|      |       |    |    |     | (vectorized) |      |
|      |       |    |    |     | default      |      |
|      | 0     |    |    |     | default      |      |
|      | 0     |    |    |     | (vectorized) |      |
|      | 0     |    |    |     | (vectorized) |      |
|      | 0     |    |    |     | (vectorized) |      |
|      | 0     |    |    |     | (vectorized) |      |



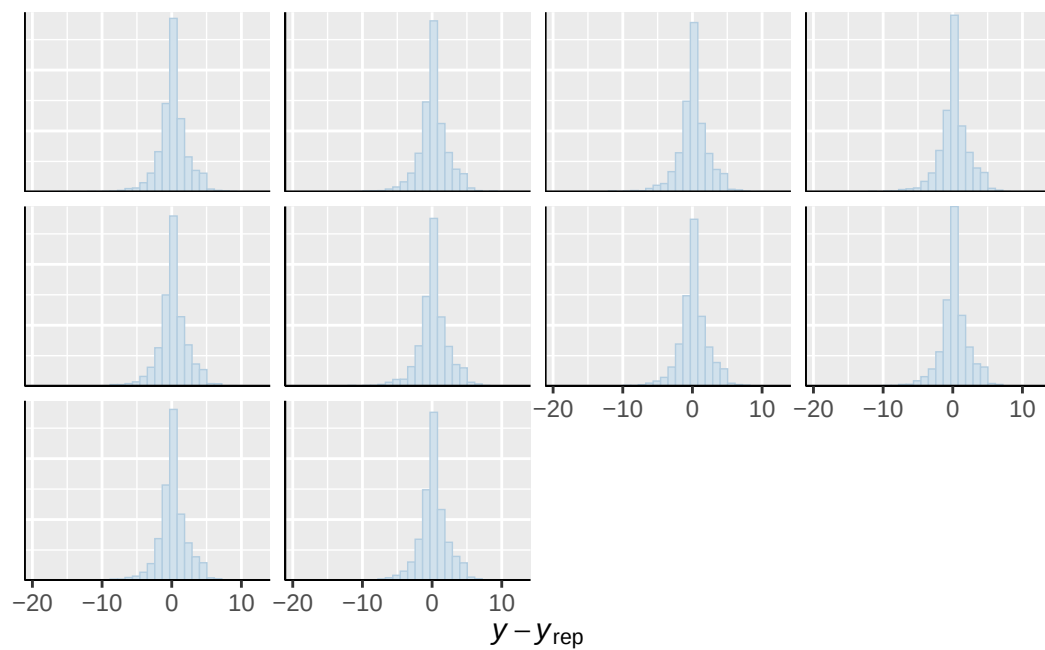


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



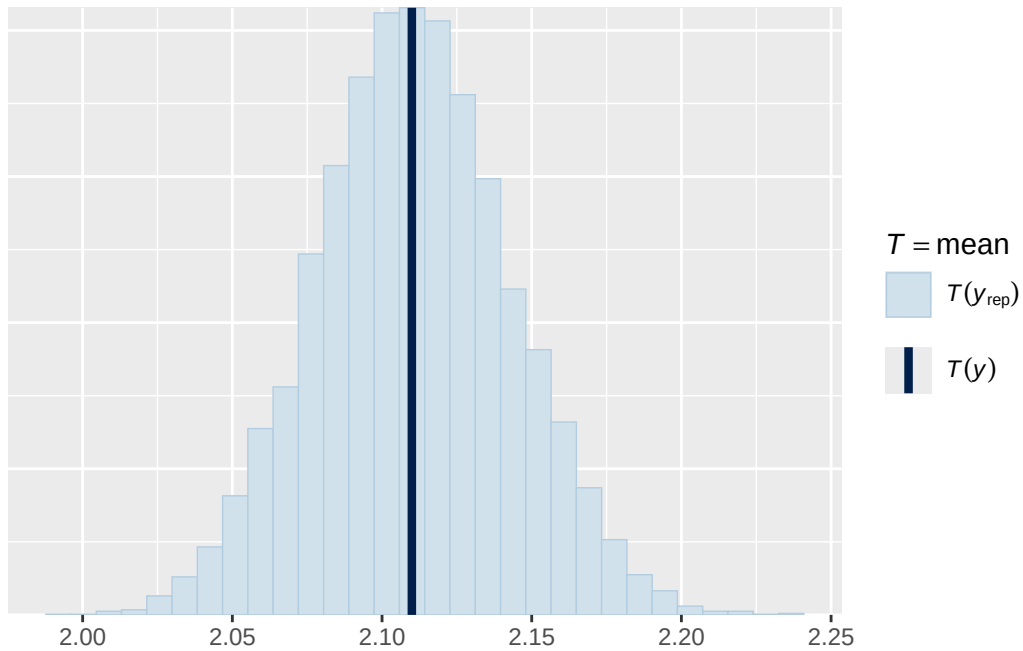


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

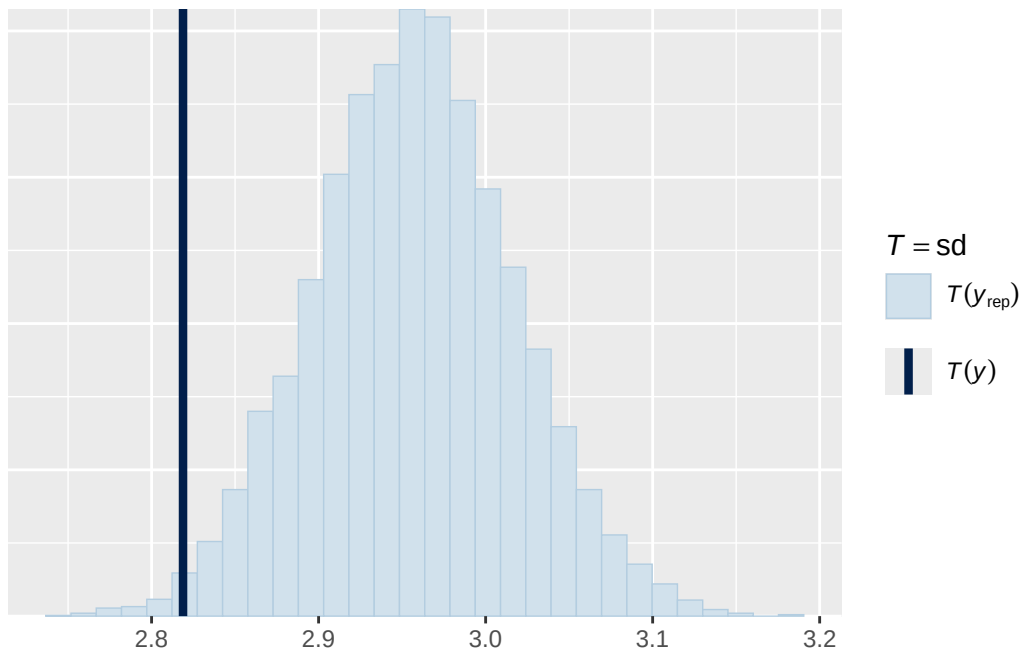


Using all posterior draws for ppc type 'stat' by default.

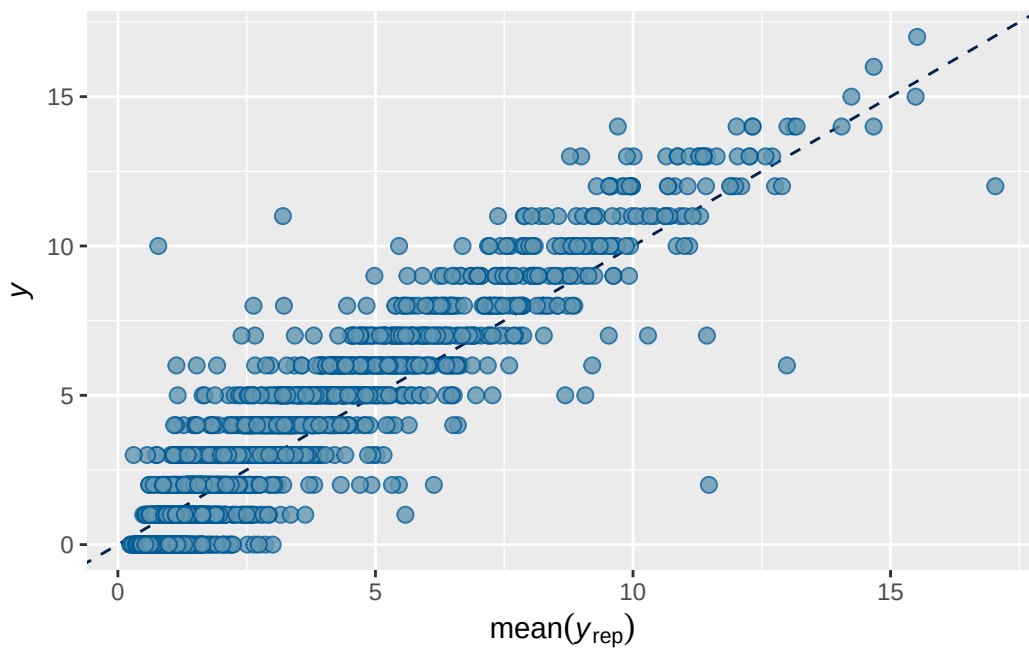
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 5B longitudinal observed output

|    | Estimate  | Est.Error   | Q2.5      | Q97.5     |
|----|-----------|-------------|-----------|-----------|
| R2 | 0.8259518 | 0.005384278 | 0.8148922 | 0.8361903 |

Family: poisson  
 Links: mu = log  
 Formula: externalising\_wave\_3 ~ breast\_development\_p\_wave\_2\_sc \* adhd\_traits\_wave\_2\_sc + age.  
 Data: obs\_df\_wide (Number of observations: 4064)  
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;  
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family\_id (Number of levels: 3530)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.955    | 0.020     | 0.915    | 0.995    | 1.002 | 929      | 2032     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.082    | 0.039     | 0.008    | 0.165    | 1.002 | 276      | 400      |

# Regression Coefficients:

|                                                      | Estimate | Est.Error |       |  |
|------------------------------------------------------|----------|-----------|-------|--|
| Intercept                                            | 0.530    | 0.037     |       |  |
| breast_development_p_wave_2_sc                       | 0.208    | 0.040     |       |  |
| adhd_traits_wave_2_sc                                | 0.598    | 0.035     |       |  |
| age_wave_2_sc                                        | -0.105   | 0.039     |       |  |
| ethnraceblack_nh                                     | -0.137   | 0.070     |       |  |
| ethnracehispanic                                     | -0.086   | 0.061     |       |  |
| ethnraceother                                        | -0.107   | 0.068     |       |  |
| inr_wave_0_sc                                        | 0.019    | 0.047     |       |  |
| externalising_wave_0_sc                              | 0.991    | 0.036     |       |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.140    | 0.070     |       |  |
|                                                      | 1-95% CI | u-95% CI  | Rhat  |  |
| Intercept                                            | 0.455    | 0.601     | 1.006 |  |
| breast_development_p_wave_2_sc                       | 0.131    | 0.289     | 1.001 |  |
| adhd_traits_wave_2_sc                                | 0.527    | 0.666     | 1.001 |  |
| age_wave_2_sc                                        | -0.184   | -0.028    | 1.001 |  |
| ethnraceblack_nh                                     | -0.275   | 0.002     | 1.003 |  |
| ethnracehispanic                                     | -0.204   | 0.036     | 1.007 |  |
| ethnraceother                                        | -0.242   | 0.028     | 1.004 |  |
| inr_wave_0_sc                                        | -0.069   | 0.112     | 1.001 |  |
| externalising_wave_0_sc                              | 0.922    | 1.060     | 1.002 |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | -0.001   | 0.276     | 1.003 |  |
|                                                      | Bulk_ESS | Tail_ESS  |       |  |
| Intercept                                            | 1045     | 1545      |       |  |
| breast_development_p_wave_2_sc                       | 1355     | 2200      |       |  |
| adhd_traits_wave_2_sc                                | 932      | 1541      |       |  |
| age_wave_2_sc                                        | 1286     | 2085      |       |  |
| ethnraceblack_nh                                     | 725      | 1255      |       |  |
| ethnracehispanic                                     | 886      | 1941      |       |  |
| ethnraceother                                        | 822      | 1297      |       |  |
| inr_wave_0_sc                                        | 820      | 1559      |       |  |
| externalising_wave_0_sc                              | 740      | 1347      |       |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 771      | 1226      |       |  |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

### Model 5B longitudinal observed diagnostics

```

prior      class
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd

coef      group resp dpar nlpar

adhd_traits_wave_2_sc
age_wave_2_sc
breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
ethnraceblack_nh
ethnracehispanic
ethnraceother
externalising_wave_0_sc
inr_wave_0_sc

family_id
Intercept family_id
site_id
Intercept site_id

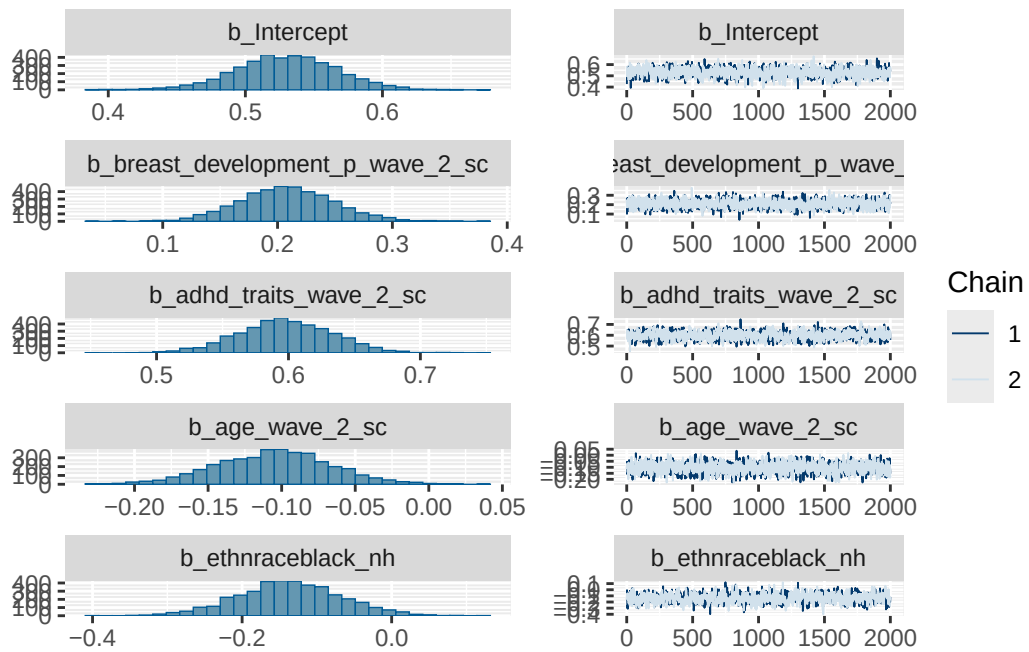
lb ub tag      source
user
(vectorized)
(vectorized)
(vectorized)
(vectorized)

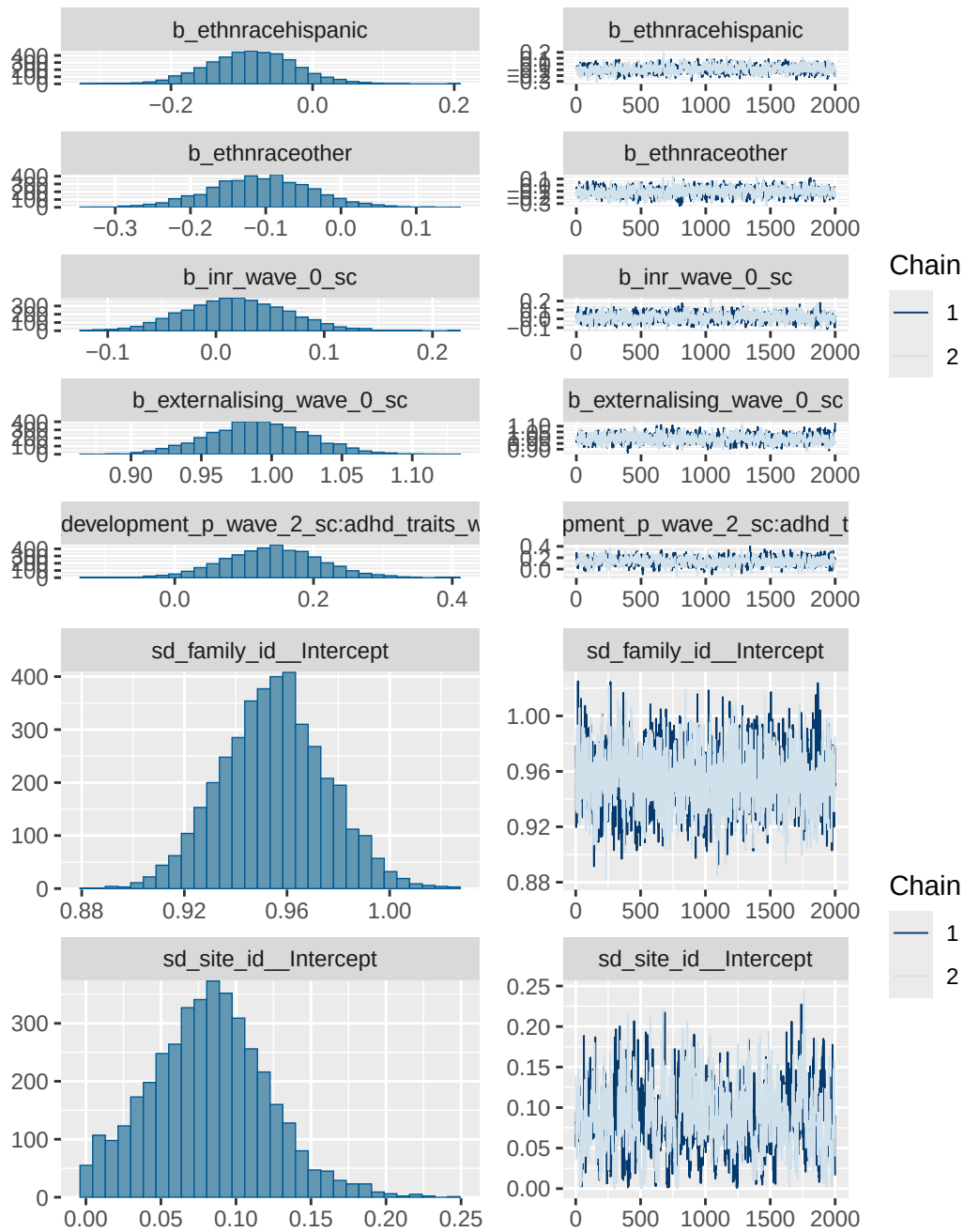
```

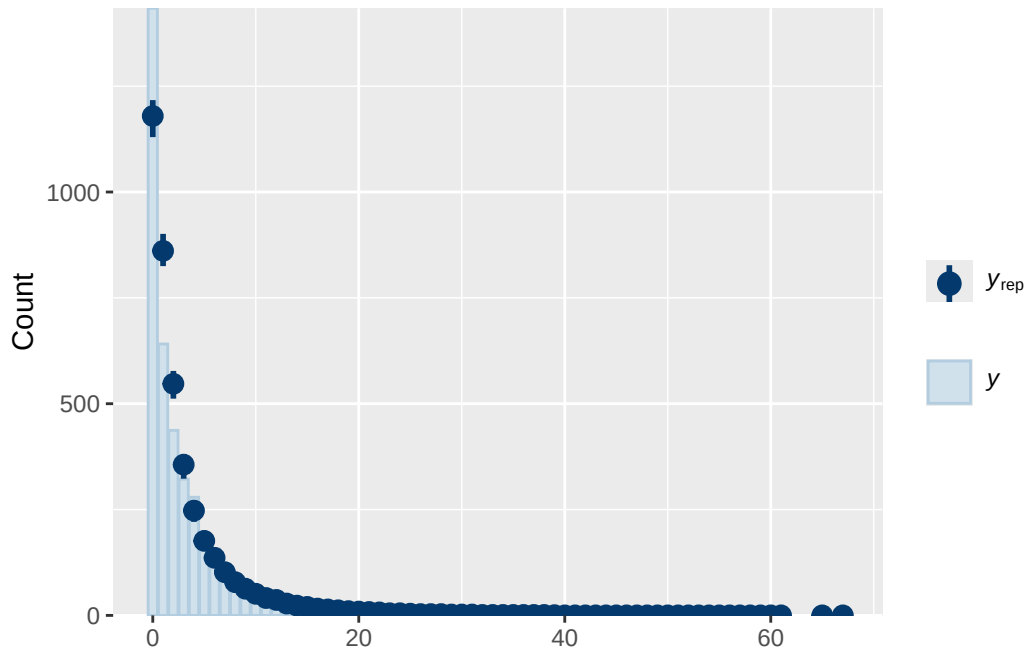
```

(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default
0      default
0      (vectorized)
0      (vectorized)
0      (vectorized)
0      (vectorized)

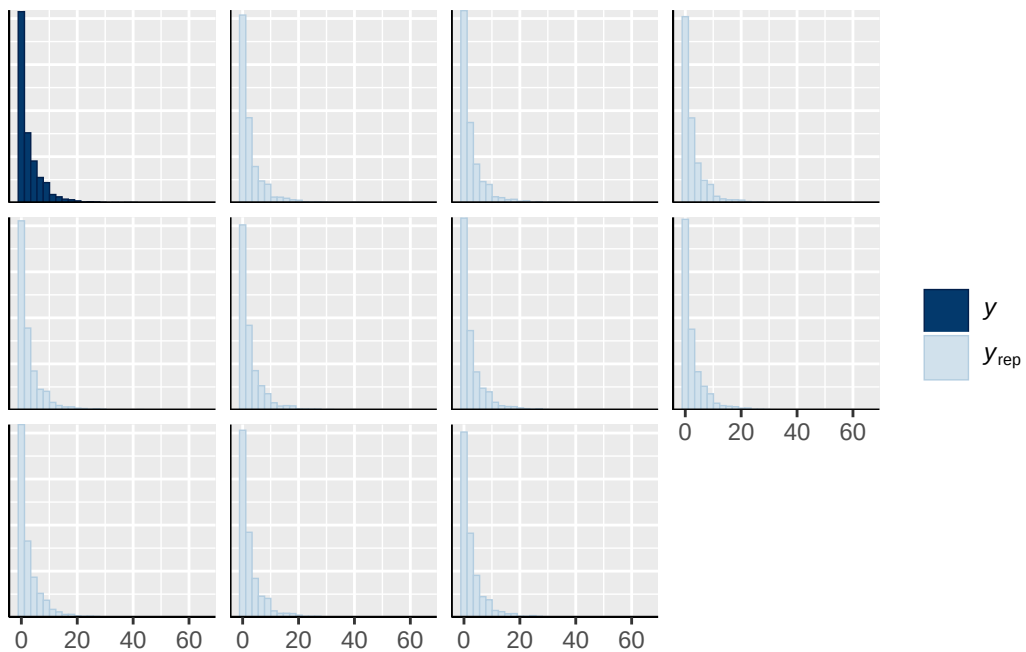
```





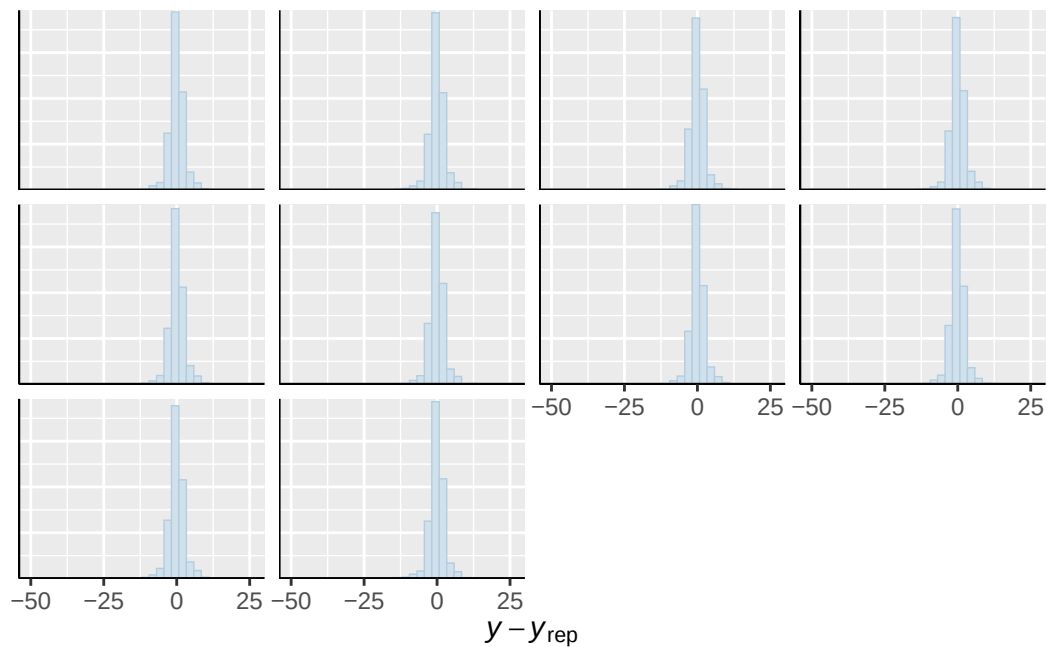


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



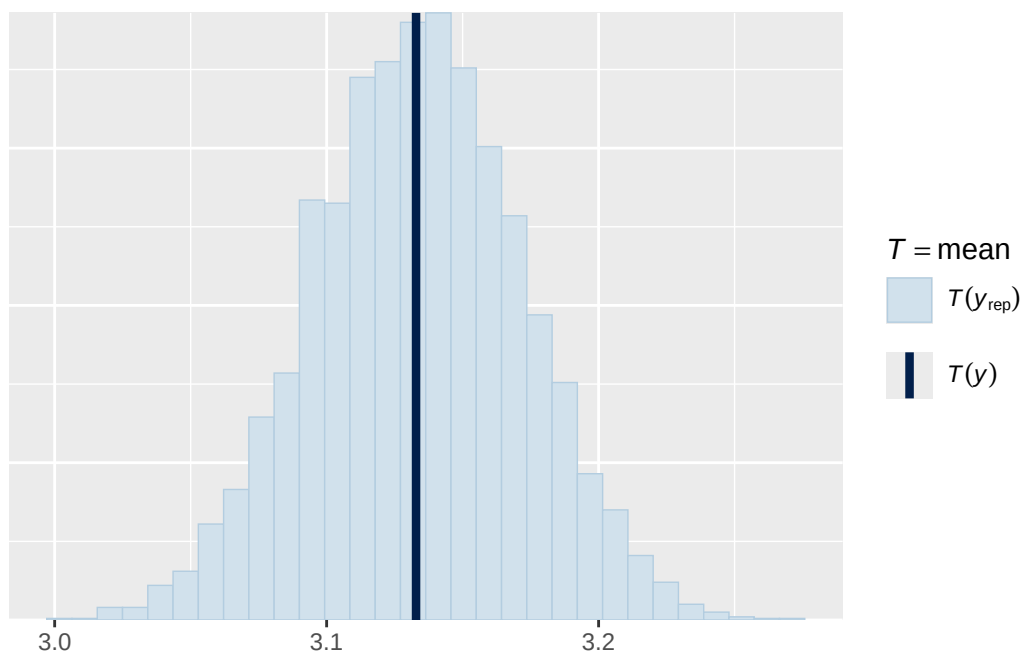
Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



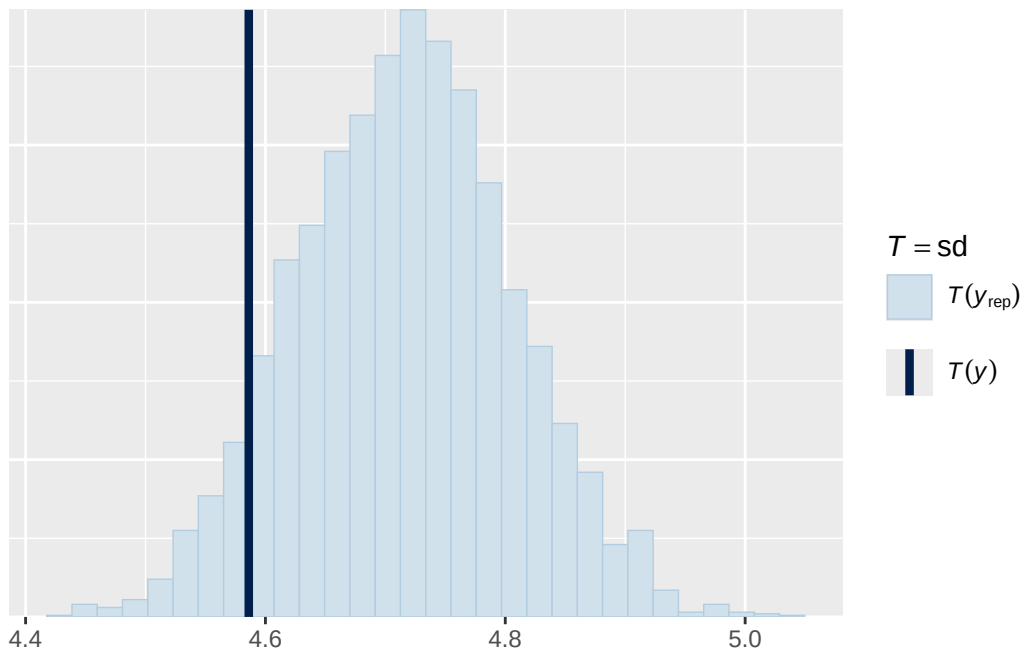


Using all posterior draws for ppc type 'stat' by default.

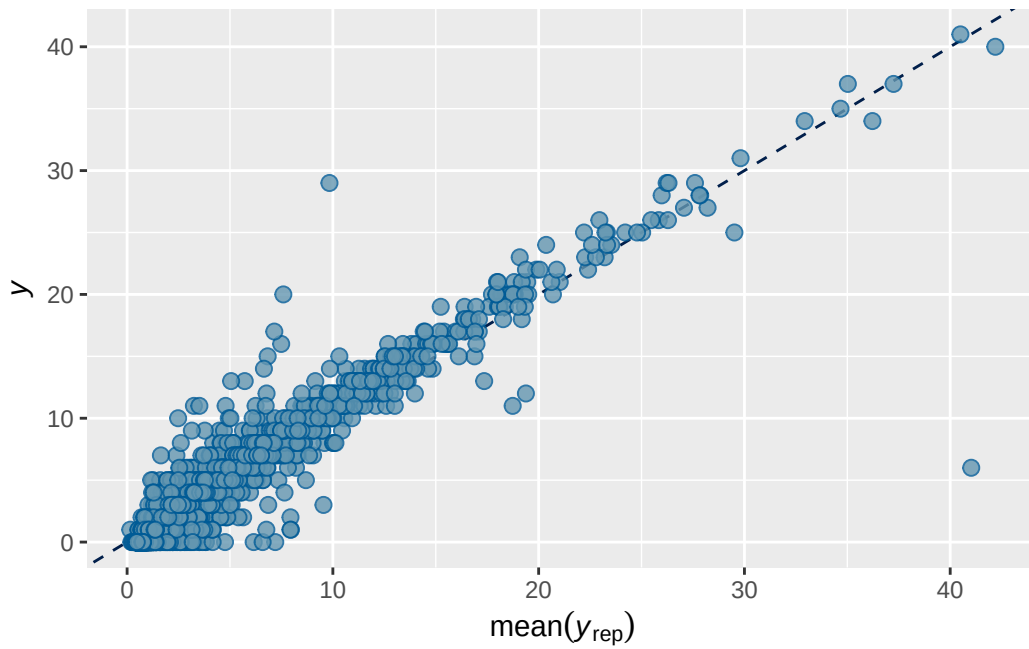
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1 longitudinal BMI output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.7360488 0.007489116 0.7209785 0.7505381

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3915)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.885    0.021   0.844   0.927 1.002    1412    2182

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.089    0.036   0.019   0.164 1.001     555     595

```

### Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS
Intercept          0.147    0.040   0.067   0.224 1.001    1879
menarche_status_p_wave_2Y 0.099    0.042   0.017   0.181 1.001    1900
age_wave_2_sc       -0.120    0.041  -0.200  -0.040 1.001    1712
ethnraceblack_nh    -0.088    0.064  -0.213   0.041 1.000    1642
ethnracehispanic    -0.047    0.057  -0.157   0.064 1.000    1851
ethnraceother        0.002    0.063  -0.119   0.133 1.001    1562
inr_wave_0_sc        0.109    0.042   0.025   0.191 1.001    1573
adhd_traits_wave_0_sc  1.331    0.033   1.266   1.395 1.006    1239
bmi_wave_0_sc        0.028    0.041  -0.050   0.107 1.002    1879

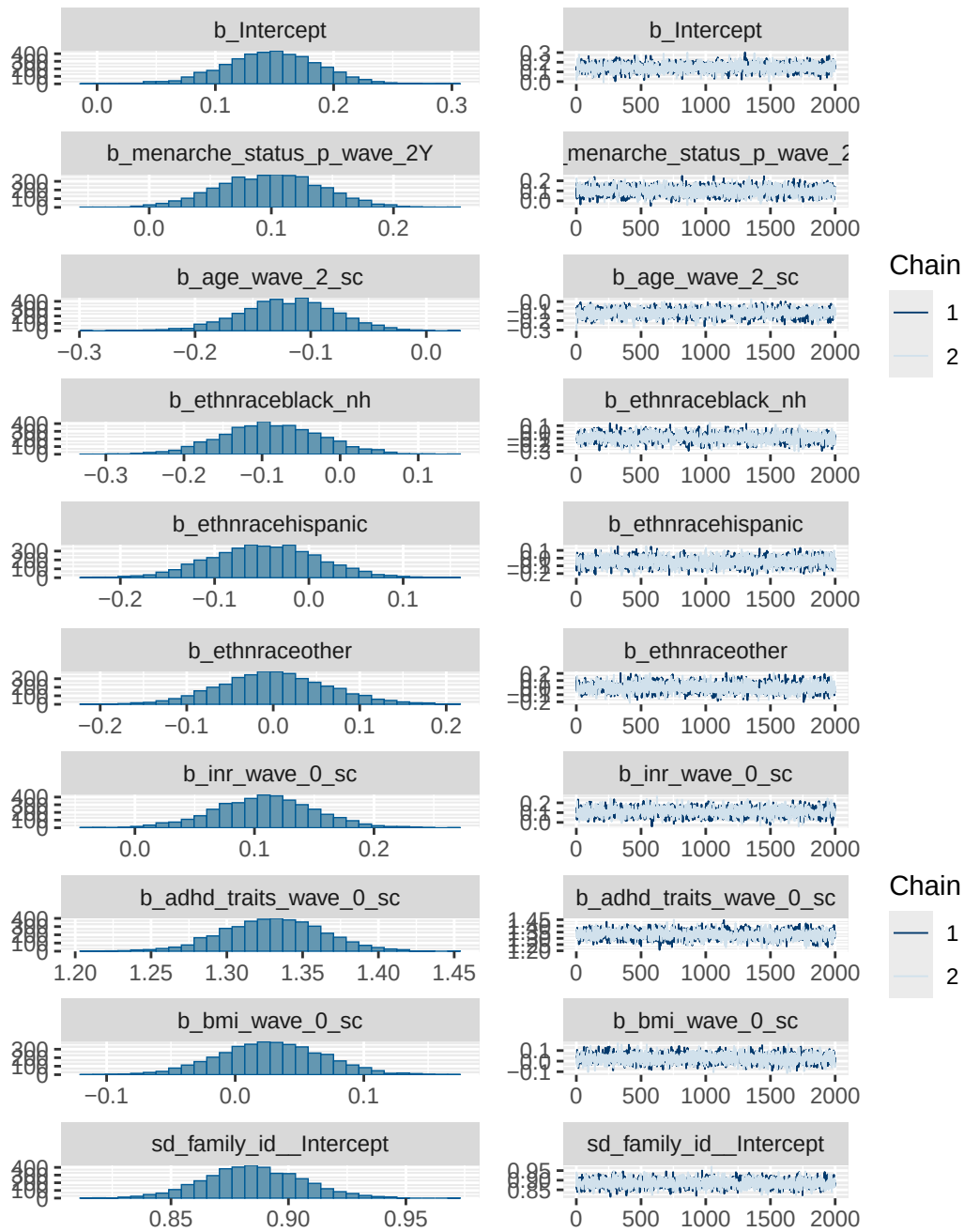
      Tail_ESS
Intercept          2160
menarche_status_p_wave_2Y 2718
age_wave_2_sc       2418
ethnraceblack_nh    2631
ethnracehispanic    2863
ethnraceother       2195
inr_wave_0_sc       2221
adhd_traits_wave_0_sc 2174
bmi_wave_0_sc       2422

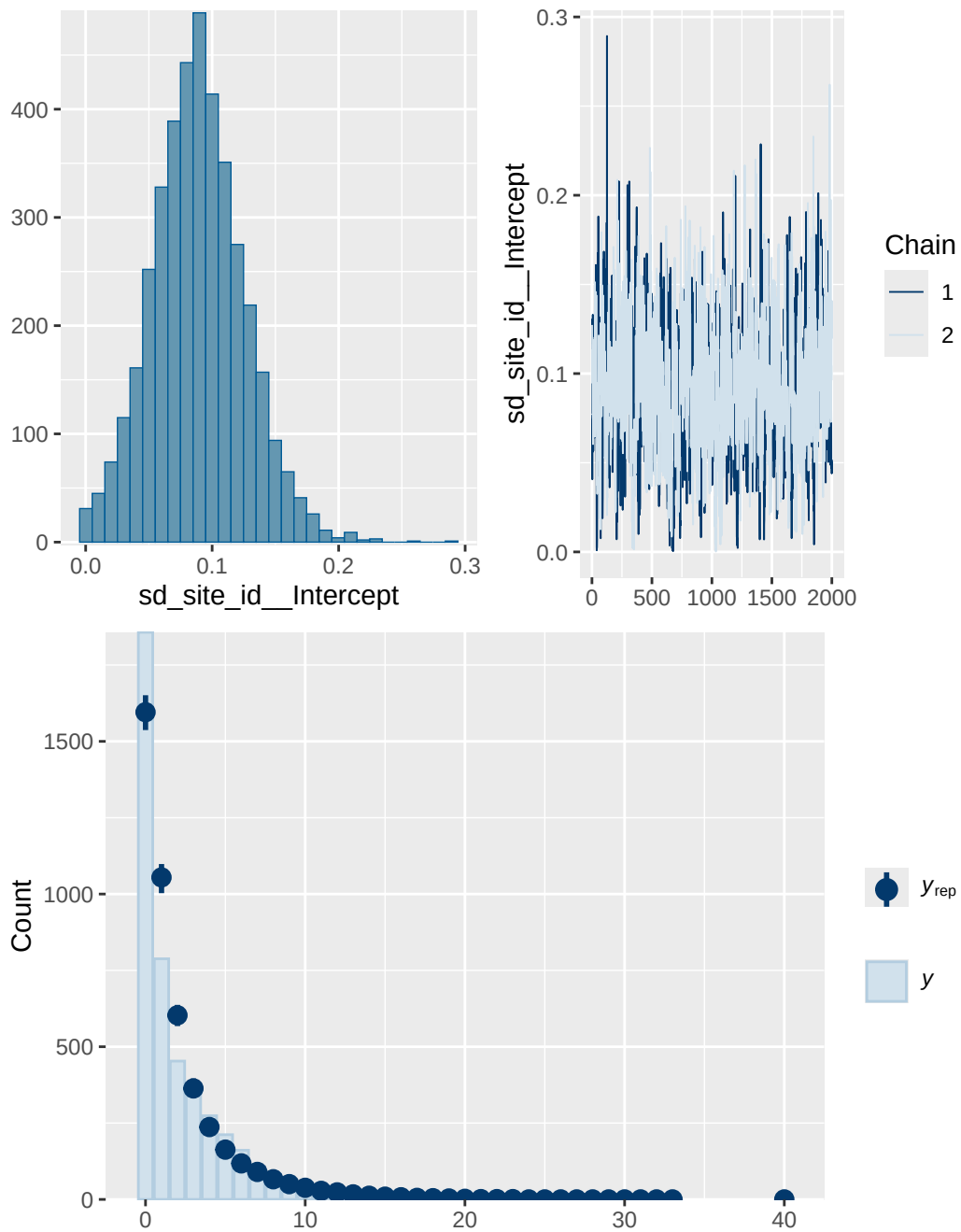
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

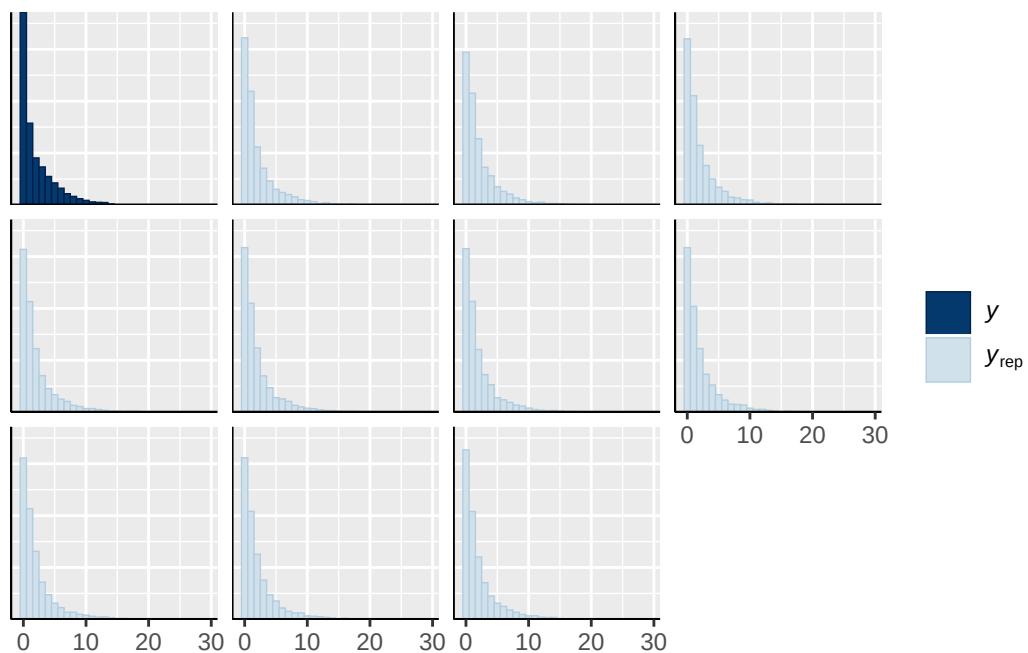
## Model 1 longitudinal BMI diagnostics

|                      | prior        | class | coef                      | group     | resp | dpar |
|----------------------|--------------|-------|---------------------------|-----------|------|------|
|                      | normal(0, 1) | b     |                           |           |      |      |
|                      | normal(0, 1) | b     | adhd_traits_wave_0_sc     |           |      |      |
|                      | normal(0, 1) | b     | age_wave_2_sc             |           |      |      |
|                      | normal(0, 1) | b     | bmi_wave_0_sc             |           |      |      |
|                      | normal(0, 1) | b     | ethnraceblack_nh          |           |      |      |
|                      | normal(0, 1) | b     | ethnracehispanic          |           |      |      |
|                      | normal(0, 1) | b     | ethnraceother             |           |      |      |
|                      | normal(0, 1) | b     | inr_wave_0_sc             |           |      |      |
|                      | normal(0, 1) | b     | menarche_status_p_wave_2Y |           |      |      |
| student_t(3, 0, 2.7) | Intercept    |       |                           |           |      |      |
| student_t(3, 0, 2.7) | sd           |       |                           |           |      |      |
| student_t(3, 0, 2.7) | sd           |       |                           | family_id |      |      |
| student_t(3, 0, 2.7) | sd           |       | Intercept                 | family_id |      |      |
| student_t(3, 0, 2.7) | sd           |       |                           | site_id   |      |      |
| student_t(3, 0, 2.7) | sd           |       | Intercept                 | site_id   |      |      |
| nlpar lb ub tag      | source       |       |                           |           |      |      |
|                      | user         |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | (vectorized) |       |                           |           |      |      |
|                      | default      |       |                           |           |      |      |
| 0                    | default      |       |                           |           |      |      |
| 0                    | (vectorized) |       |                           |           |      |      |
| 0                    | (vectorized) |       |                           |           |      |      |
| 0                    | (vectorized) |       |                           |           |      |      |
| 0                    | (vectorized) |       |                           |           |      |      |

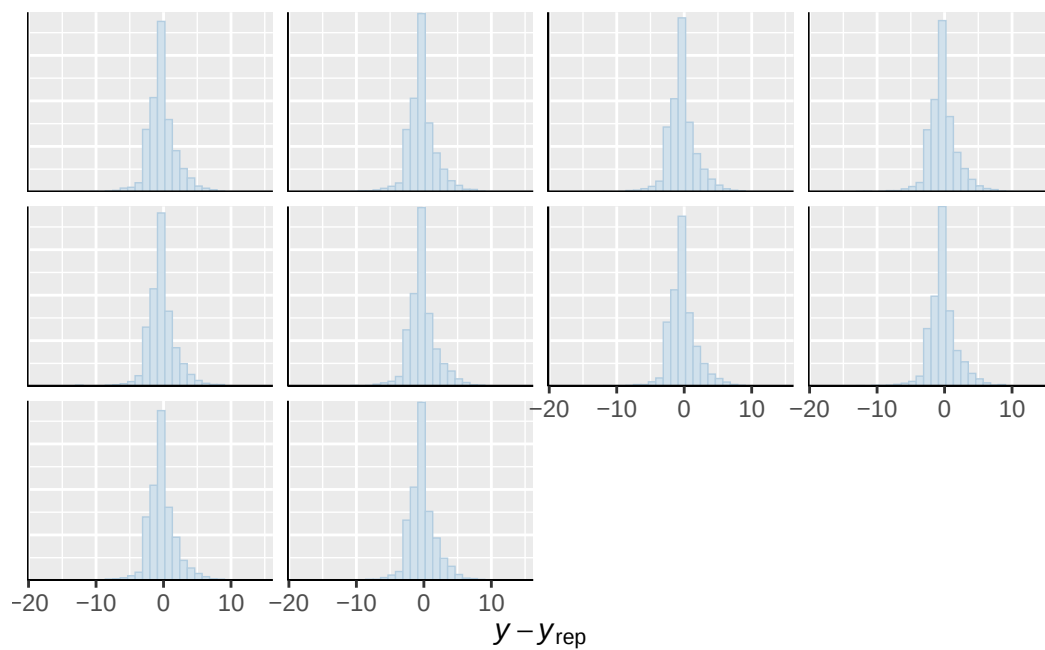




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

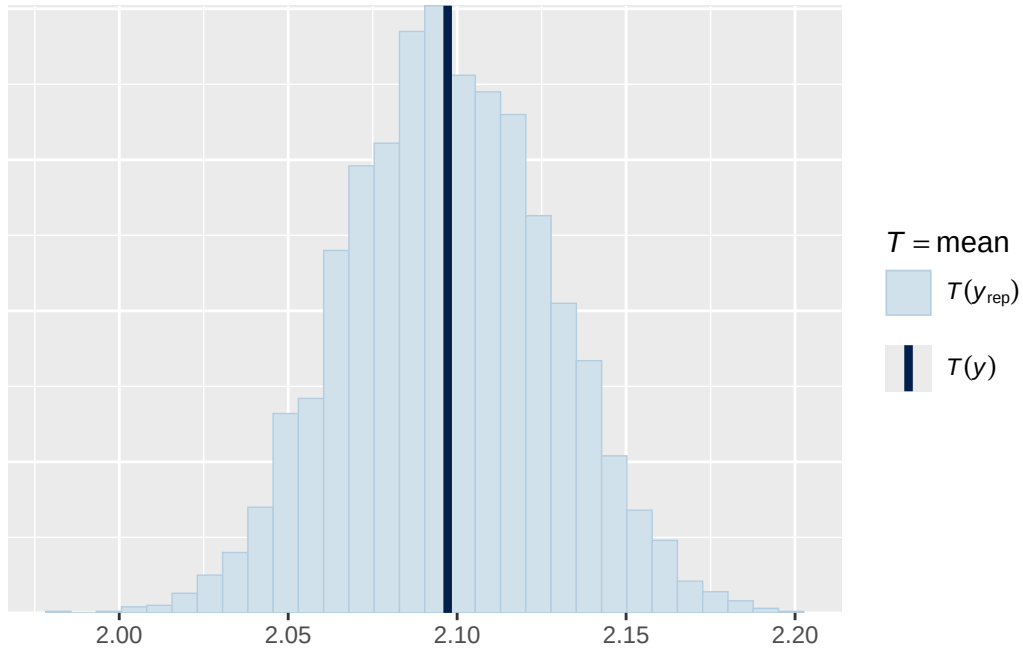


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

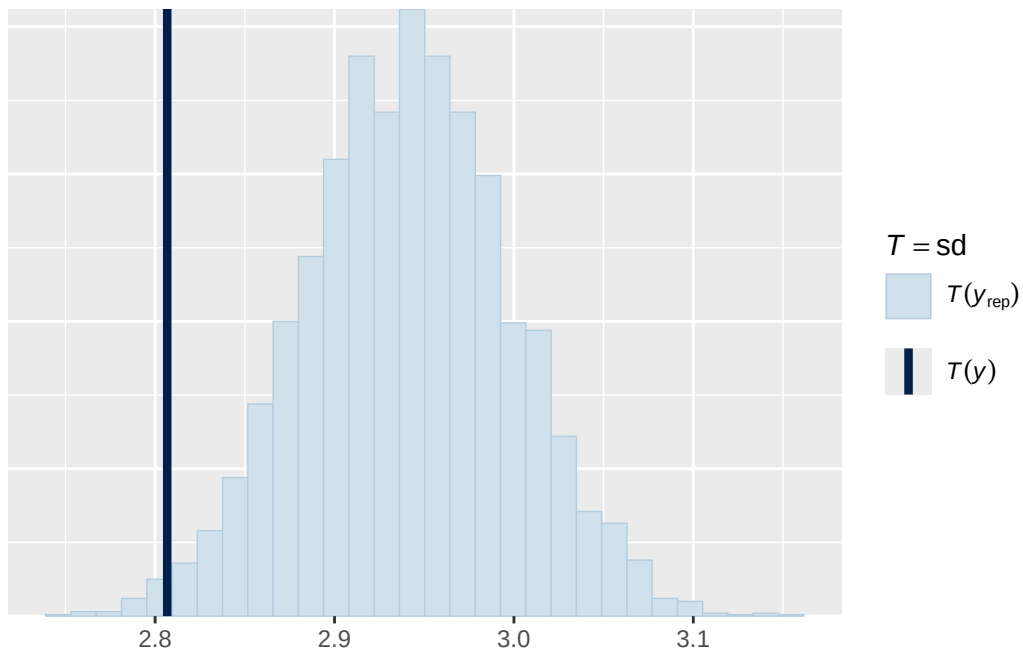


Using all posterior draws for ppc type 'stat' by default.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

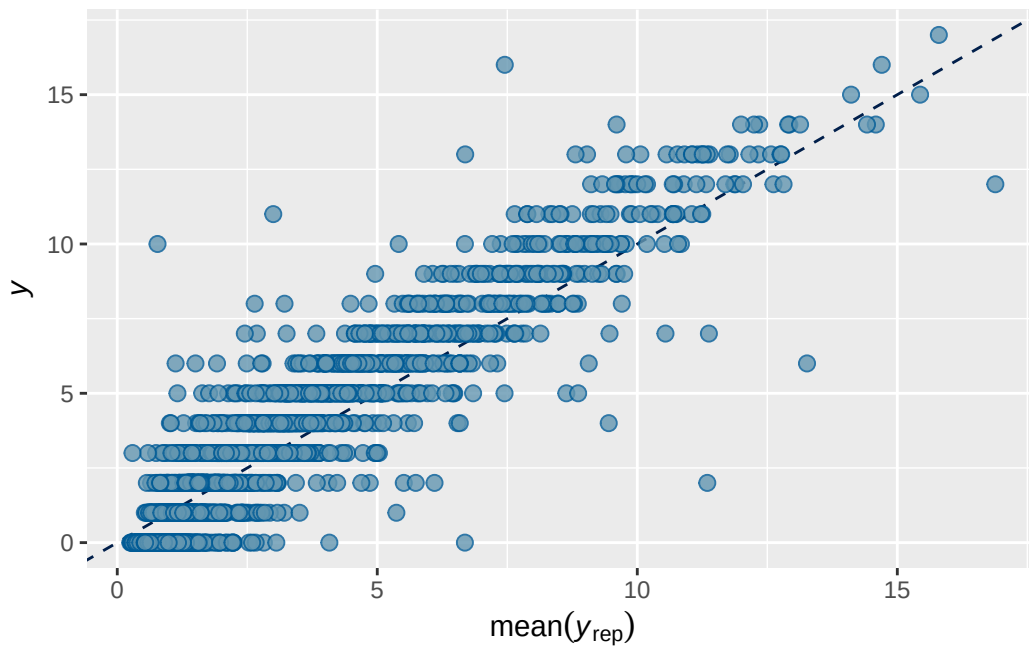


Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.





Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1B longitudinal BMI output

|    | Estimate  | Est.Error   | Q2.5     | Q97.5     |
|----|-----------|-------------|----------|-----------|
| R2 | 0.7358724 | 0.007396846 | 0.721296 | 0.7497154 |

```
Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~family\_id (Number of levels: 3928)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.885    | 0.021     | 0.846    | 0.929    | 1.000 | 1313     | 1736     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.092    | 0.037     | 0.021    | 0.169    | 1.009 | 609      | 679      |

# Regression Coefficients:

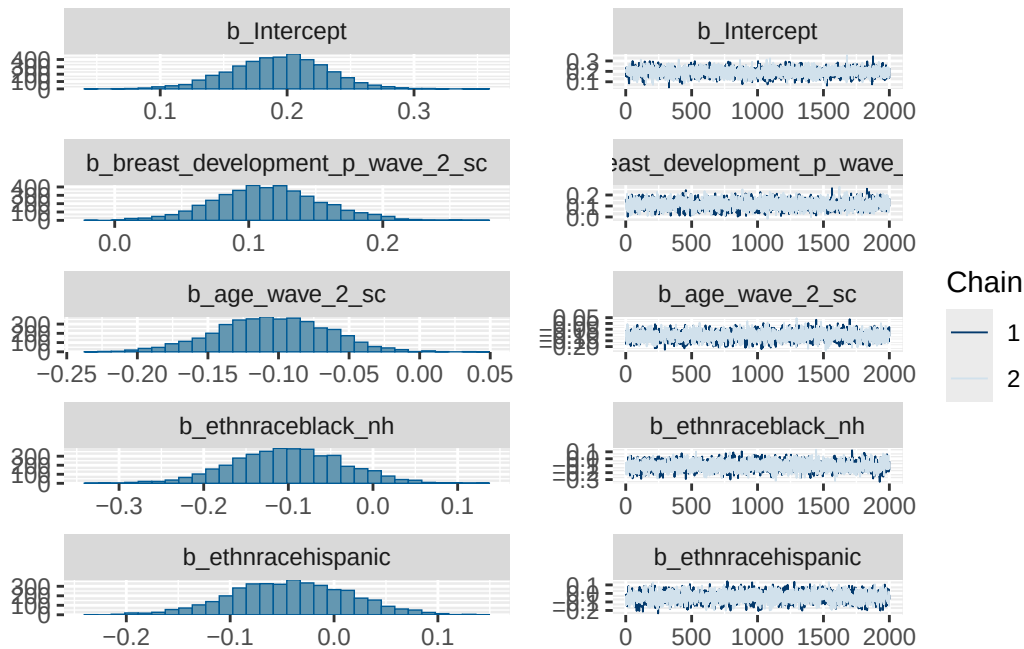
|                                | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  |
|--------------------------------|----------|-----------|----------|----------|-------|
| Intercept                      | 0.193    | 0.038     | 0.116    | 0.265    | 1.000 |
| breast_development_p_wave_2_sc | 0.115    | 0.040     | 0.038    | 0.196    | 1.001 |
| age_wave_2_sc                  | -0.105   | 0.039     | -0.183   | -0.031   | 1.000 |
| ethnraceblack_nh               | -0.100   | 0.065     | -0.227   | 0.025    | 1.001 |
| ethnracehispanic               | -0.042   | 0.057     | -0.156   | 0.071    | 1.001 |
| ethnraceother                  | -0.011   | 0.062     | -0.131   | 0.112    | 1.001 |
| inr_wave_0_sc                  | 0.092    | 0.042     | 0.011    | 0.174    | 1.000 |
| adhd_traits_wave_0_sc          | 1.322    | 0.032     | 1.258    | 1.386    | 1.001 |
| bmi_wave_0_sc                  | 0.042    | 0.041     | -0.041   | 0.121    | 1.000 |
|                                | Bulk_ESS | Tail_ESS  |          |          |       |
| Intercept                      | 2353     | 2745      |          |          |       |
| breast_development_p_wave_2_sc | 2280     | 2800      |          |          |       |
| age_wave_2_sc                  | 1932     | 2678      |          |          |       |
| ethnraceblack_nh               | 1930     | 2513      |          |          |       |
| ethnracehispanic               | 1778     | 2742      |          |          |       |
| ethnraceother                  | 1938     | 2569      |          |          |       |
| inr_wave_0_sc                  | 1850     | 2596      |          |          |       |
| adhd_traits_wave_0_sc          | 1449     | 2200      |          |          |       |
| bmi_wave_0_sc                  | 1985     | 2574      |          |          |       |

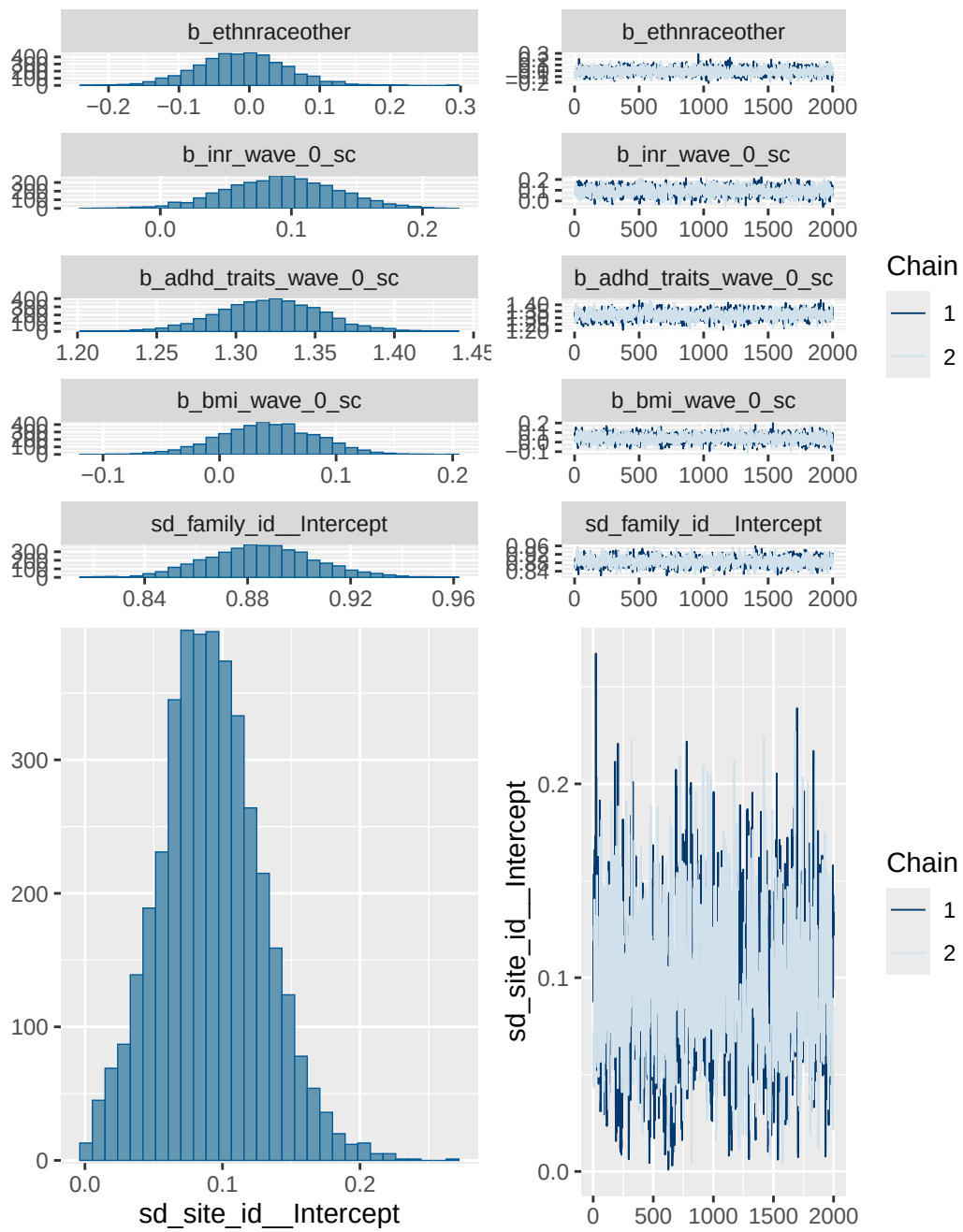
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

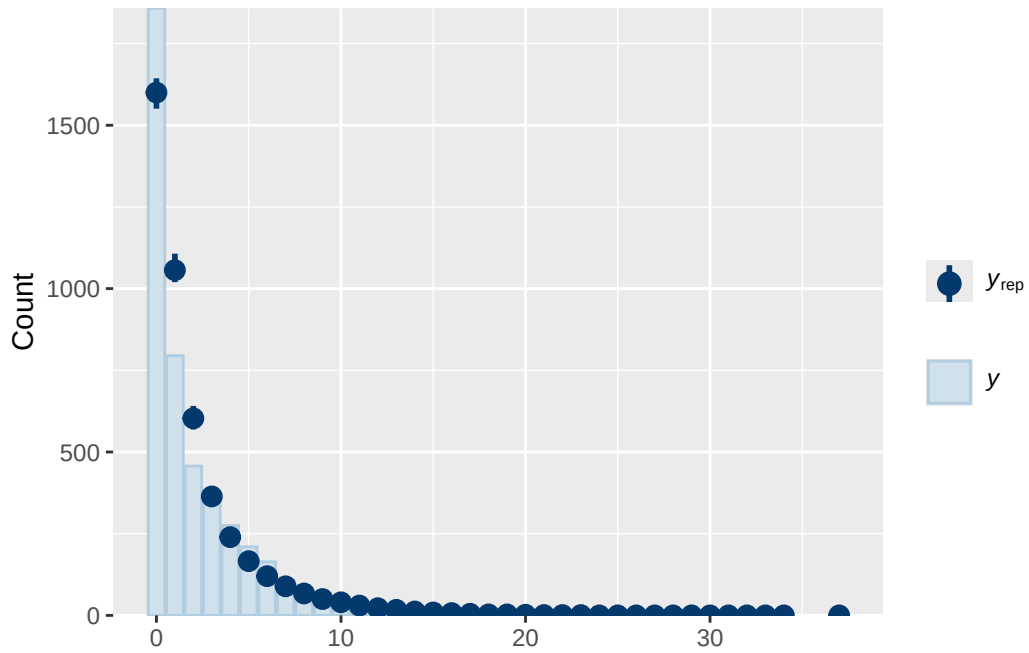
## Model 1B longitudinal BMI diagnostics

| prior                | class     | coef                           | group resp |
|----------------------|-----------|--------------------------------|------------|
| normal(0, 1)         | b         |                                |            |
| normal(0, 1)         | b         | adhd_traits_wave_0_sc          |            |
| normal(0, 1)         | b         | age_wave_2_sc                  |            |
| normal(0, 1)         | b         | bmi_wave_0_sc                  |            |
| normal(0, 1)         | b         | breast_development_p_wave_2_sc |            |
| normal(0, 1)         | b         | ethnraceblack_nh               |            |
| normal(0, 1)         | b         | ethnracehispanic               |            |
| normal(0, 1)         | b         | ethnraceother                  |            |
| normal(0, 1)         | b         | inr_wave_0_sc                  |            |
| student_t(3, 0, 2.7) | Intercept |                                |            |
| student_t(3, 0, 2.7) | sd        |                                |            |
| student_t(3, 0, 2.7) | sd        |                                | family_id  |

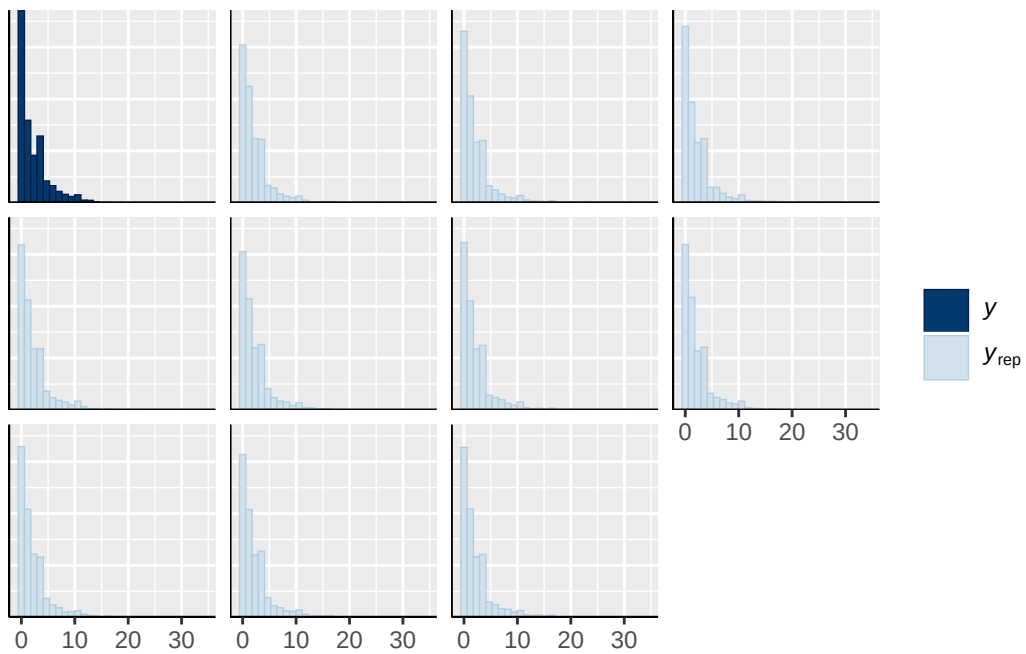
|                      |              |           |           |
|----------------------|--------------|-----------|-----------|
| student_t(3, 0, 2.7) | sd           | Intercept | family_id |
| student_t(3, 0, 2.7) | sd           |           | site_id   |
| student_t(3, 0, 2.7) | sd           | Intercept | site_id   |
| dpar nlpar lb ub tag | source       |           |           |
|                      | user         |           |           |
|                      | (vectorized) |           |           |
|                      | (vectorized) |           |           |
|                      | (vectorized) |           |           |
|                      | (vectorized) |           |           |
|                      | (vectorized) |           |           |
|                      | (vectorized) |           |           |
|                      | (vectorized) |           |           |
|                      | (vectorized) |           |           |
|                      | default      |           |           |
| 0                    | default      |           |           |
| 0                    | (vectorized) |           |           |
| 0                    | (vectorized) |           |           |
| 0                    | (vectorized) |           |           |
| 0                    | (vectorized) |           |           |



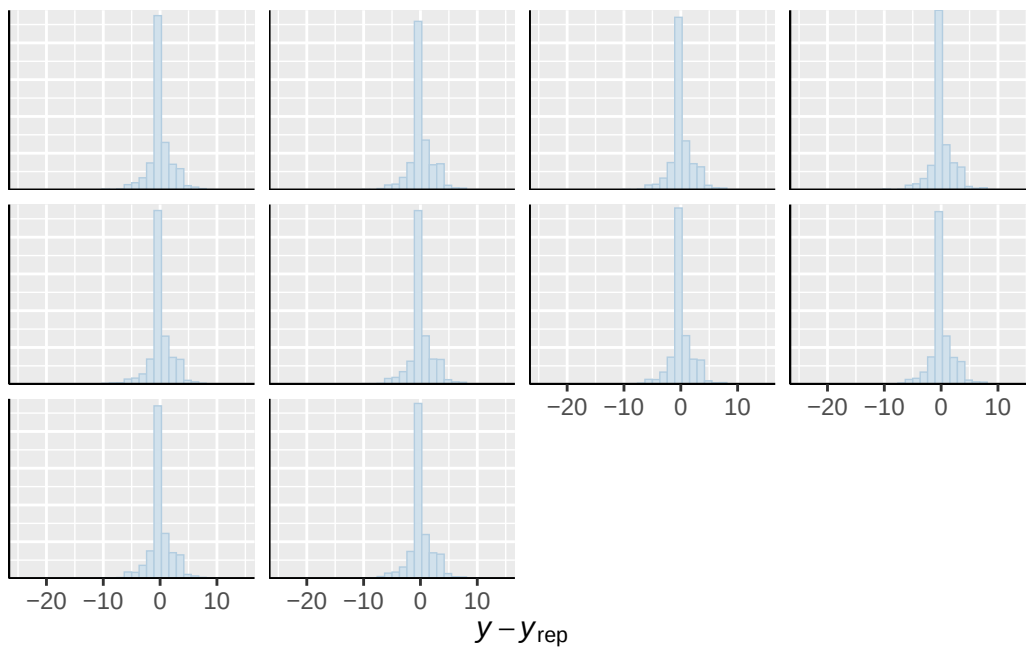




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

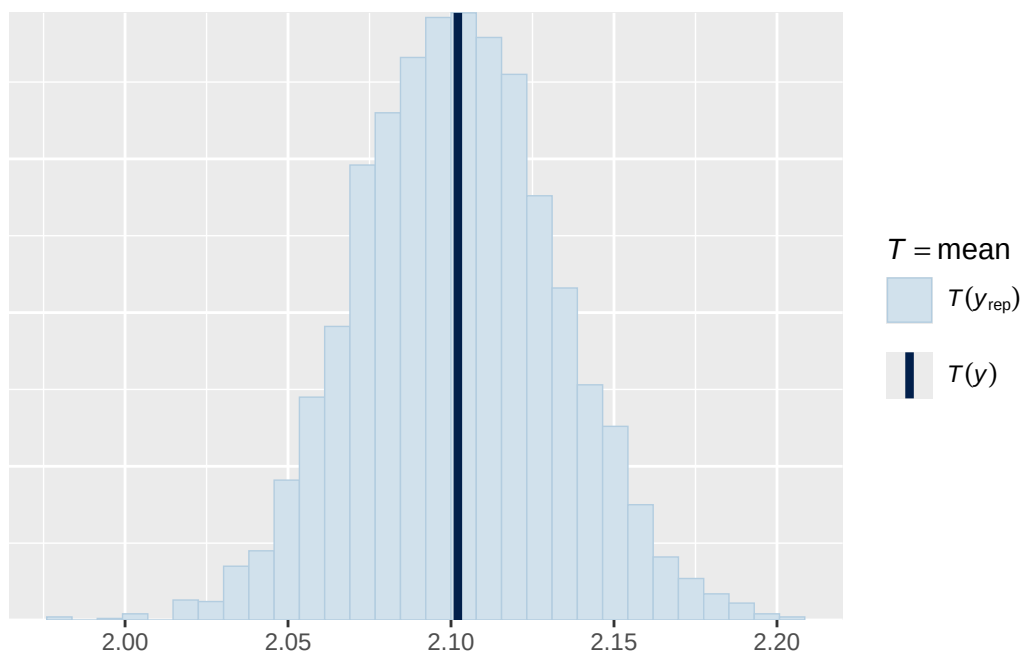


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

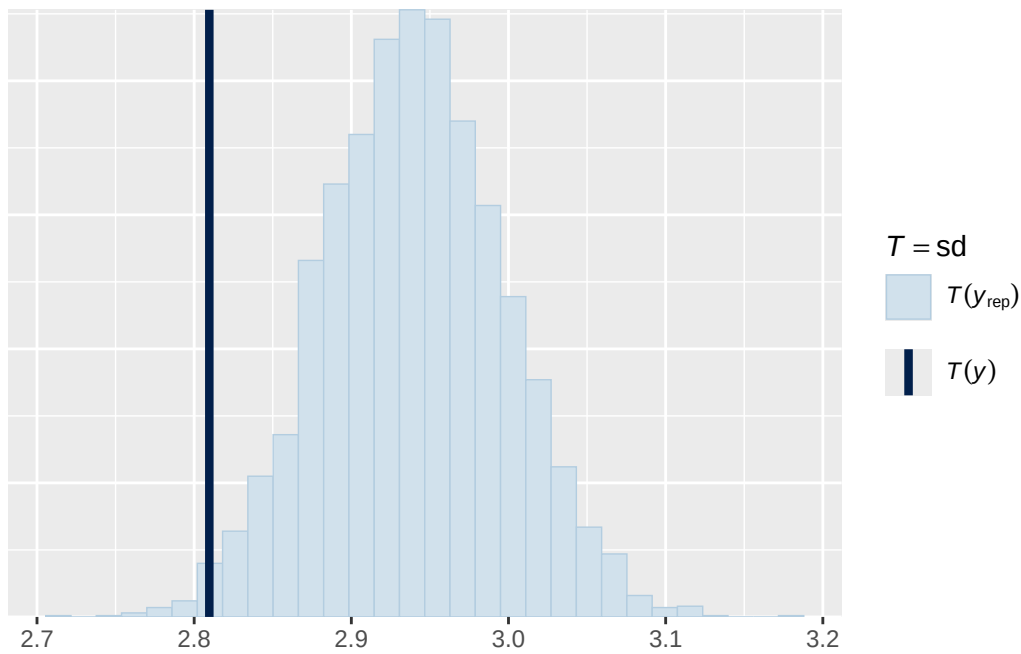


Using all posterior draws for ppc type 'stat' by default.

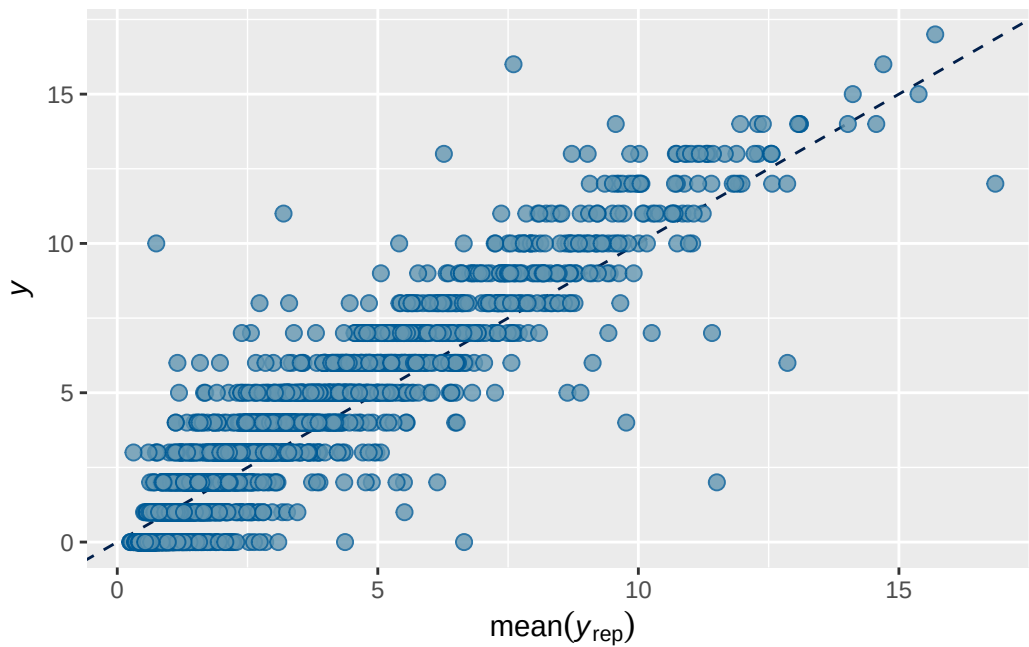
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 5B longitudinal BMI output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8235553 0.005162047 0.8131239 0.8331237

```

```

Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.986    0.020   0.948   1.025 1.000    2542    3803

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.067    0.038   0.004   0.146 1.000     717    1701

```

### Regression Coefficients:

|                                                      | Estimate | Est.Error |       |  |  |
|------------------------------------------------------|----------|-----------|-------|--|--|
| Intercept                                            | 0.520    | 0.035     |       |  |  |
| breast_development_p_wave_2_sc                       | 0.154    | 0.039     |       |  |  |
| adhd_traits_wave_2_sc                                | 0.591    | 0.034     |       |  |  |
| age_wave_2_sc                                        | -0.118   | 0.038     |       |  |  |
| ethnraceblack_nh                                     | -0.173   | 0.067     |       |  |  |
| ethnracehispanic                                     | -0.118   | 0.058     |       |  |  |
| ethnraceother                                        | -0.087   | 0.063     |       |  |  |
| inr_wave_0_sc                                        | 0.017    | 0.043     |       |  |  |
| externalising_wave_0_sc                              | 0.908    | 0.034     |       |  |  |
| bmi_wave_0_sc                                        | 0.159    | 0.040     |       |  |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.170    | 0.066     |       |  |  |
|                                                      | 1-95% CI | u-95% CI  | Rhat  |  |  |
| Intercept                                            | 0.450    | 0.587     | 1.000 |  |  |
| breast_development_p_wave_2_sc                       | 0.078    | 0.231     | 1.000 |  |  |
| adhd_traits_wave_2_sc                                | 0.525    | 0.657     | 1.000 |  |  |
| age_wave_2_sc                                        | -0.190   | -0.042    | 1.000 |  |  |
| ethnraceblack_nh                                     | -0.305   | -0.042    | 1.000 |  |  |
| ethnracehispanic                                     | -0.229   | -0.003    | 1.000 |  |  |
| ethnraceother                                        | -0.209   | 0.035     | 1.001 |  |  |



|                                                      |          |          |       |
|------------------------------------------------------|----------|----------|-------|
| inr_wave_0_sc                                        | -0.071   | 0.102    | 1.000 |
| externalising_wave_0_sc                              | 0.841    | 0.975    | 1.000 |
| bmi_wave_0_sc                                        | 0.082    | 0.239    | 1.000 |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.043    | 0.302    | 1.000 |
|                                                      | Bulk_ESS | Tail_ESS |       |
| Intercept                                            | 4214     | 5044     |       |
| breast_development_p_wave_2_sc                       | 3976     | 5238     |       |
| adhd_traits_wave_2_sc                                | 3207     | 4994     |       |
| age_wave_2_sc                                        | 4181     | 5455     |       |
| ethnraceblack_nh                                     | 3215     | 4931     |       |
| ethnracehispanic                                     | 3137     | 4979     |       |
| ethnraceother                                        | 3458     | 4916     |       |
| inr_wave_0_sc                                        | 3600     | 4754     |       |
| externalising_wave_0_sc                              | 2728     | 4140     |       |
| bmi_wave_0_sc                                        | 3302     | 4481     |       |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 2872     | 4850     |       |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

## Model 5B longitudinal BMI diagnostics

| prior                | class     |      |       |      |            |
|----------------------|-----------|------|-------|------|------------|
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| normal(0, 1)         | b         |      |       |      |            |
| student_t(3, 0, 2.9) | Intercept |      |       |      |            |
| student_t(3, 0, 2.9) | sd        |      |       |      |            |
| student_t(3, 0, 2.9) | sd        |      |       |      |            |
| student_t(3, 0, 2.9) | sd        |      |       |      |            |
| student_t(3, 0, 2.9) | sd        |      |       |      |            |
| student_t(3, 0, 2.9) | sd        |      |       |      |            |
|                      |           | coef | group | resp | dpar nlpar |

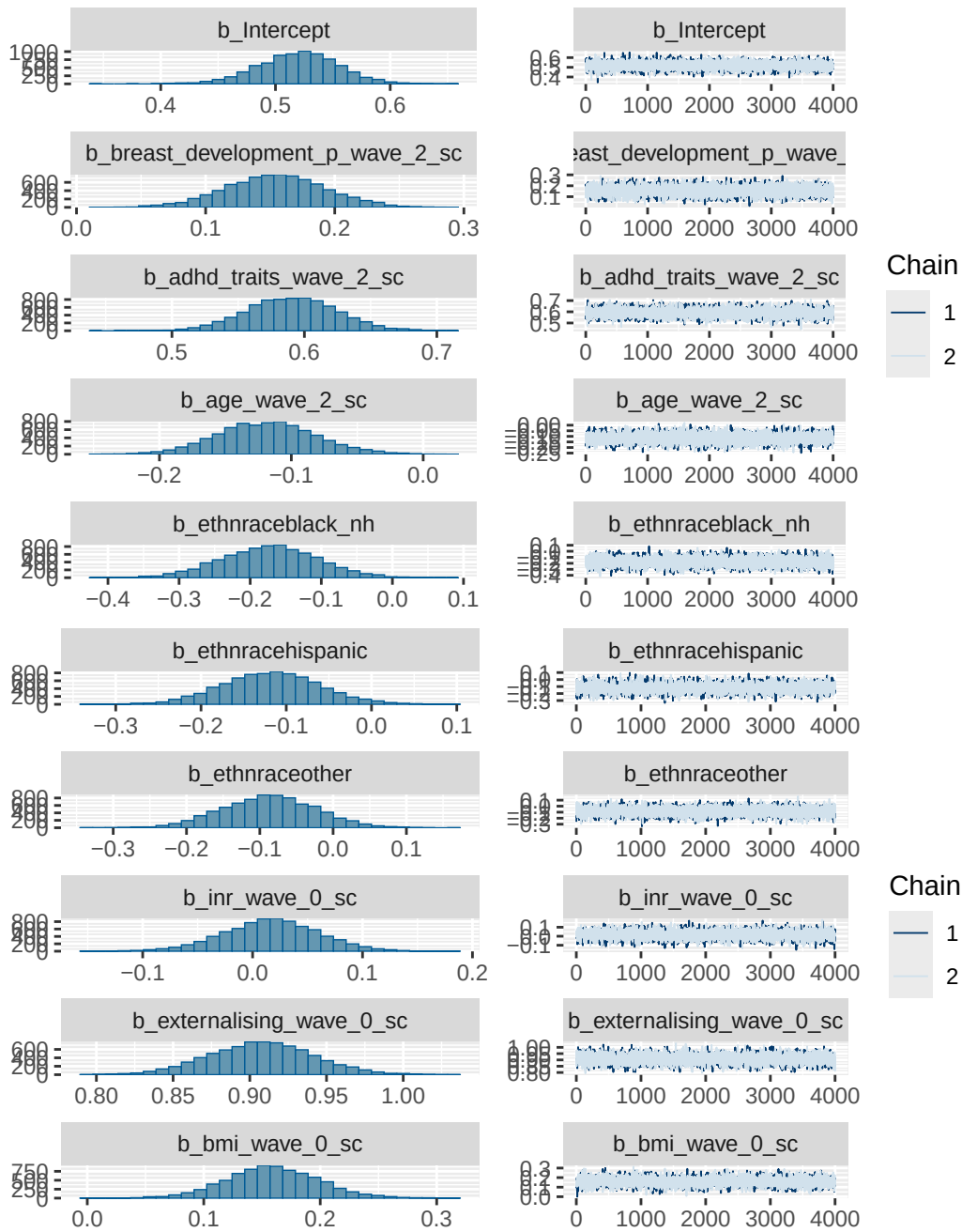
```

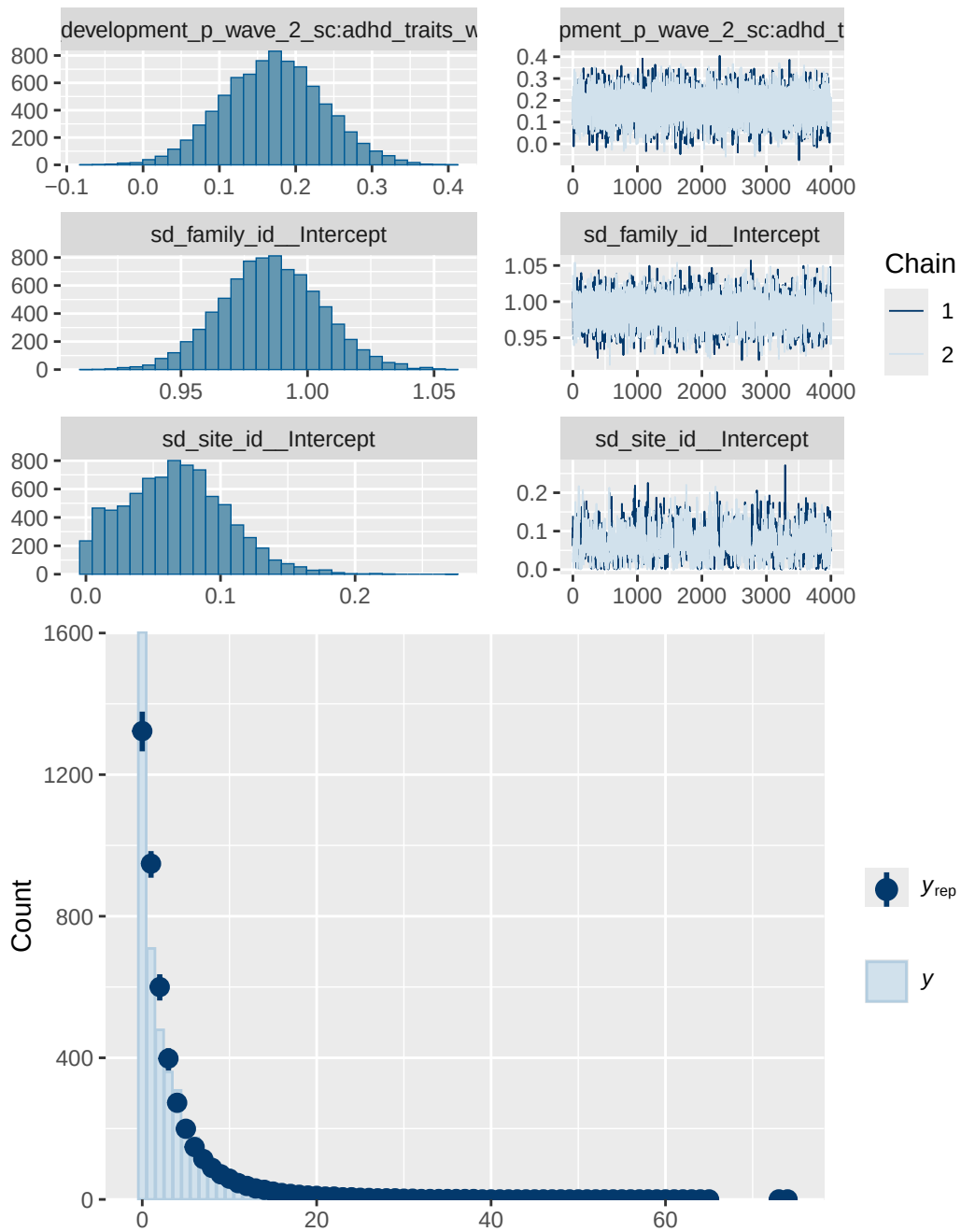
                                adhd_traits_wave_2_sc
                                age_wave_2_sc
                                bmi_wave_0_sc
                                breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
                                ethnraceblack_nh
                                ethnracehispanic
                                ethnraceother
                                externalising_wave_0_sc
                                inr_wave_0_sc

                                family_id
                                Intercept family_id
                                site_id
                                Intercept site_id

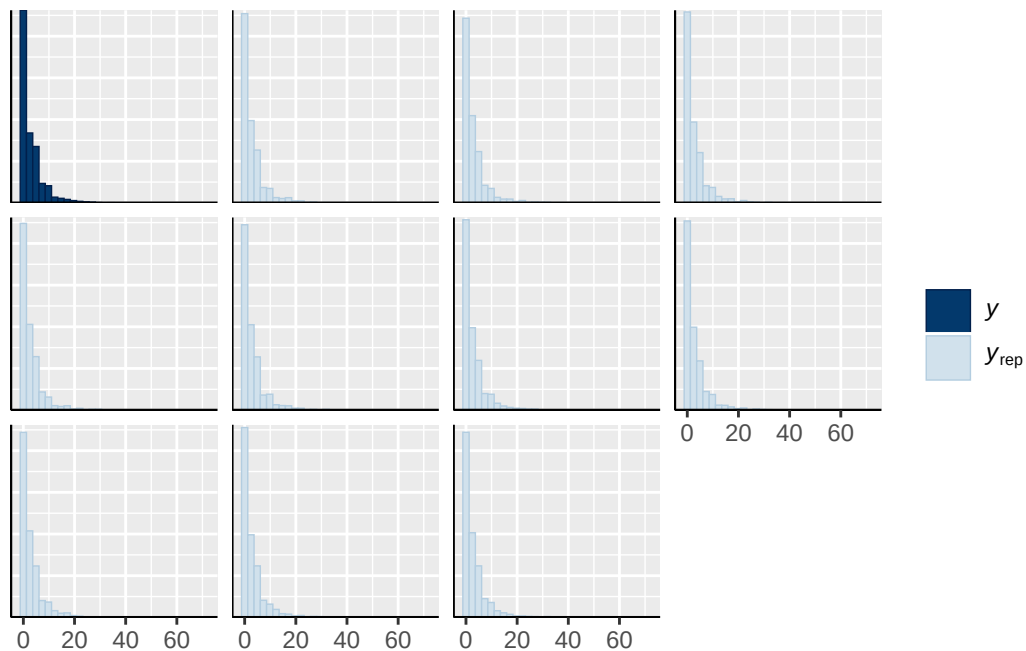
lb ub tag      source
                                user
                                (vectorized)
                                (vectorized)
                                (vectorized)
                                (vectorized)
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0          (vectorized)
0          (vectorized)
0          (vectorized)

```

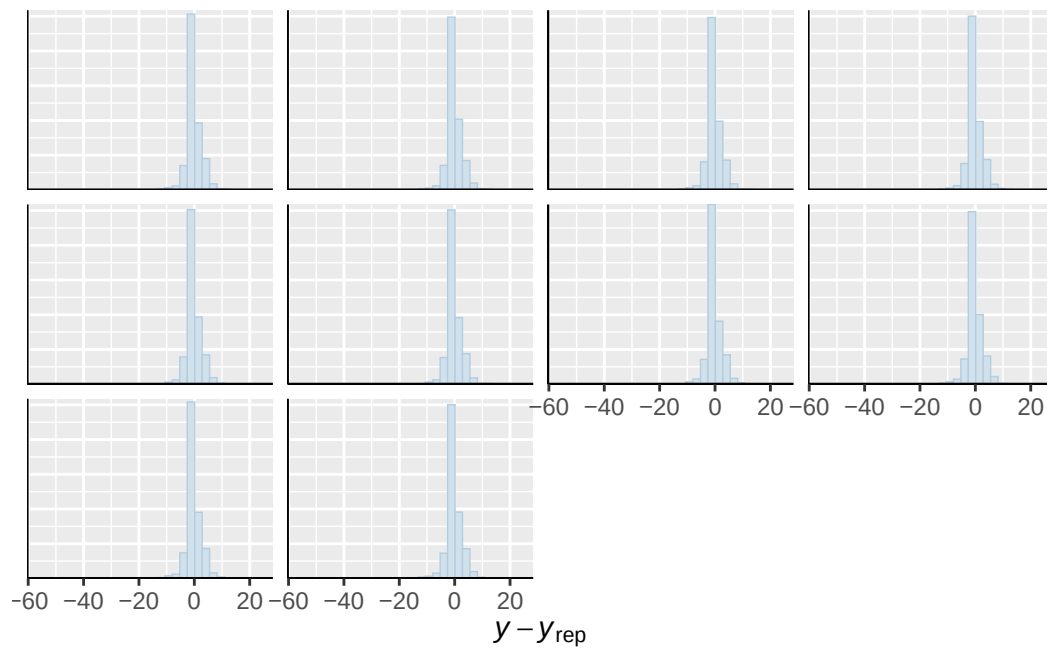




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

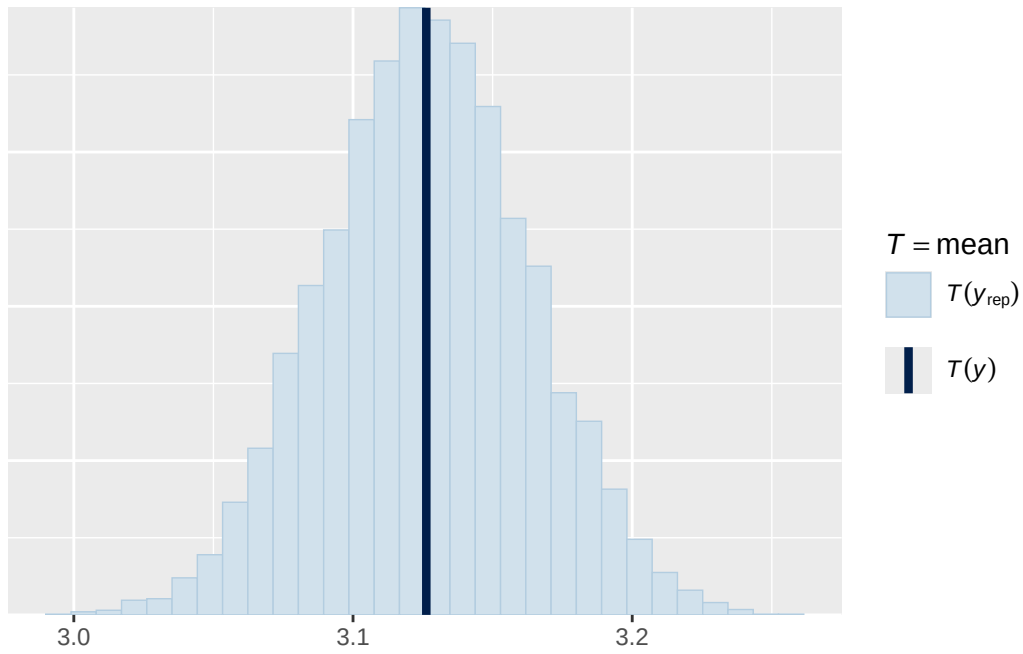


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

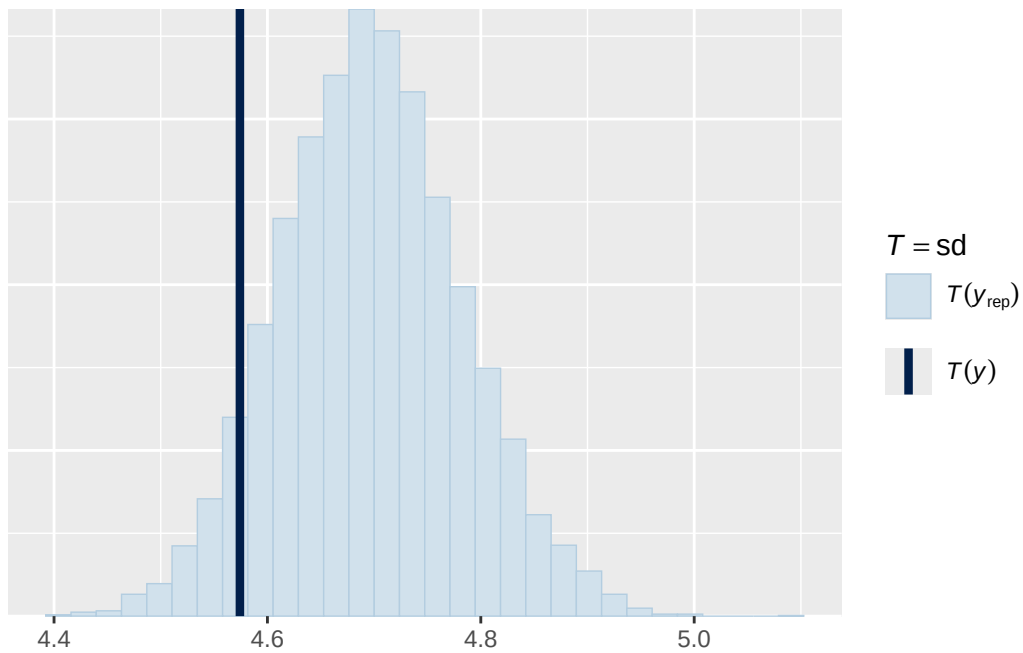


Using all posterior draws for ppc type 'stat' by default.

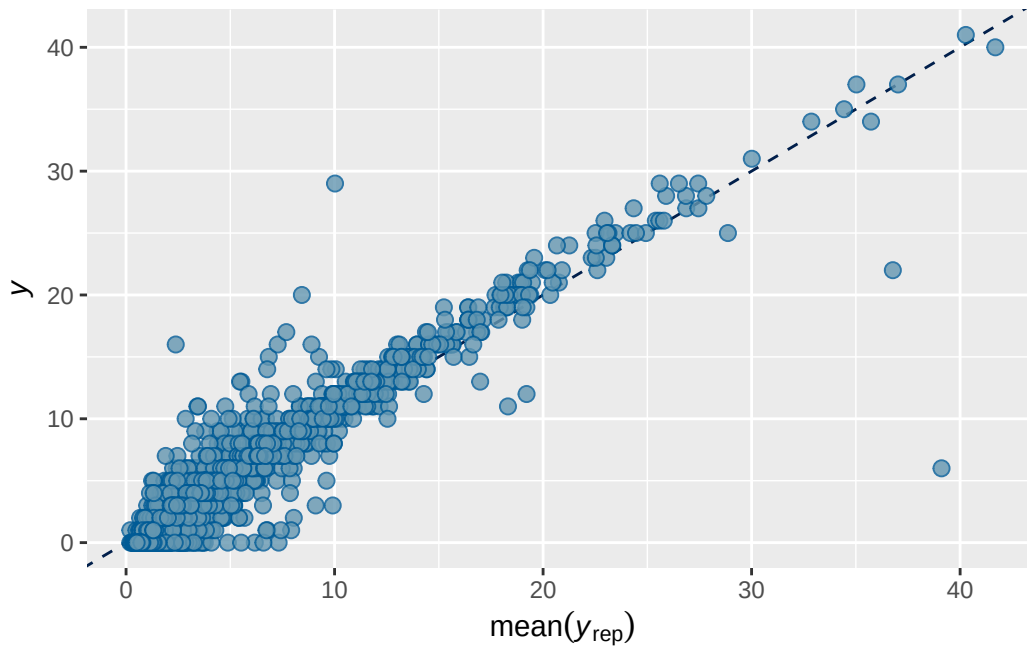
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1 longitudinal stimulant output

|    | Estimate  | Est.Error   | Q2.5      | Q97.5     |
|----|-----------|-------------|-----------|-----------|
| R2 | 0.7394205 | 0.007405415 | 0.7247865 | 0.7533296 |

```
Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~family\_id (Number of levels: 3915)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.871    | 0.021     | 0.832    | 0.913    | 1.001 | 1580     | 2613     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.082    | 0.036     | 0.012    | 0.158    | 1.001 | 663      | 964      |

# Regression Coefficients:

|                           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS |
|---------------------------|----------|-----------|----------|----------|-------|----------|
| Intercept                 | 0.112    | 0.040     | 0.033    | 0.191    | 1.001 | 3038     |
| menarche_status_p_wave_2Y | 0.101    | 0.041     | 0.021    | 0.182    | 1.000 | 3011     |
| age_wave_2_sc             | -0.119   | 0.041     | -0.201   | -0.038   | 1.000 | 3452     |
| ethnraceblack_nh          | -0.060   | 0.063     | -0.183   | 0.062    | 1.001 | 2584     |
| ethnracehispanic          | -0.026   | 0.059     | -0.140   | 0.091    | 1.000 | 2499     |
| ethnraceother             | -0.003   | 0.061     | -0.121   | 0.121    | 1.000 | 3267     |
| inr_wave_0_sc             | 0.080    | 0.041     | -0.001   | 0.161    | 1.001 | 2776     |
| adhd_traits_wave_0_sc     | 1.229    | 0.034     | 1.164    | 1.298    | 1.001 | 2386     |
| stimulant_status_wave_3Y  | 0.592    | 0.068     | 0.458    | 0.722    | 1.001 | 2203     |
|                           | Tail_ESS |           |          |          |       |          |
| Intercept                 | 3069     |           |          |          |       |          |
| menarche_status_p_wave_2Y | 3263     |           |          |          |       |          |
| age_wave_2_sc             | 3191     |           |          |          |       |          |
| ethnraceblack_nh          | 3439     |           |          |          |       |          |
| ethnracehispanic          | 2786     |           |          |          |       |          |
| ethnraceother             | 3443     |           |          |          |       |          |
| inr_wave_0_sc             | 3077     |           |          |          |       |          |
| adhd_traits_wave_0_sc     | 2675     |           |          |          |       |          |
| stimulant_status_wave_3Y  | 2754     |           |          |          |       |          |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

## Model 1 stimulant diagnostics

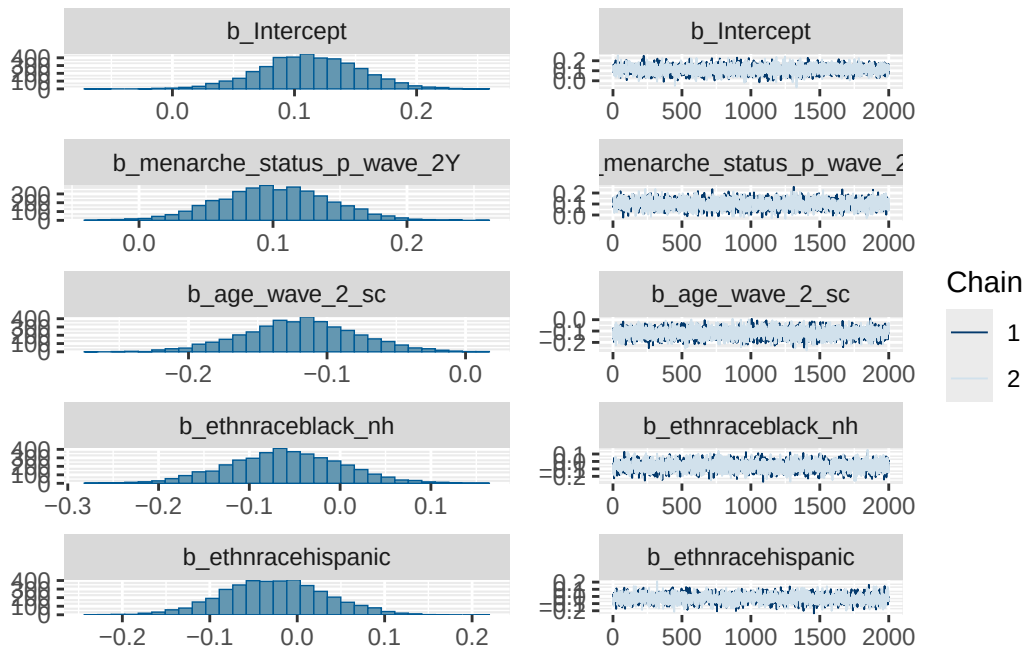
| prior                | class     | coef                      | group     | resp | dpar |
|----------------------|-----------|---------------------------|-----------|------|------|
| normal(0, 1)         | b         |                           |           |      |      |
| normal(0, 1)         | b         | adhd_traits_wave_0_sc     |           |      |      |
| normal(0, 1)         | b         | age_wave_2_sc             |           |      |      |
| normal(0, 1)         | b         | ethnraceblack_nh          |           |      |      |
| normal(0, 1)         | b         | ethnracehispanic          |           |      |      |
| normal(0, 1)         | b         | ethnraceother             |           |      |      |
| normal(0, 1)         | b         | inr_wave_0_sc             |           |      |      |
| normal(0, 1)         | b         | menarche_status_p_wave_2Y |           |      |      |
| normal(0, 1)         | b         | stimulant_status_wave_3Y  |           |      |      |
| student_t(3, 0, 2.7) | Intercept |                           |           |      |      |
| student_t(3, 0, 2.7) | sd        |                           |           |      |      |
| student_t(3, 0, 2.7) | sd        |                           | family_id |      |      |

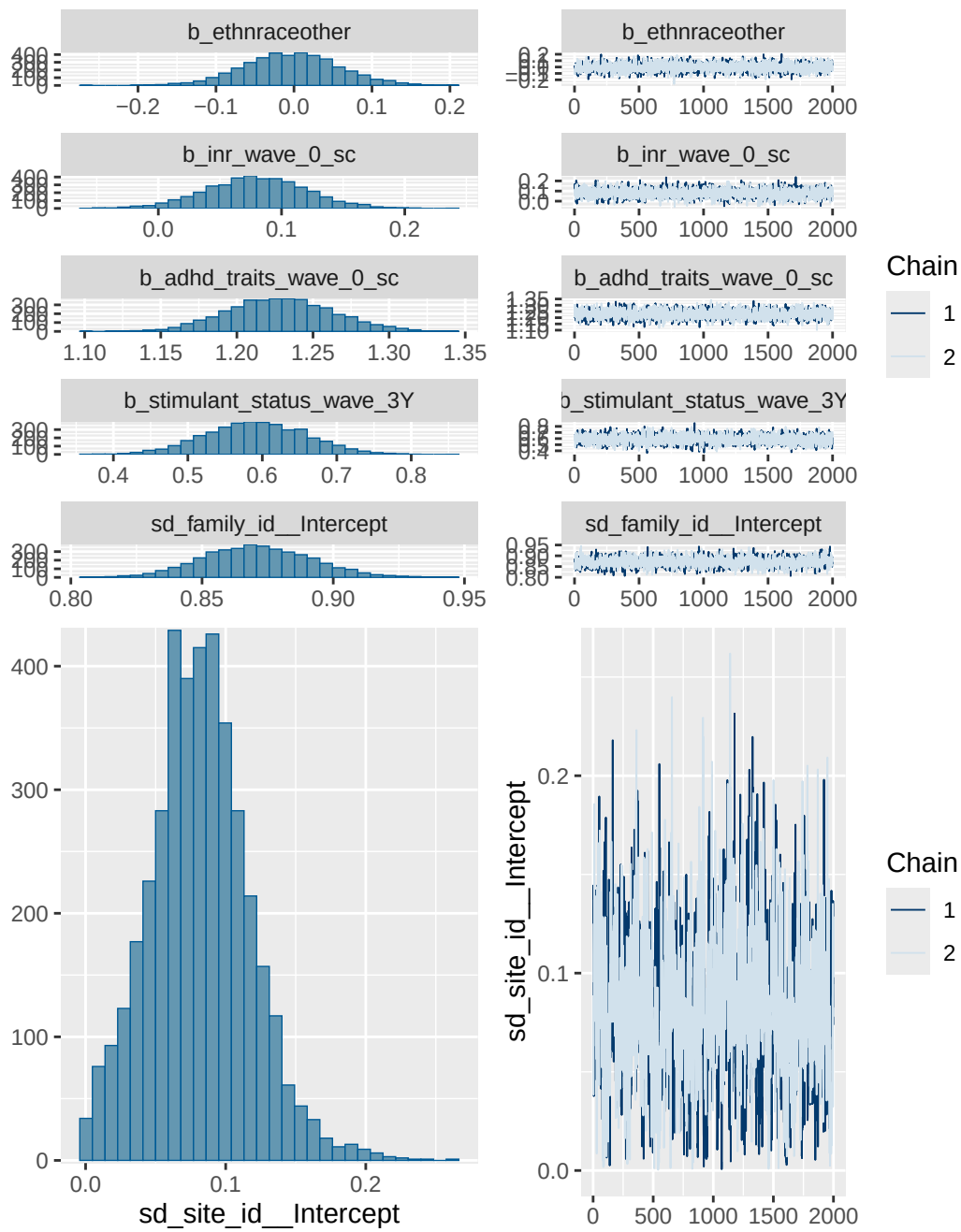


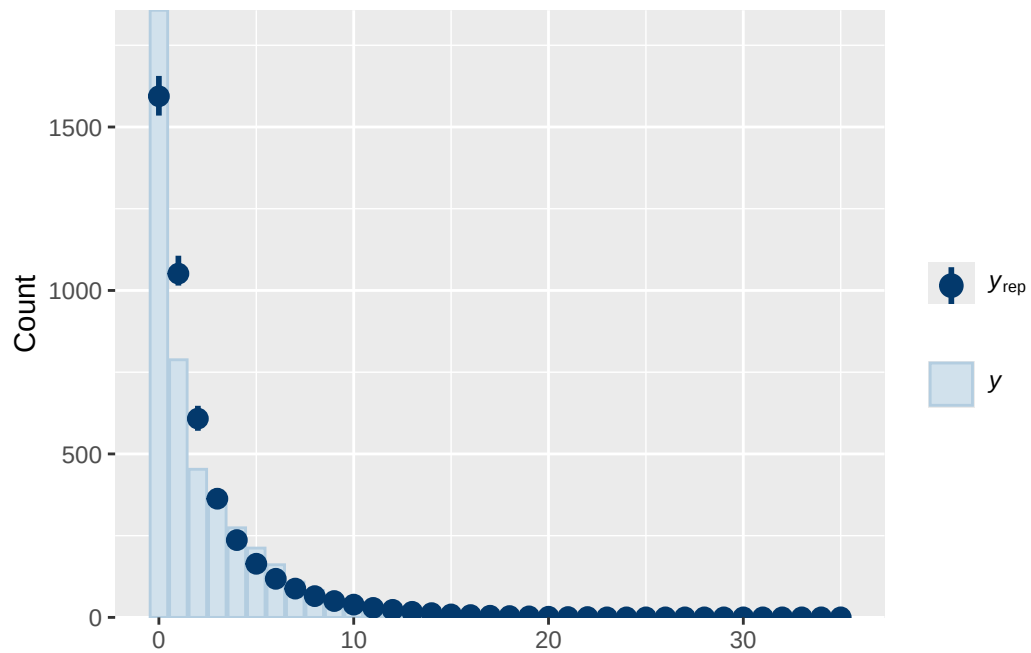
```

student_t(3, 0, 2.7)      sd      Intercept family_id
student_t(3, 0, 2.7)      sd      site_id
student_t(3, 0, 2.7)      sd      Intercept site_id
nlpar lb ub tag          source
                        user
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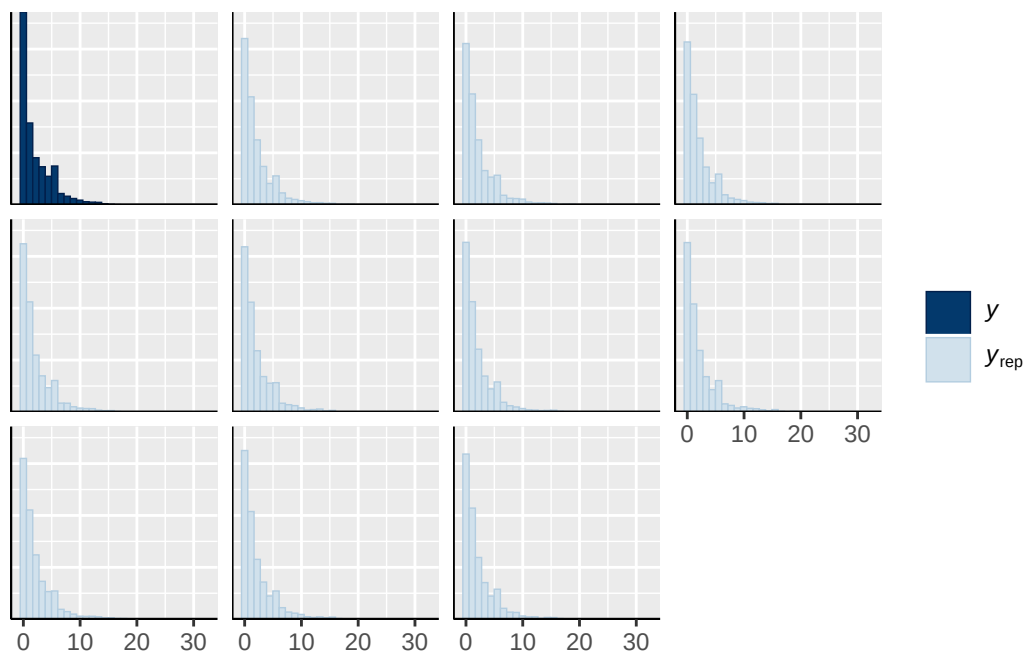
```



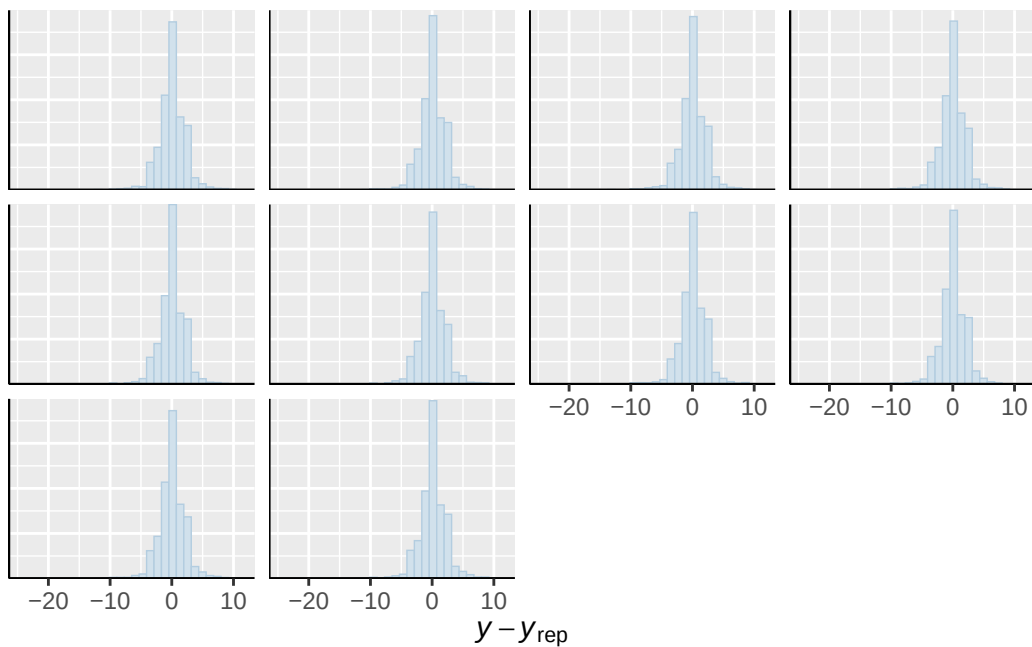




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

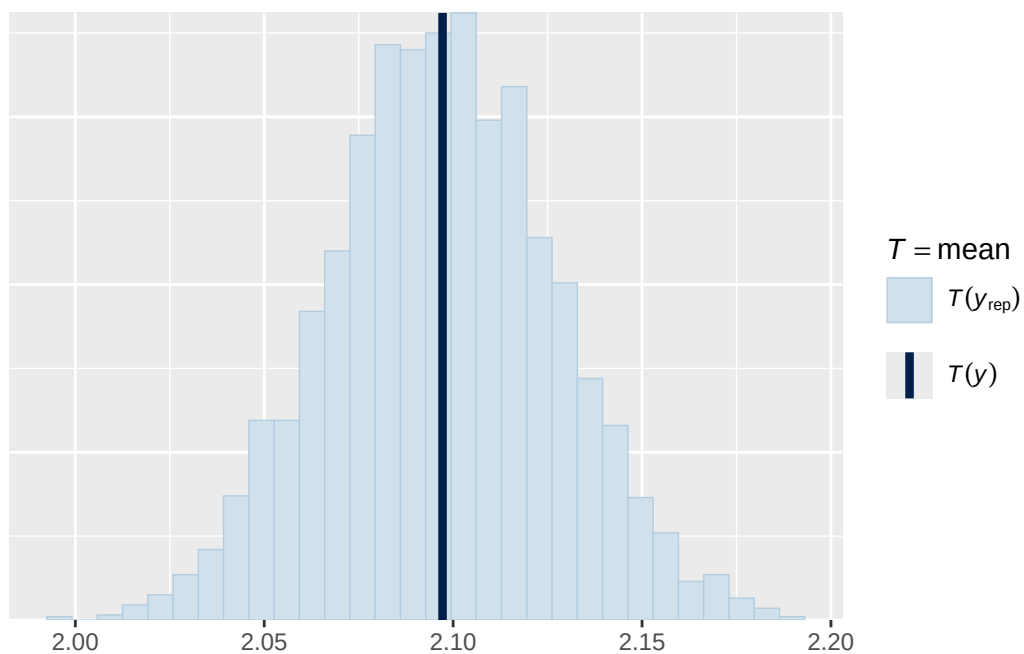


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

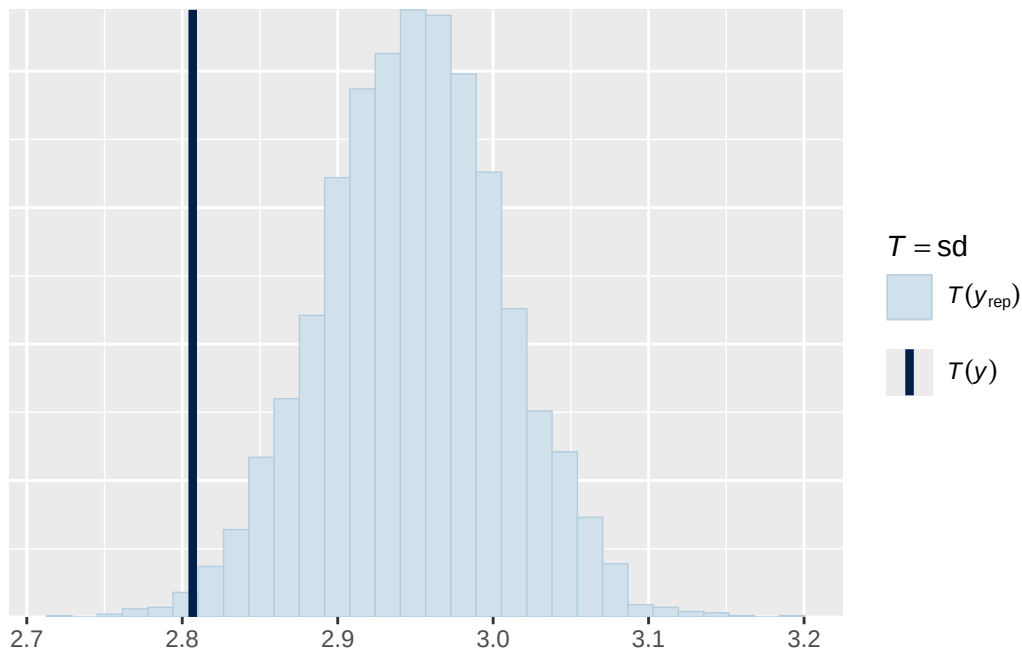


Using all posterior draws for ppc type 'stat' by default.

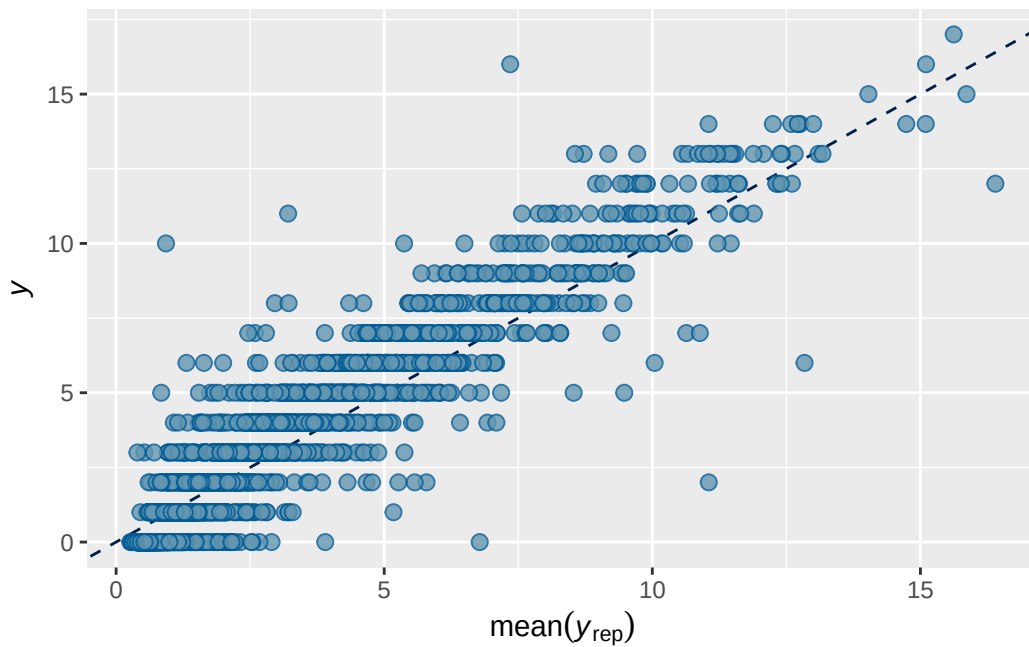
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1B longitudinal stimulant output

Estimate Est.Error Q2.5 Q97.5  
R2 0.7392941 0.007295095 0.7243178 0.752978

Family: poisson  
Links: mu = log  
Formula: adhd\_traits\_wave\_3 ~ breast\_development\_p\_wave\_2\_sc + age\_wave\_2\_sc + ethnrace + inr\_wave\_0\_sc  
Data: analytic\_df\_wide (Number of observations: 4515)  
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;  
total post-warmup draws = 4000

### Multilevel Hyperparameters:

~family\_id (Number of levels: 3928)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.871    | 0.021     | 0.831    | 0.913    | 1.001 | 1124     | 2045     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.086    | 0.036     | 0.017    | 0.162    | 1.004 | 473      | 604      |

### Regression Coefficients:

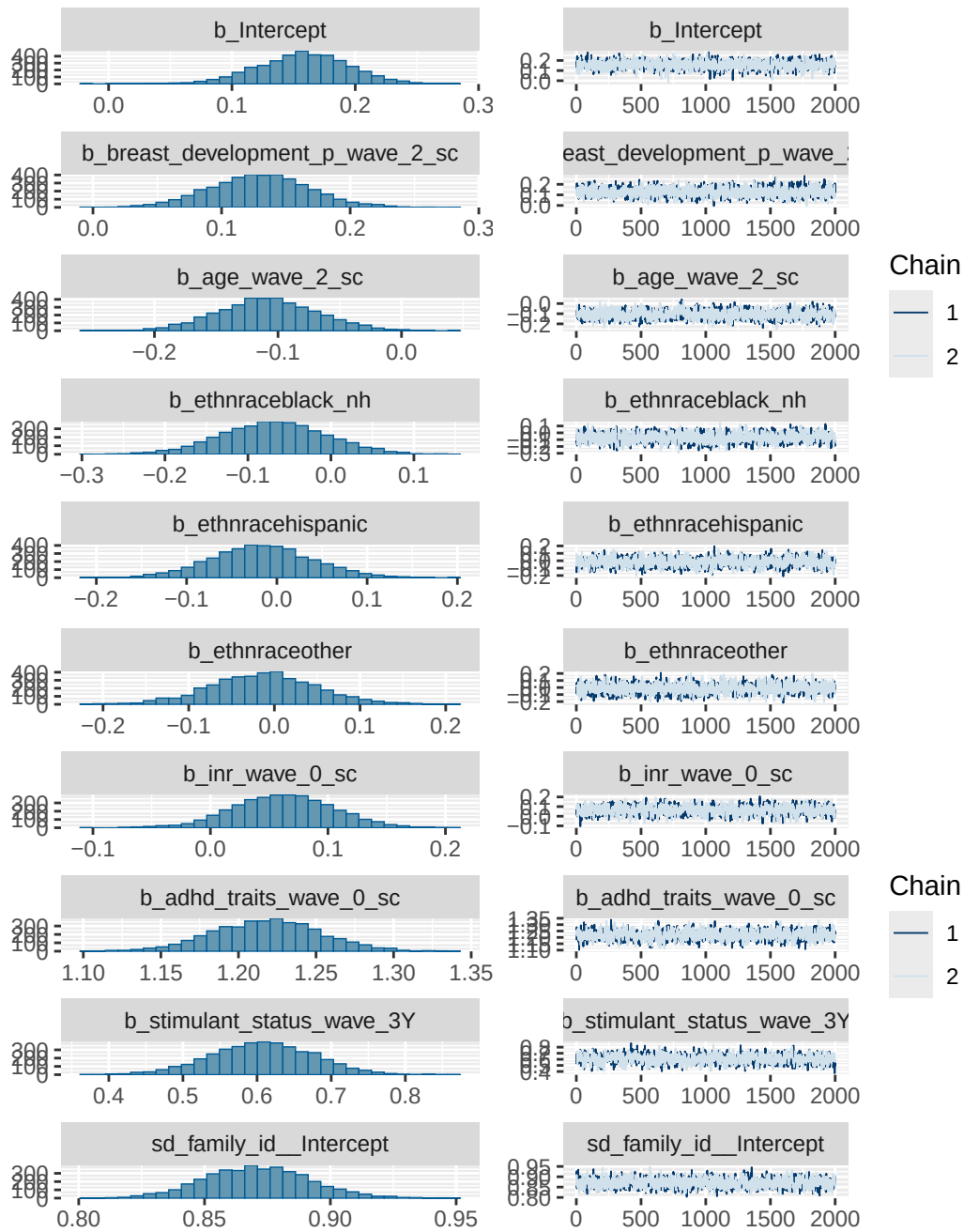
|                                | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  |
|--------------------------------|----------|-----------|----------|----------|-------|
| Intercept                      | 0.158    | 0.037     | 0.083    | 0.229    | 1.001 |
| breast_development_p_wave_2_sc | 0.130    | 0.039     | 0.055    | 0.207    | 1.001 |
| age_wave_2_sc                  | -0.108   | 0.040     | -0.185   | -0.029   | 1.002 |
| ethnraceblack_nh               | -0.068   | 0.063     | -0.194   | 0.058    | 1.000 |
| ethnracehispanic               | -0.017   | 0.056     | -0.127   | 0.094    | 1.002 |
| ethnraceother                  | -0.013   | 0.062     | -0.136   | 0.112    | 1.002 |
| inr_wave_0_sc                  | 0.062    | 0.042     | -0.021   | 0.143    | 1.000 |
| adhd_traits_wave_0_sc          | 1.218    | 0.035     | 1.151    | 1.286    | 1.000 |
| stimulant_status_wave_3Y       | 0.607    | 0.069     | 0.468    | 0.740    | 1.003 |

|                                | Bulk_ESS | Tail_ESS |
|--------------------------------|----------|----------|
| Intercept                      | 1655     | 2118     |
| breast_development_p_wave_2_sc | 1639     | 2517     |
| age_wave_2_sc                  | 1140     | 2018     |
| ethnraceblack_nh               | 1398     | 2258     |
| ethnracehispanic               | 1093     | 2113     |
| ethnraceother                  | 1478     | 2384     |
| inr_wave_0_sc                  | 1366     | 2232     |
| adhd_traits_wave_0_sc          | 1023     | 1712     |
| stimulant_status_wave_3Y       | 969      | 1728     |

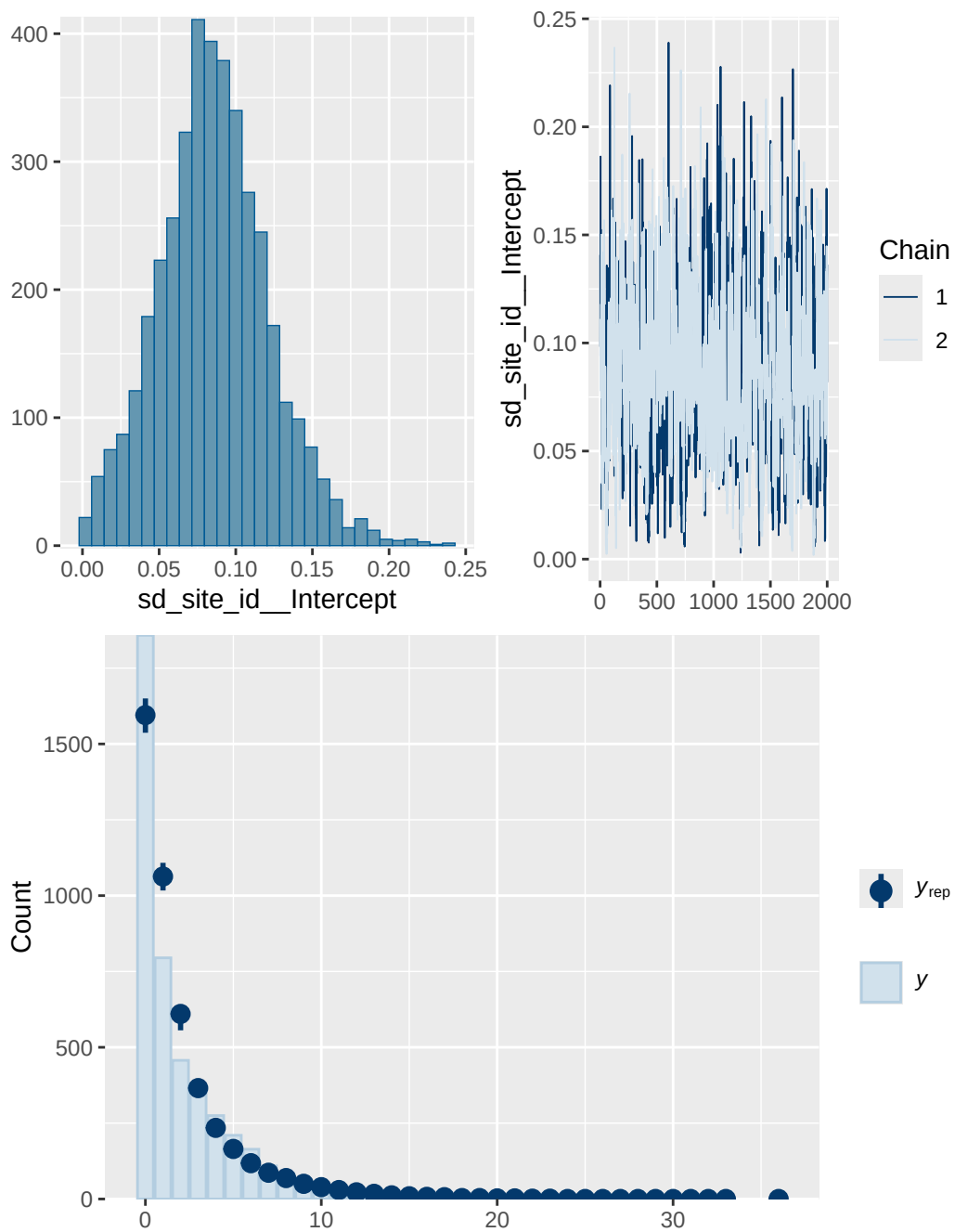
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

## Model 1B longitudinal stimulant diagnostics

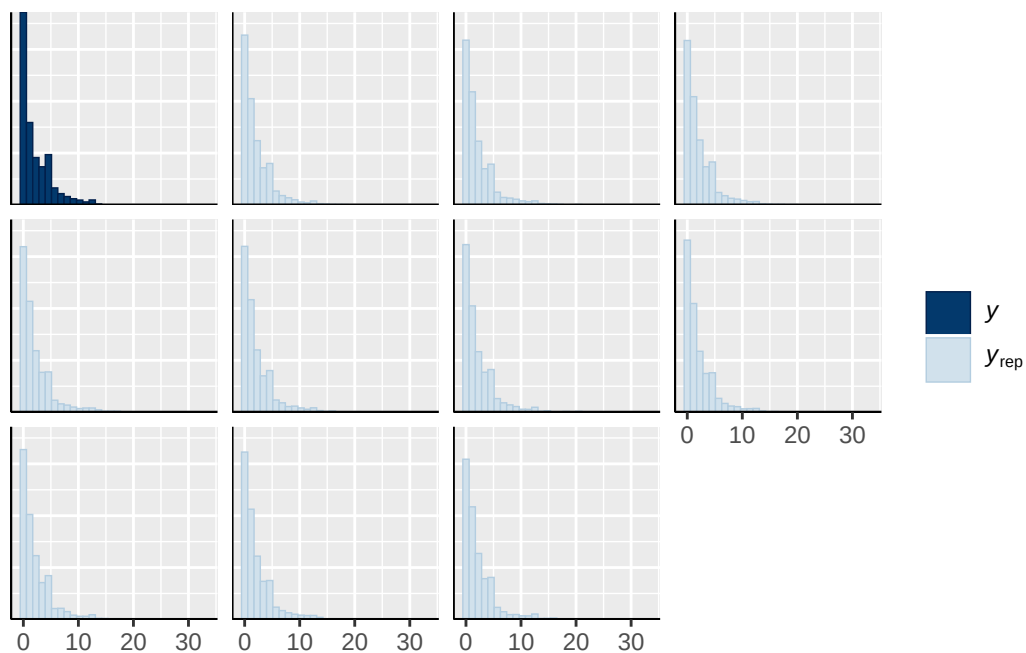
|                      | prior        | class        | coef                           | group     | resp |
|----------------------|--------------|--------------|--------------------------------|-----------|------|
|                      | normal(0, 1) | b            |                                |           |      |
|                      | normal(0, 1) | b            | adhd_traits_wave_0_sc          |           |      |
|                      | normal(0, 1) | b            | age_wave_2_sc                  |           |      |
|                      | normal(0, 1) | b            | breast_development_p_wave_2_sc |           |      |
|                      | normal(0, 1) | b            | ethnraceblack_nh               |           |      |
|                      | normal(0, 1) | b            | ethnracehispanic               |           |      |
|                      | normal(0, 1) | b            | ethnraceother                  |           |      |
|                      | normal(0, 1) | b            | inr_wave_0_sc                  |           |      |
|                      | normal(0, 1) | b            | stimulant_status_wave_3Y       |           |      |
| student_t(3, 0, 2.7) | Intercept    |              |                                |           |      |
| student_t(3, 0, 2.7) | sd           |              |                                |           |      |
| student_t(3, 0, 2.7) | sd           |              |                                | family_id |      |
| student_t(3, 0, 2.7) | sd           |              | Intercept                      | family_id |      |
| student_t(3, 0, 2.7) | sd           |              |                                | site_id   |      |
| student_t(3, 0, 2.7) | sd           |              | Intercept                      | site_id   |      |
| dpar nlpar lb ub tag |              | source       |                                |           |      |
|                      |              | user         |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | (vectorized) |                                |           |      |
|                      |              | default      |                                |           |      |
| 0                    |              | default      |                                |           |      |
| 0                    |              | (vectorized) |                                |           |      |
| 0                    |              | (vectorized) |                                |           |      |
| 0                    |              | (vectorized) |                                |           |      |
| 0                    |              | (vectorized) |                                |           |      |



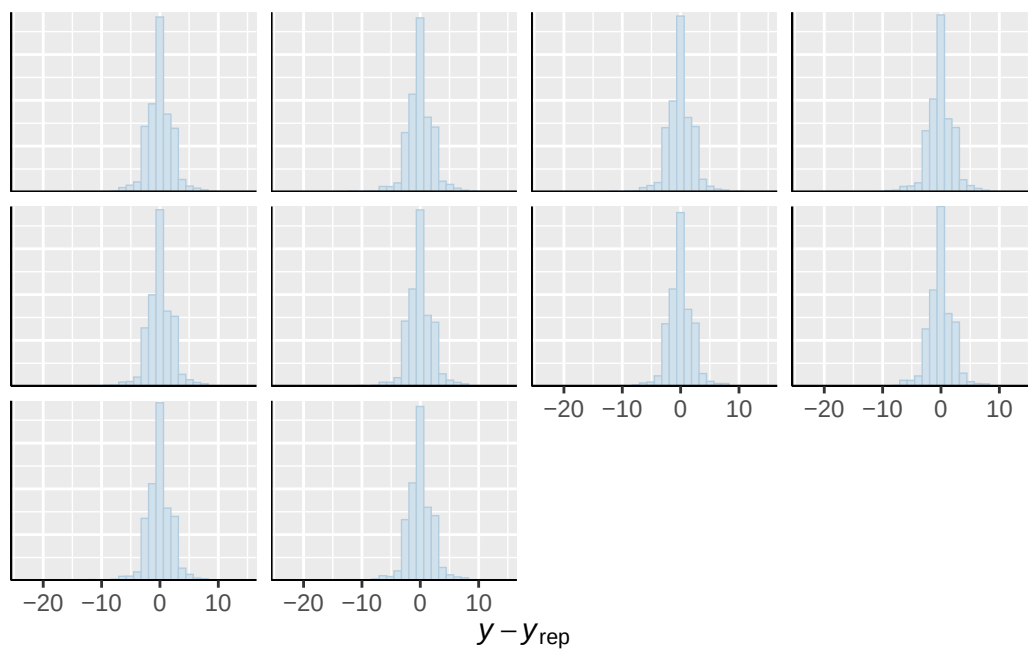




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

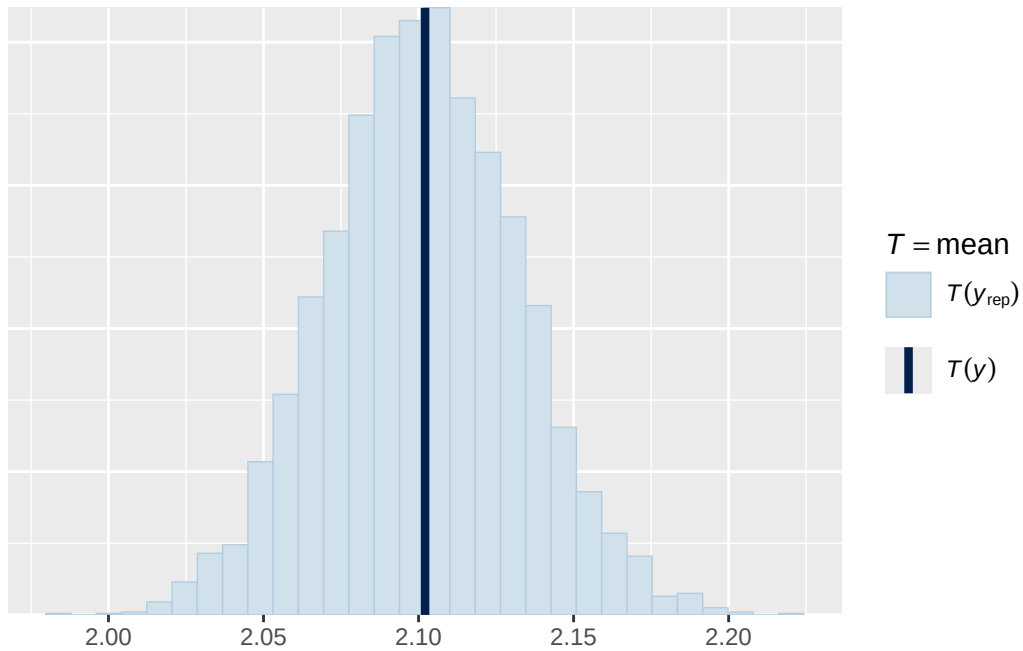


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

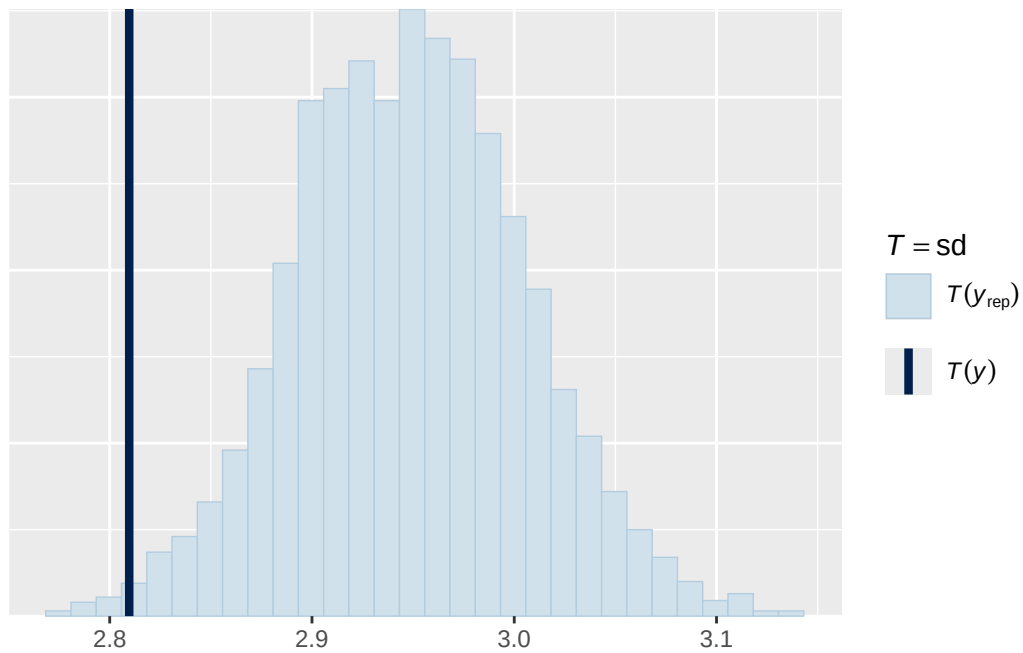


Using all posterior draws for ppc type 'stat' by default.

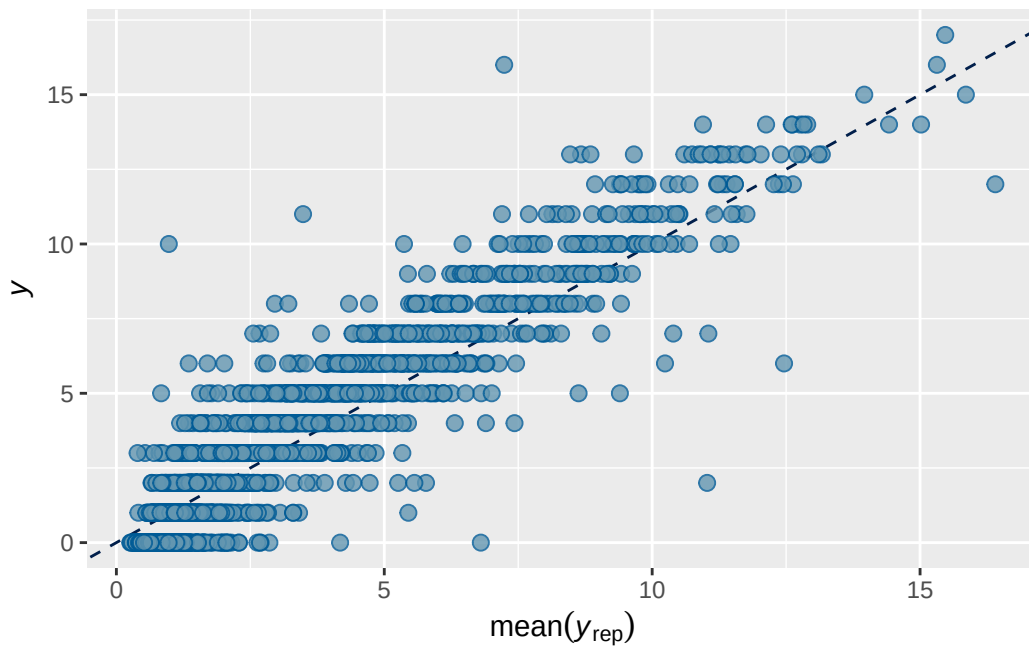
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 5B longitudinal stimulant output

|    | Estimate  | Est.Error   | Q2.5      | Q97.5     |
|----|-----------|-------------|-----------|-----------|
| R2 | 0.8239053 | 0.005205048 | 0.8133844 | 0.8337296 |

Family: poisson  
 Links: mu = log  
 Formula: externalising\_wave\_3 ~ breast\_development\_p\_wave\_2\_sc \* adhd\_traits\_wave\_2\_sc + age.  
 Data: analytic\_df\_wide (Number of observations: 4515)  
 Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;  
 total post-warmup draws = 8000

Multilevel Hyperparameters:

~family\_id (Number of levels: 3928)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.986    | 0.020     | 0.946    | 1.026    | 1.001 | 2381     | 4669     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | 1-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.065    | 0.037     | 0.005    | 0.145    | 1.009 | 611      | 1517     |

# Regression Coefficients:

|                                                      | Estimate | Est.Error |       |  |
|------------------------------------------------------|----------|-----------|-------|--|
| Intercept                                            | 0.503    | 0.035     |       |  |
| breast_development_p_wave_2_sc                       | 0.188    | 0.038     |       |  |
| adhd_traits_wave_2_sc                                | 0.570    | 0.036     |       |  |
| age_wave_2_sc                                        | -0.119   | 0.038     |       |  |
| ethnraceblack_nh                                     | -0.129   | 0.066     |       |  |
| ethnracehispanic                                     | -0.096   | 0.057     |       |  |
| ethnraceother                                        | -0.082   | 0.064     |       |  |
| inr_wave_0_sc                                        | -0.004   | 0.042     |       |  |
| externalising_wave_0_sc                              | 0.914    | 0.035     |       |  |
| stimulant_status_wave_3Y                             | 0.111    | 0.074     |       |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.175    | 0.065     |       |  |
|                                                      | 1-95% CI | u-95% CI  | Rhat  |  |
| Intercept                                            | 0.435    | 0.571     | 1.001 |  |
| breast_development_p_wave_2_sc                       | 0.114    | 0.264     | 1.001 |  |
| adhd_traits_wave_2_sc                                | 0.498    | 0.641     | 1.001 |  |
| age_wave_2_sc                                        | -0.193   | -0.045    | 1.000 |  |
| ethnraceblack_nh                                     | -0.260   | 0.000     | 1.003 |  |
| ethnracehispanic                                     | -0.206   | 0.015     | 1.000 |  |
| ethnraceother                                        | -0.209   | 0.044     | 1.001 |  |
| inr_wave_0_sc                                        | -0.086   | 0.079     | 1.001 |  |
| externalising_wave_0_sc                              | 0.846    | 0.982     | 1.001 |  |
| stimulant_status_wave_3Y                             | -0.033   | 0.257     | 1.001 |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.044    | 0.304     | 1.000 |  |
|                                                      | Bulk_ESS | Tail_ESS  |       |  |
| Intercept                                            | 4138     | 4789      |       |  |
| breast_development_p_wave_2_sc                       | 4101     | 5676      |       |  |
| adhd_traits_wave_2_sc                                | 2777     | 4361      |       |  |
| age_wave_2_sc                                        | 3692     | 5187      |       |  |
| ethnraceblack_nh                                     | 2311     | 4174      |       |  |
| ethnracehispanic                                     | 3236     | 4672      |       |  |
| ethnraceother                                        | 2893     | 4509      |       |  |
| inr_wave_0_sc                                        | 3251     | 5037      |       |  |
| externalising_wave_0_sc                              | 2506     | 3980      |       |  |
| stimulant_status_wave_3Y                             | 2319     | 3887      |       |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 2940     | 4482      |       |  |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

### Model 5B longitudinal stimulant diagnostics

```

prior      class
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd

coef      group resp dpar nlpar

adhd_traits_wave_2_sc
age_wave_2_sc
breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
ethnraceblack_nh
ethnracehispanic
ethnraceother
externalising_wave_0_sc
inr_wave_0_sc
stimulant_status_wave_3Y

family_id
Intercept family_id
site_id
Intercept site_id

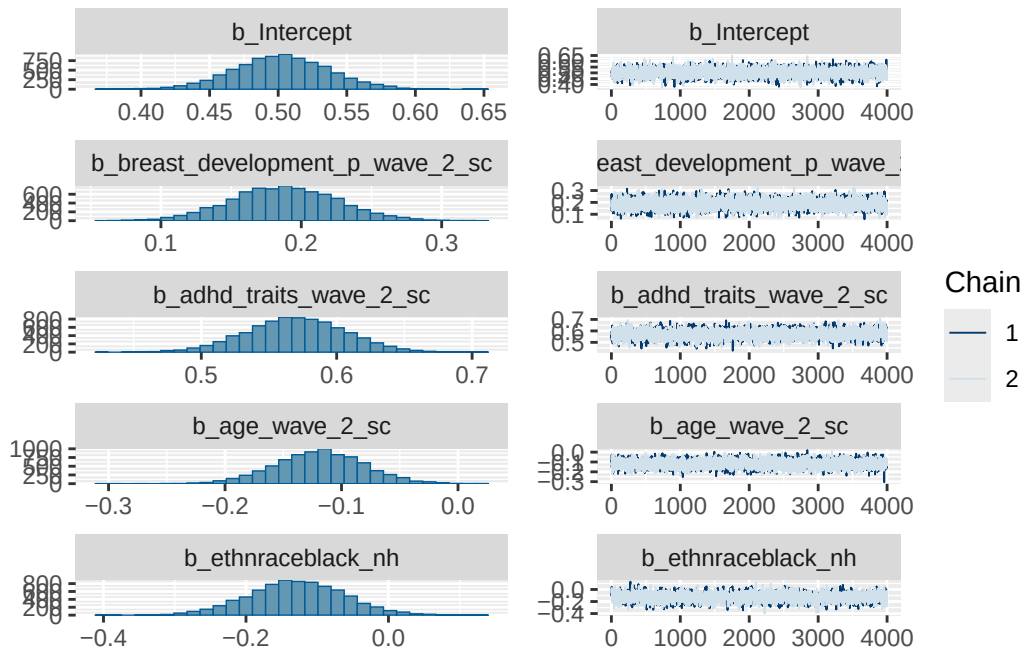
lb ub tag      source
user
(vectorized)
(vectorized)

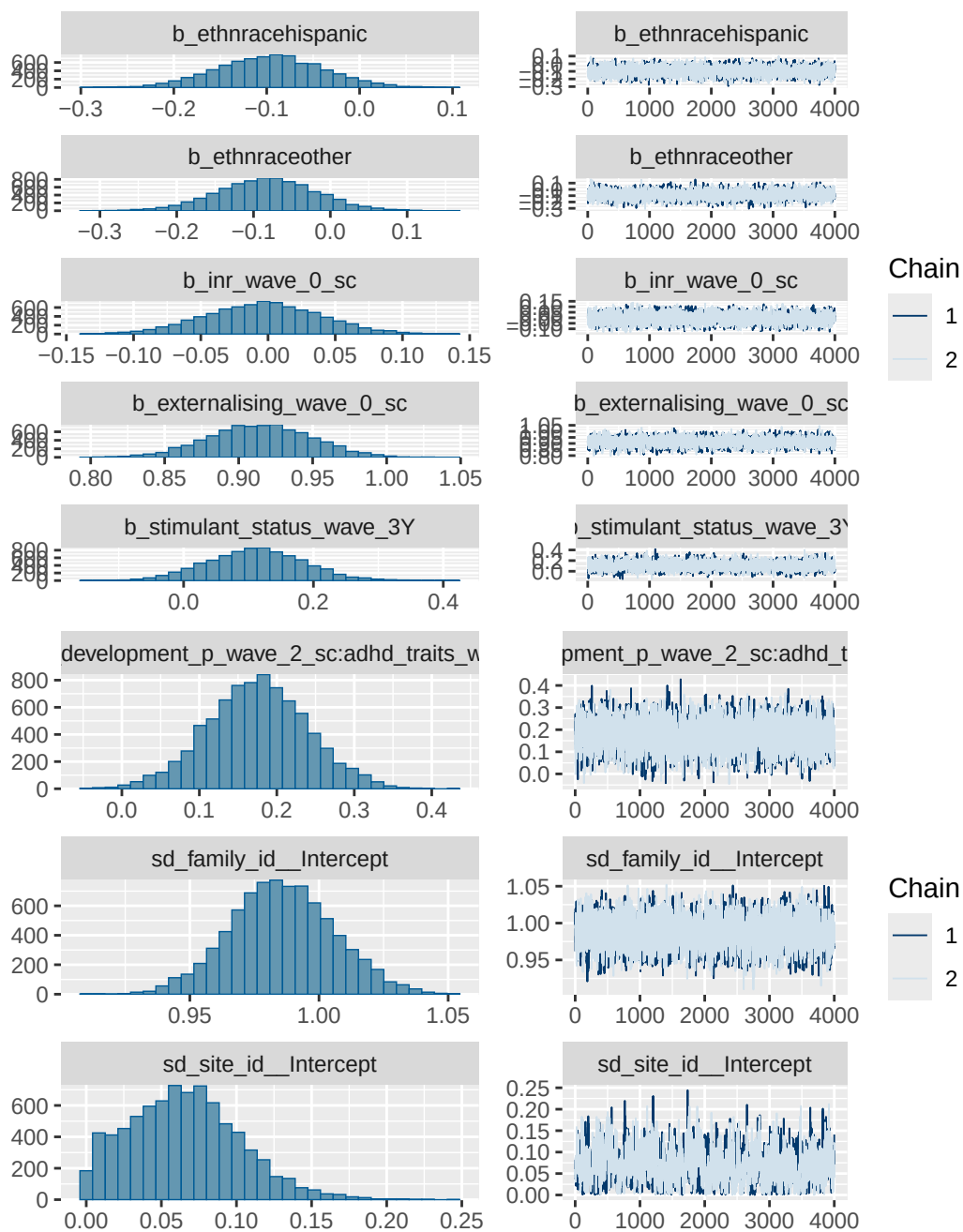
```

```

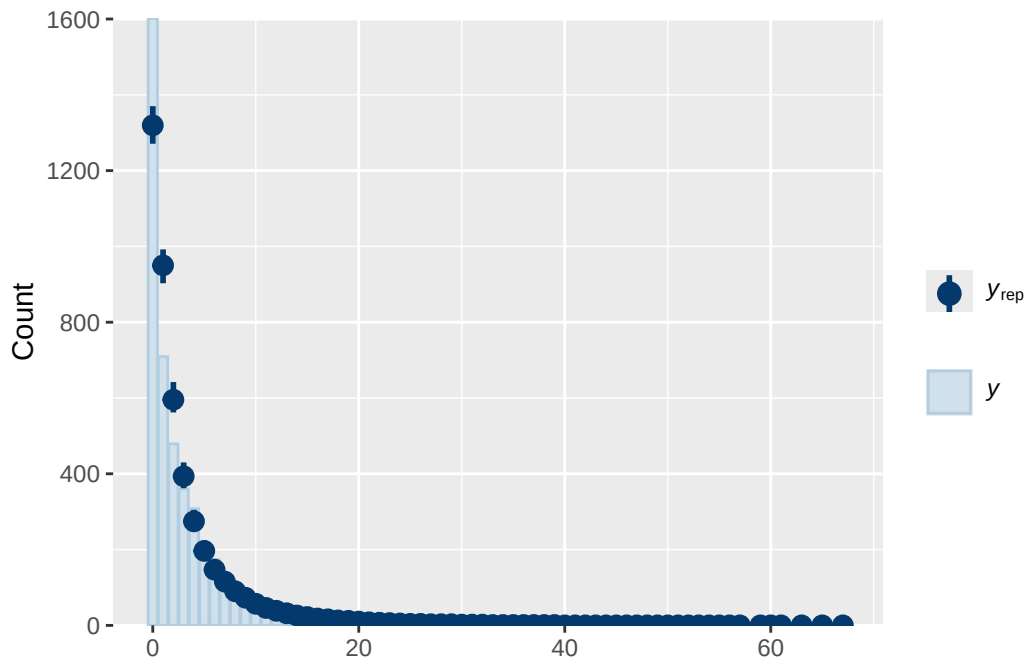
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default
0      default
0      (vectorized)
0      (vectorized)
0      (vectorized)
0      (vectorized)

```

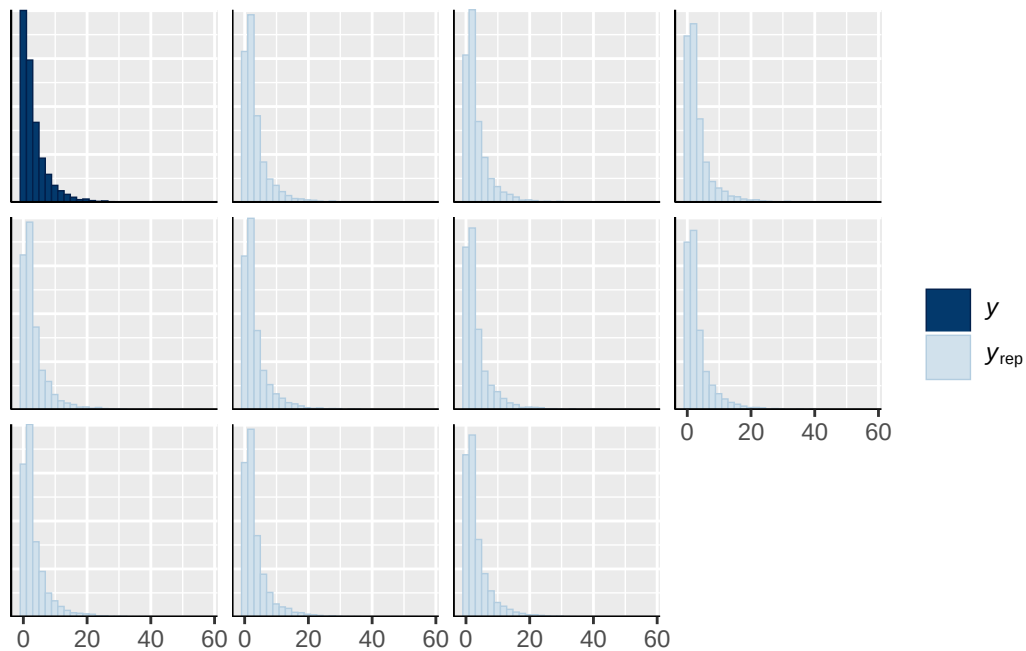




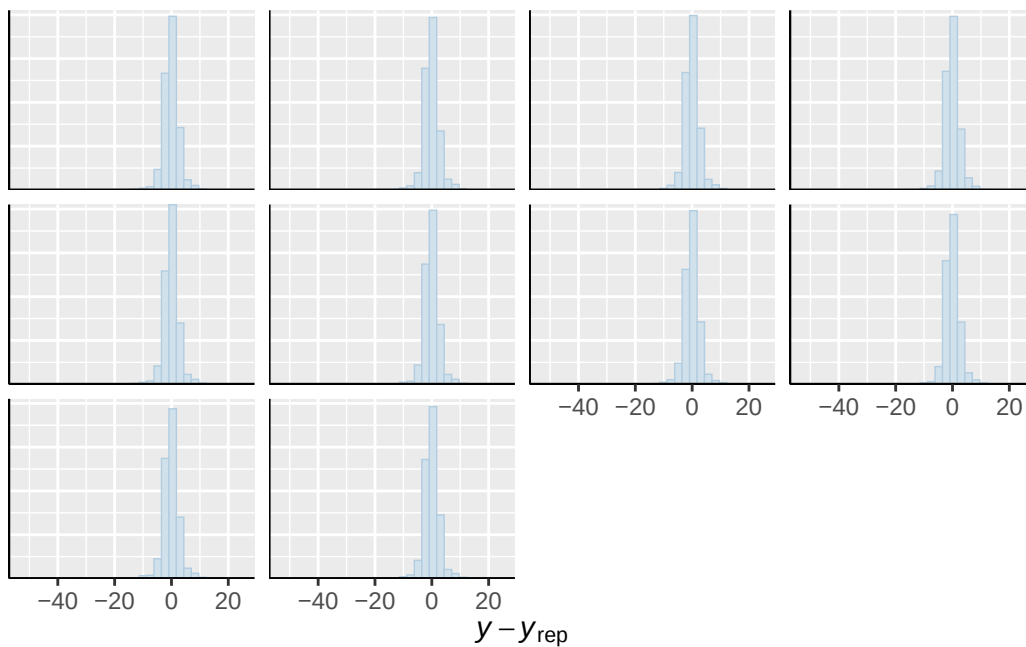




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

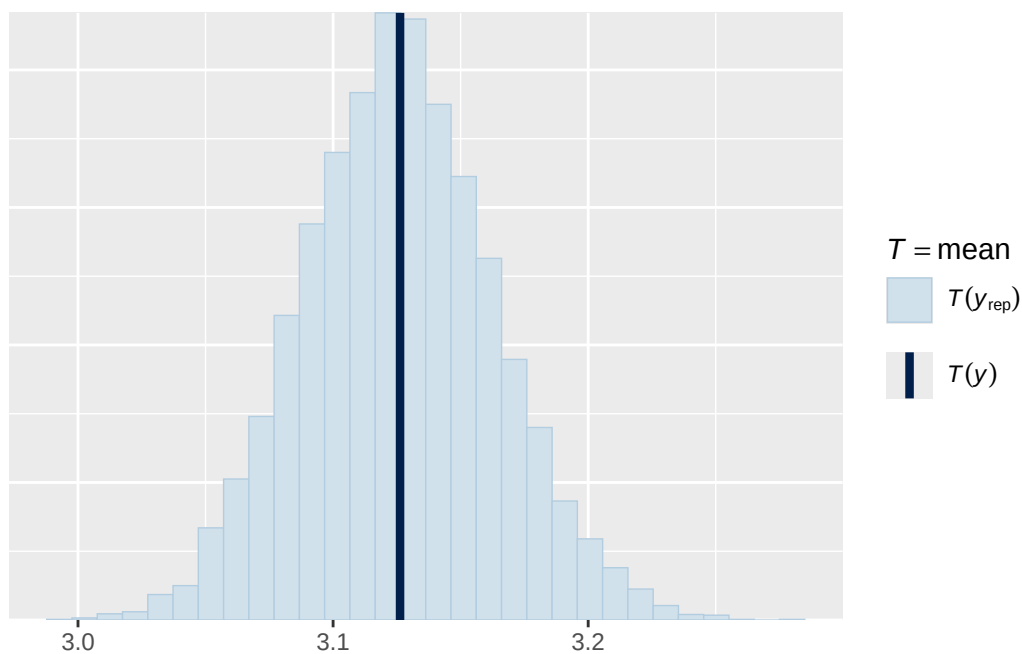


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

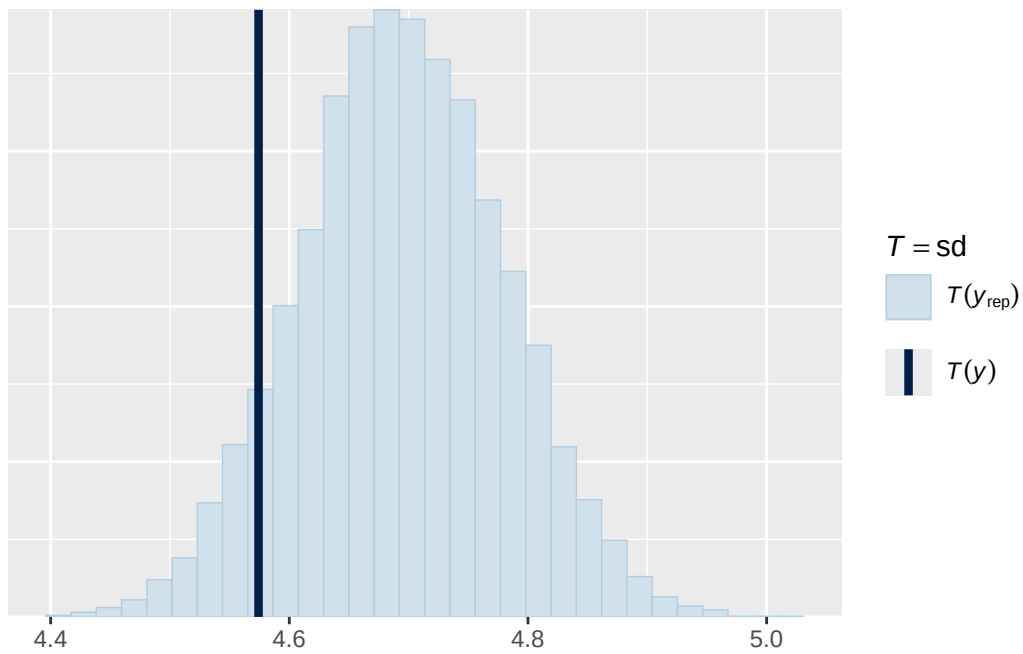


Using all posterior draws for ppc type 'stat' by default.

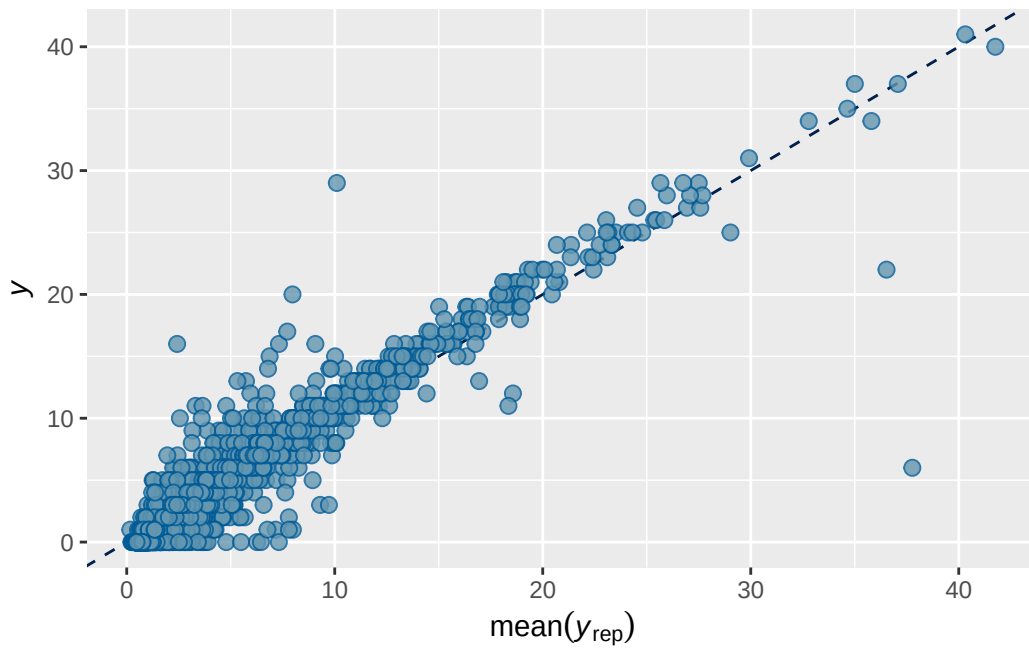
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1 longitudinal prior 1 output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.7348938 0.007531416 0.7199932 0.7493823

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3915)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.881    0.020   0.841   0.922 1.007     937    1457

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.092    0.036   0.020   0.165 1.007     504     518

```

### Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS
Intercept          0.150    0.039   0.072   0.224 1.003     1400
menarche_status_p_wave_2Y 0.099    0.040   0.022   0.178 1.001     1665
age_wave_2_sc       -0.113    0.040  -0.191  -0.038 1.001     1723
ethnraceblack_nh    -0.077    0.063  -0.197   0.049 1.001     1514
ethnracehispanic    -0.042    0.054  -0.146   0.066 1.001     1381
ethnraceother        0.003    0.060  -0.116   0.121 1.001     1324
inr_wave_0_sc        0.100    0.041   0.018   0.182 1.000     1489
adhd_traits_wave_0_sc  1.307    0.033   1.242   1.374 1.001     1073
      Tail_ESS
Intercept          2033
menarche_status_p_wave_2Y 2492
age_wave_2_sc       2637
ethnraceblack_nh    2412
ethnracehispanic    2077
ethnraceother        1677
inr_wave_0_sc        2112
adhd_traits_wave_0_sc 1886

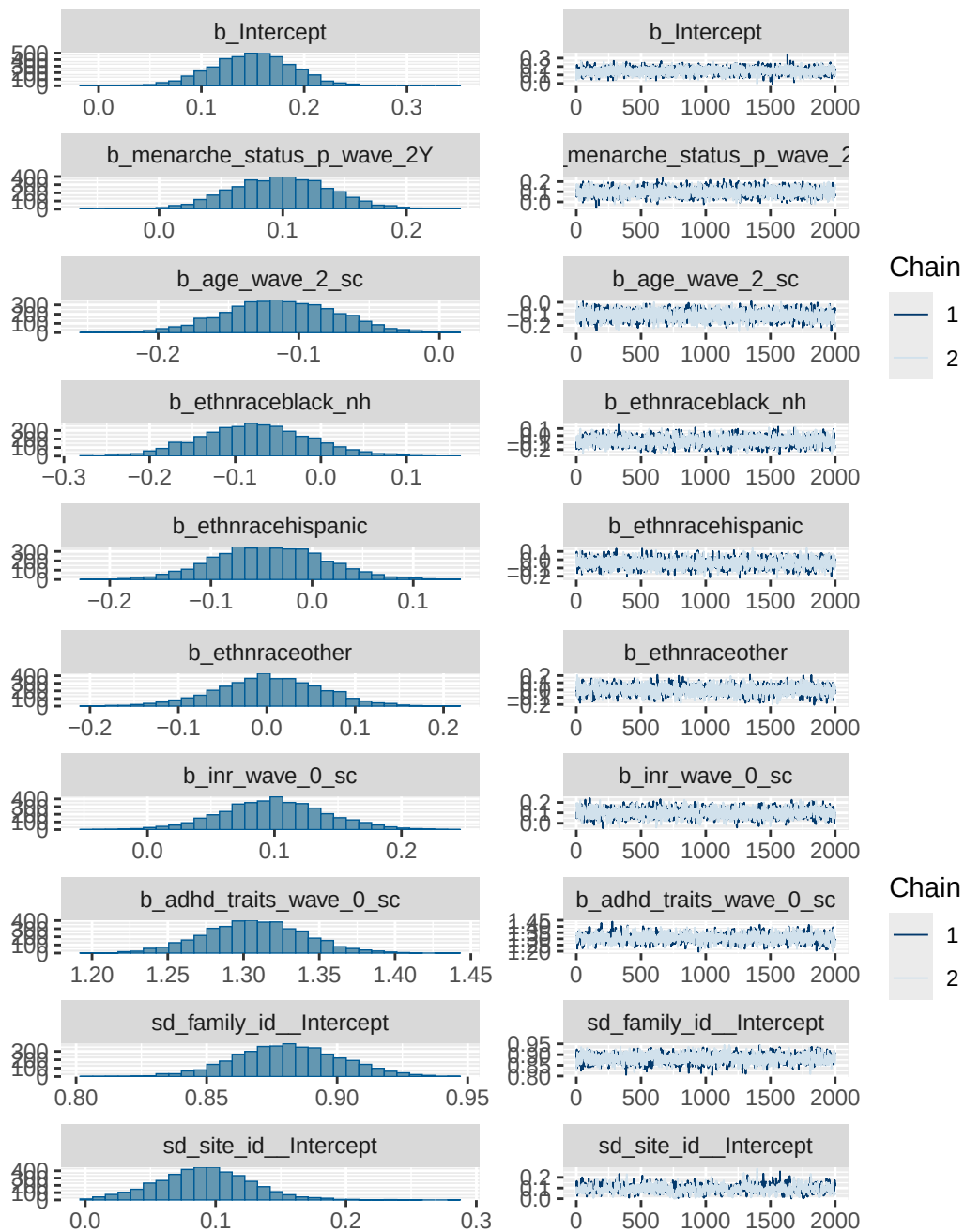
```

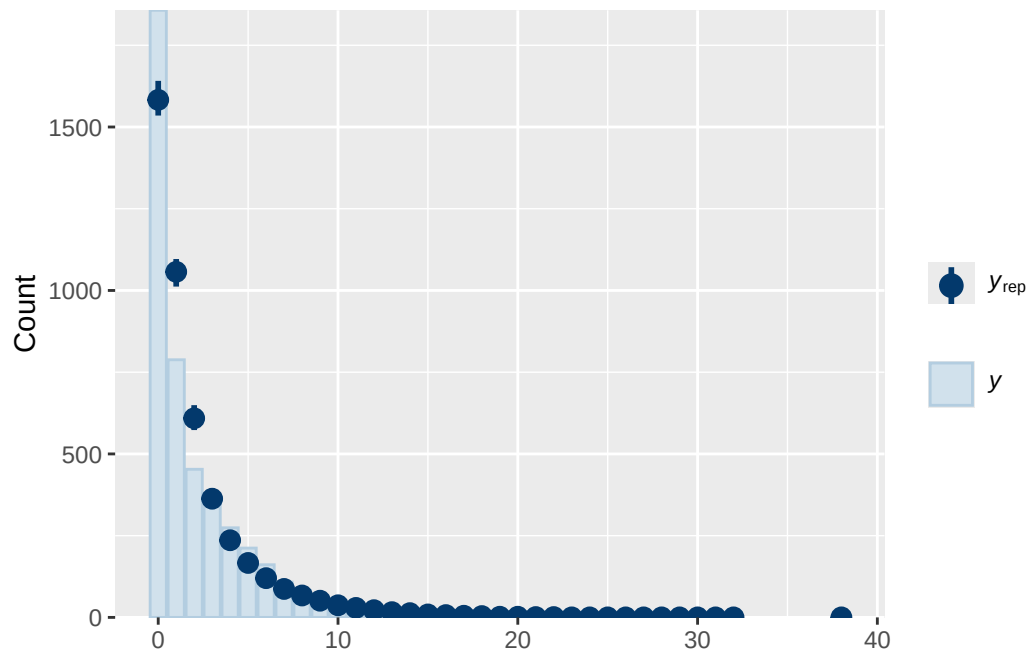
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS

and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

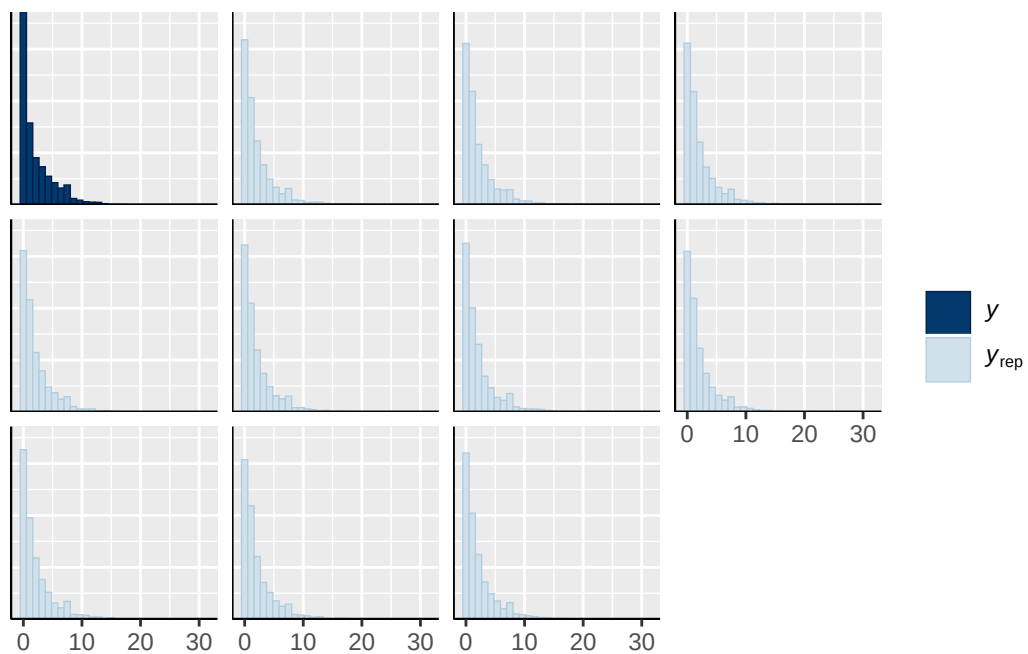
## Model 1 longitudinal prior 1 diagnostics

|                      | prior           | class | coef                      | group        | resp | dpar |
|----------------------|-----------------|-------|---------------------------|--------------|------|------|
|                      | normal(0, 0.25) | b     |                           |              |      |      |
|                      | normal(0, 0.25) | b     | adhd_traits_wave_0_sc     |              |      |      |
|                      | normal(0, 0.25) | b     | age_wave_2_sc             |              |      |      |
|                      | normal(0, 0.25) | b     | ethnraceblack_nh          |              |      |      |
|                      | normal(0, 0.25) | b     | ethnracehispanic          |              |      |      |
|                      | normal(0, 0.25) | b     | ethnraceother             |              |      |      |
|                      | normal(0, 0.25) | b     | inr_wave_0_sc             |              |      |      |
|                      | normal(0, 0.25) | b     | menarche_status_p_wave_2Y |              |      |      |
| student_t(3, 0, 2.7) | Intercept       |       |                           |              |      |      |
| student_t(3, 0, 2.7) | sd              |       |                           |              |      |      |
| student_t(3, 0, 2.7) | sd              |       |                           | family_id    |      |      |
| student_t(3, 0, 2.7) | sd              |       | Intercept                 | family_id    |      |      |
| student_t(3, 0, 2.7) | sd              |       |                           | site_id      |      |      |
| student_t(3, 0, 2.7) | sd              |       | Intercept                 | site_id      |      |      |
| nlpar                | lb              | ub    | tag                       | source       |      |      |
|                      |                 |       |                           | user         |      |      |
|                      |                 |       |                           | (vectorized) |      |      |
|                      |                 |       |                           | (vectorized) |      |      |
|                      |                 |       |                           | (vectorized) |      |      |
|                      |                 |       |                           | (vectorized) |      |      |
|                      |                 |       |                           | (vectorized) |      |      |
|                      |                 |       |                           | (vectorized) |      |      |
|                      |                 |       |                           | (vectorized) |      |      |
|                      |                 |       |                           | default      |      |      |
| 0                    |                 |       |                           | default      |      |      |
| 0                    |                 |       |                           | (vectorized) |      |      |
| 0                    |                 |       |                           | (vectorized) |      |      |
| 0                    |                 |       |                           | (vectorized) |      |      |
| 0                    |                 |       |                           | (vectorized) |      |      |

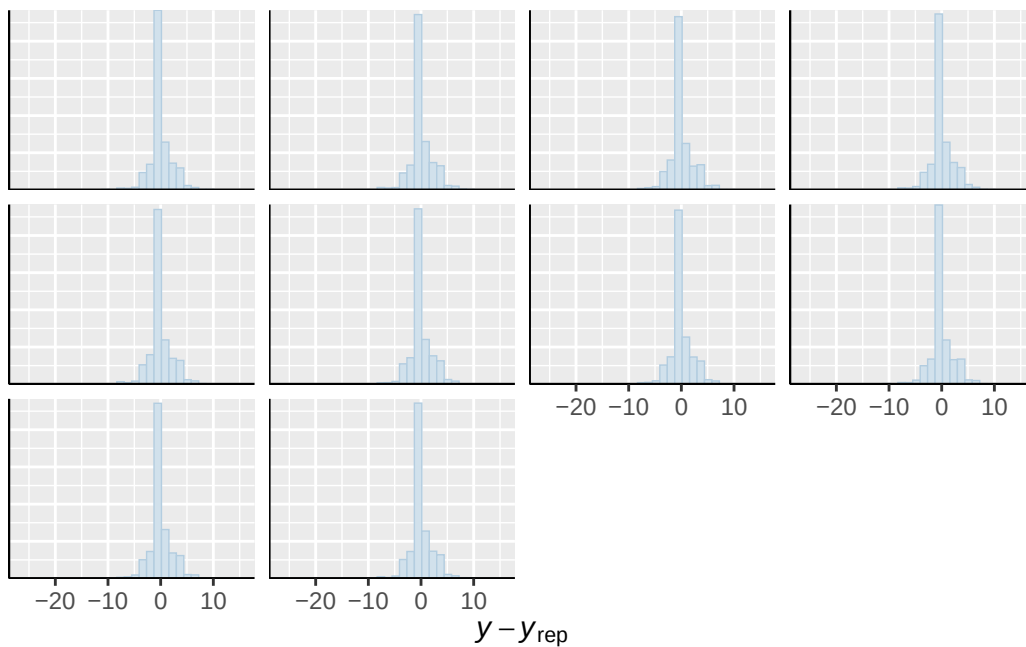




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

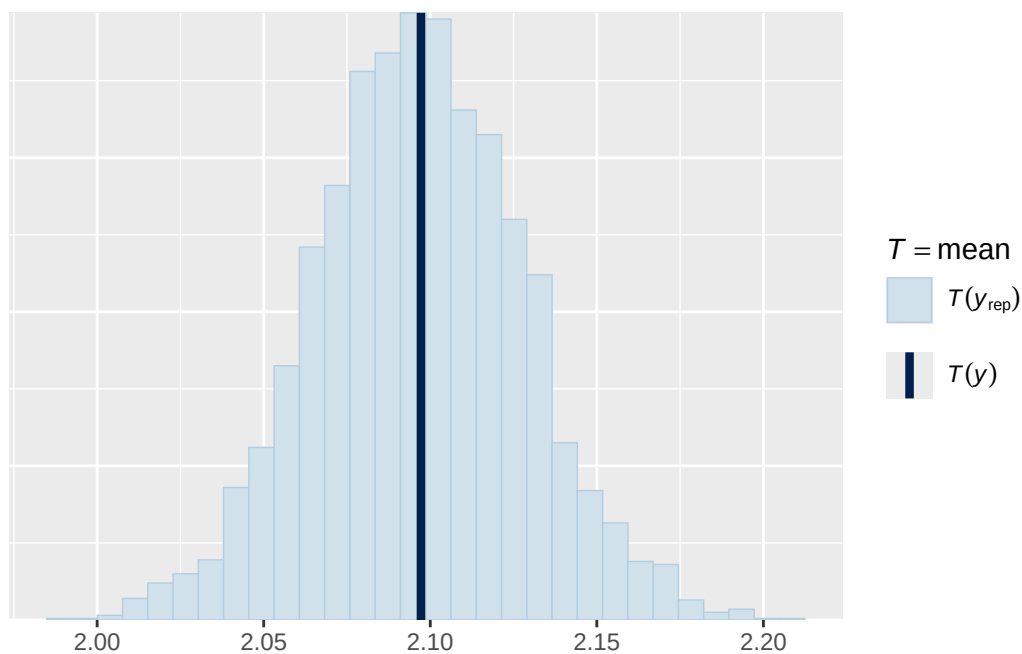


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



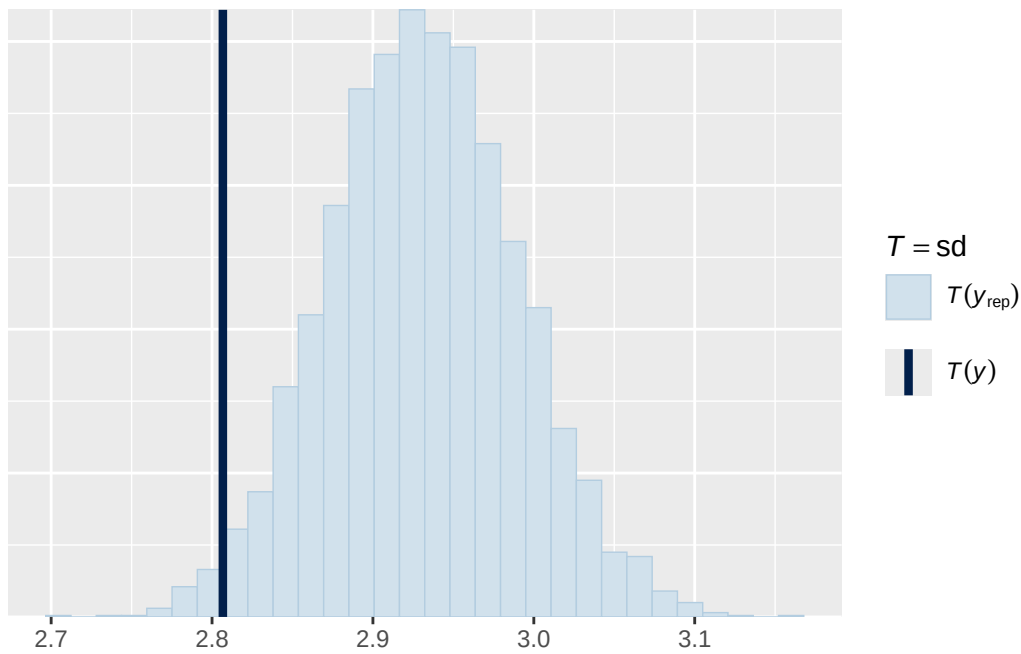
Using all posterior draws for ppc type 'stat' by default.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

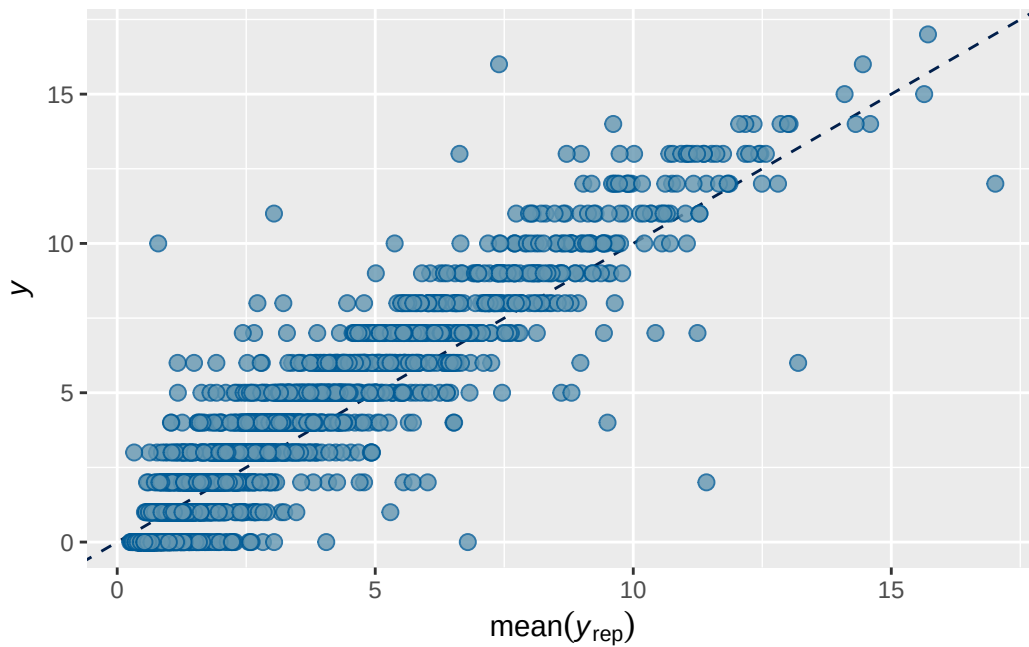




Using all posterior draws for ppc type 'stat' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1B longitudinal prior 1 output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.7345825 0.007551707 0.7193221 0.7491023

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + inr_wave_0_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.883    0.021   0.844   0.924 1.001    1351    2068

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.093    0.035   0.024   0.166 1.004     427     447

```

### Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat
Intercept      0.193    0.036   0.118   0.264 1.001
breast_development_p_wave_2_sc 0.118    0.040   0.039   0.194 1.002
age_wave_2_sc   -0.101    0.039  -0.176  -0.024 1.002
ethnraceblack_nh -0.082    0.062  -0.205   0.039 1.000
ethnracehispanic -0.031    0.055  -0.135   0.081 1.001
ethnraceother   -0.007    0.059  -0.125   0.107 1.001
inr_wave_0_sc    0.084    0.041   0.001   0.162 1.001
adhd_traits_wave_0_sc 1.300    0.033   1.233   1.366 1.001

      Bulk_ESS Tail_ESS
Intercept      1961    2153
breast_development_p_wave_2_sc 1659    2617
age_wave_2_sc   1453    2377
ethnraceblack_nh 1491    2487
ethnracehispanic 1159    1633
ethnraceother   1565    2433
inr_wave_0_sc    1462    2222
adhd_traits_wave_0_sc 1081    2297

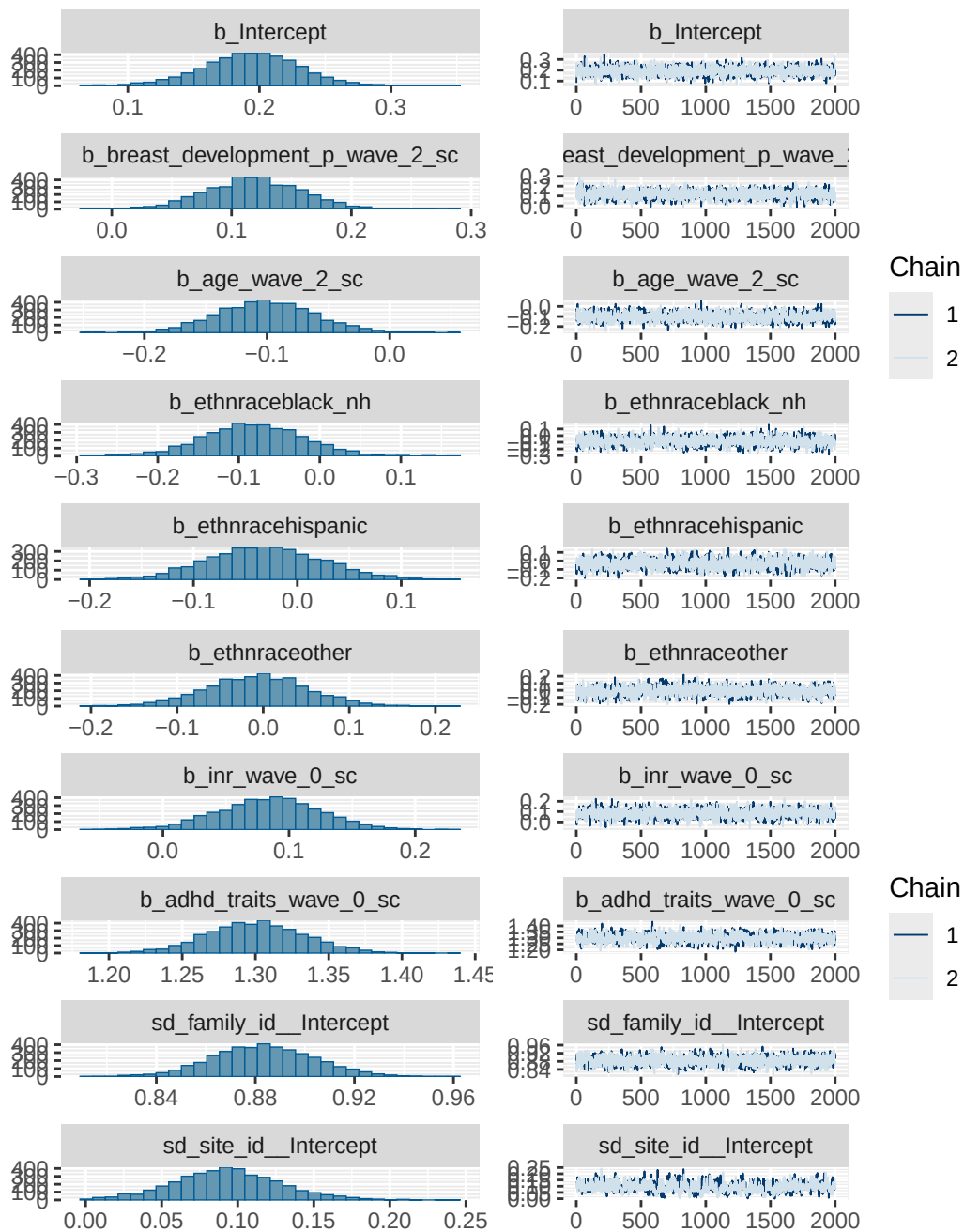
```

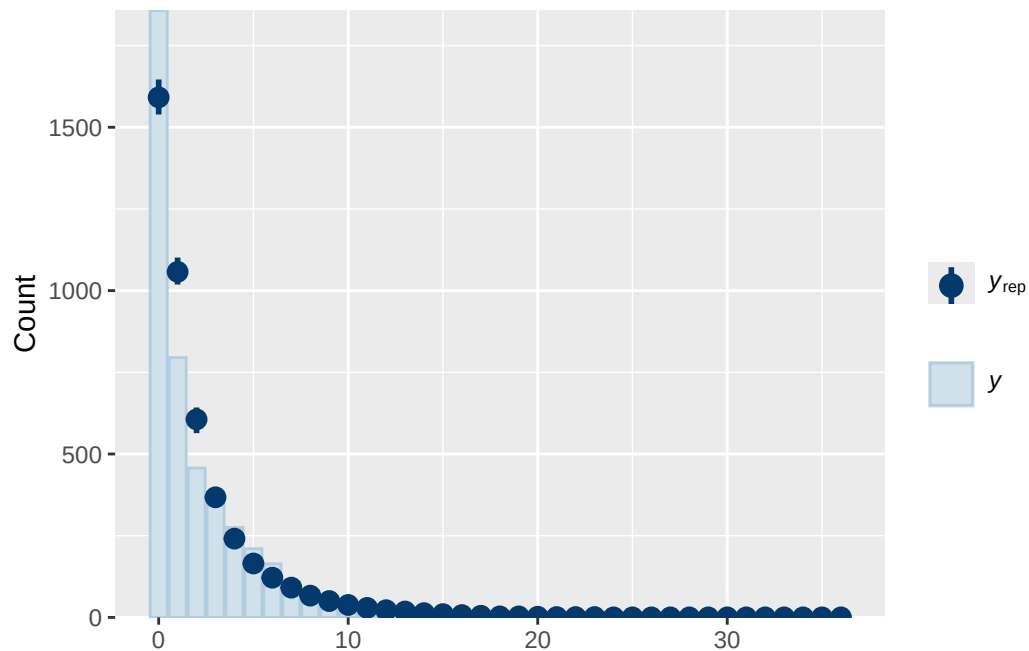
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS

and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

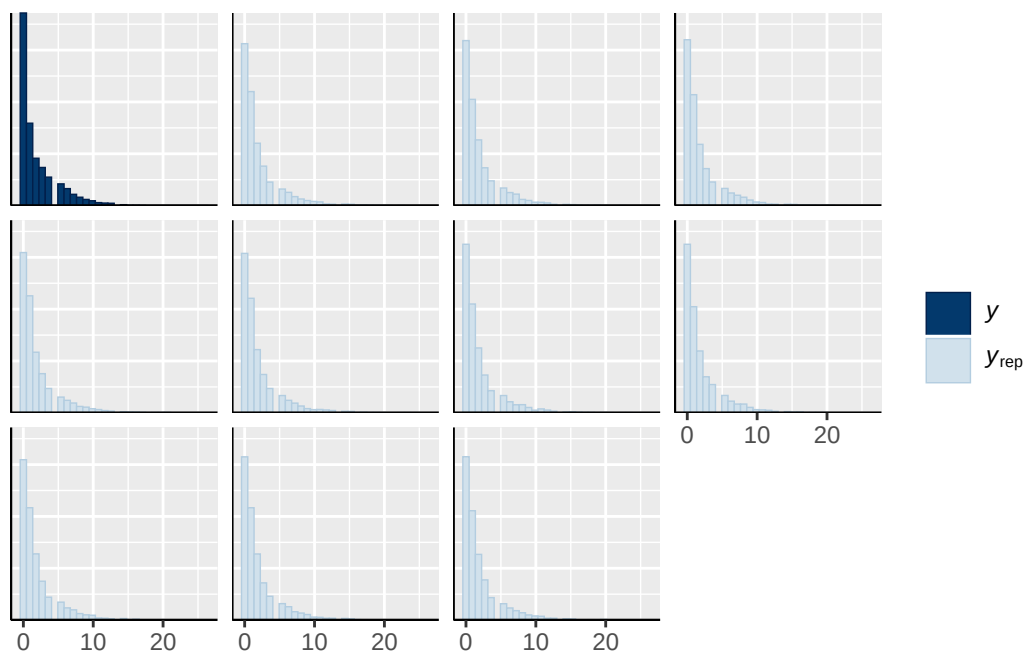
## Model 1B longitudinal prior 1 diagnostics

|                      | prior           | class        | coef                           | group     | resp |
|----------------------|-----------------|--------------|--------------------------------|-----------|------|
|                      | normal(0, 0.25) | b            |                                |           |      |
|                      | normal(0, 0.25) | b            | adhd_traits_wave_0_sc          |           |      |
|                      | normal(0, 0.25) | b            | age_wave_2_sc                  |           |      |
|                      | normal(0, 0.25) | b            | breast_development_p_wave_2_sc |           |      |
|                      | normal(0, 0.25) | b            | ethnraceblack_nh               |           |      |
|                      | normal(0, 0.25) | b            | ethnracehispanic               |           |      |
|                      | normal(0, 0.25) | b            | ethnraceother                  |           |      |
|                      | normal(0, 0.25) | b            | inr_wave_0_sc                  |           |      |
| student_t(3, 0, 2.7) | Intercept       |              |                                |           |      |
| student_t(3, 0, 2.7) | sd              |              |                                |           |      |
| student_t(3, 0, 2.7) | sd              |              |                                | family_id |      |
| student_t(3, 0, 2.7) | sd              |              | Intercept                      | family_id |      |
| student_t(3, 0, 2.7) | sd              |              |                                | site_id   |      |
| student_t(3, 0, 2.7) | sd              |              | Intercept                      | site_id   |      |
| dpar nlpar lb ub tag |                 | source       |                                |           |      |
|                      |                 | user         |                                |           |      |
|                      |                 | (vectorized) |                                |           |      |
|                      |                 | (vectorized) |                                |           |      |
|                      |                 | (vectorized) |                                |           |      |
|                      |                 | (vectorized) |                                |           |      |
|                      |                 | (vectorized) |                                |           |      |
|                      |                 | (vectorized) |                                |           |      |
|                      |                 | (vectorized) |                                |           |      |
|                      |                 | default      |                                |           |      |
| 0                    |                 | default      |                                |           |      |
| 0                    |                 | (vectorized) |                                |           |      |
| 0                    |                 | (vectorized) |                                |           |      |
| 0                    |                 | (vectorized) |                                |           |      |
| 0                    |                 | (vectorized) |                                |           |      |





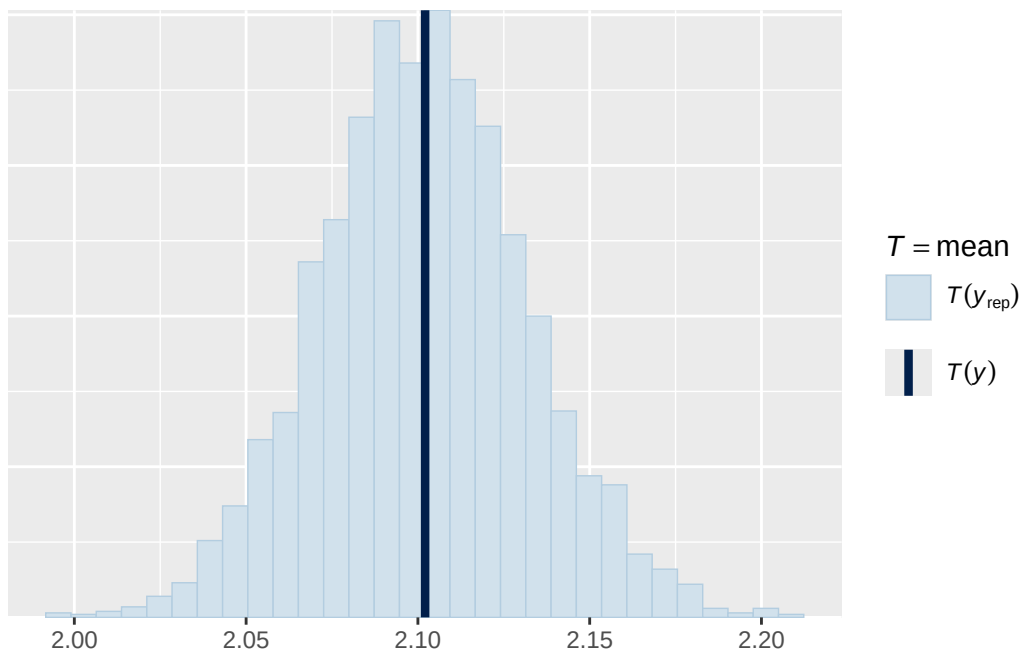
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



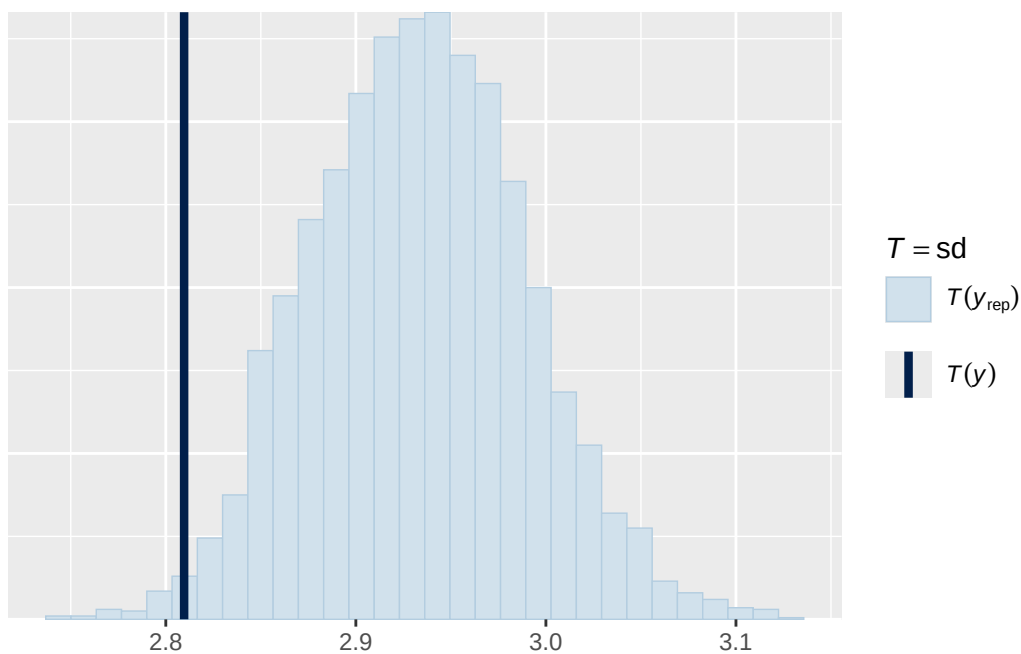
Using all posterior draws for ppc type 'stat' by default.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest.

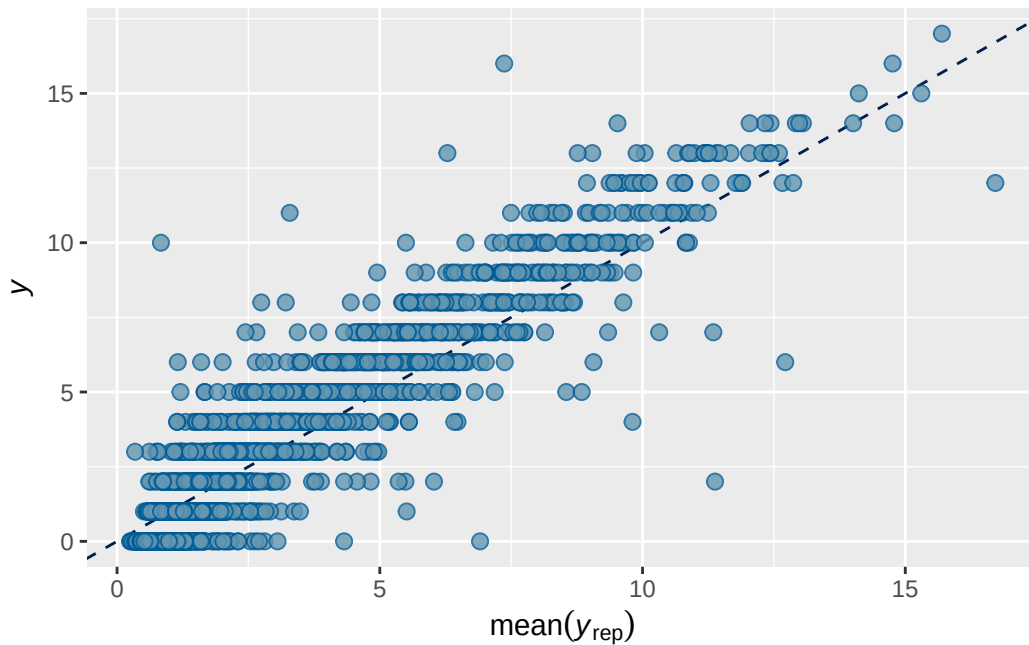
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 5B longitudinal prior 1 output

```

      Estimate  Est.Error    Q2.5    Q97.5
R2 0.8229693 0.005223904 0.8125308 0.8329675

```

Family: poisson

Links: mu = log

Formula: externalising\_wave\_3 ~ breast\_development\_p\_wave\_2\_sc \* adhd\_traits\_wave\_2\_sc + age

Data: analytic\_df\_wide (Number of observations: 4515)

Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;  
total post-warmup draws = 8000

Multilevel Hyperparameters:

~family\_id (Number of levels: 3928)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.983    | 0.020     | 0.946    | 1.023    | 1.002 | 1810     | 3364     |

~site\_id (Number of levels: 22)

|               | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat  | Bulk_ESS | Tail_ESS |
|---------------|----------|-----------|----------|----------|-------|----------|----------|
| sd(Intercept) | 0.067    | 0.037     | 0.005    | 0.147    | 1.002 | 549      | 1105     |

Regression Coefficients:

|                                                      | Estimate | Est.Error |       |  |
|------------------------------------------------------|----------|-----------|-------|--|
| Intercept                                            | 0.508    | 0.035     |       |  |
| breast_development_p_wave_2_sc                       | 0.185    | 0.038     |       |  |
| adhd_traits_wave_2_sc                                | 0.583    | 0.034     |       |  |
| age_wave_2_sc                                        | -0.114   | 0.036     |       |  |
| ethnraceblack_nh                                     | -0.121   | 0.065     |       |  |
| ethnracehispanic                                     | -0.090   | 0.056     |       |  |
| ethnraceother                                        | -0.072   | 0.062     |       |  |
| inr_wave_0_sc                                        | 0.000    | 0.042     |       |  |
| externalising_wave_0_sc                              | 0.903    | 0.034     |       |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.159    | 0.063     |       |  |
|                                                      | 1-95% CI | u-95% CI  | Rhat  |  |
| Intercept                                            | 0.437    | 0.577     | 1.001 |  |
| breast_development_p_wave_2_sc                       | 0.111    | 0.260     | 1.001 |  |
| adhd_traits_wave_2_sc                                | 0.515    | 0.651     | 1.002 |  |
| age_wave_2_sc                                        | -0.186   | -0.043    | 1.001 |  |
| ethnraceblack_nh                                     | -0.251   | 0.008     | 1.000 |  |
| ethnracehispanic                                     | -0.199   | 0.020     | 1.001 |  |
| ethnraceother                                        | -0.192   | 0.049     | 1.002 |  |
| inr_wave_0_sc                                        | -0.081   | 0.085     | 1.001 |  |
| externalising_wave_0_sc                              | 0.838    | 0.968     | 1.000 |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.033    | 0.282     | 1.001 |  |
|                                                      | Bulk_ESS | Tail_ESS  |       |  |
| Intercept                                            | 1540     | 2546      |       |  |
| breast_development_p_wave_2_sc                       | 2252     | 3825      |       |  |
| adhd_traits_wave_2_sc                                | 1485     | 2841      |       |  |
| age_wave_2_sc                                        | 2118     | 4282      |       |  |
| ethnraceblack_nh                                     | 1606     | 2644      |       |  |
| ethnracehispanic                                     | 1581     | 2946      |       |  |
| ethnraceother                                        | 1532     | 2744      |       |  |
| inr_wave_0_sc                                        | 1856     | 3055      |       |  |
| externalising_wave_0_sc                              | 1264     | 2791      |       |  |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 1515     | 3545      |       |  |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

## Model 5B longitudinal prior 1 diagnostics

|                 |       |
|-----------------|-------|
| prior           | class |
| normal(0, 0.25) | b     |

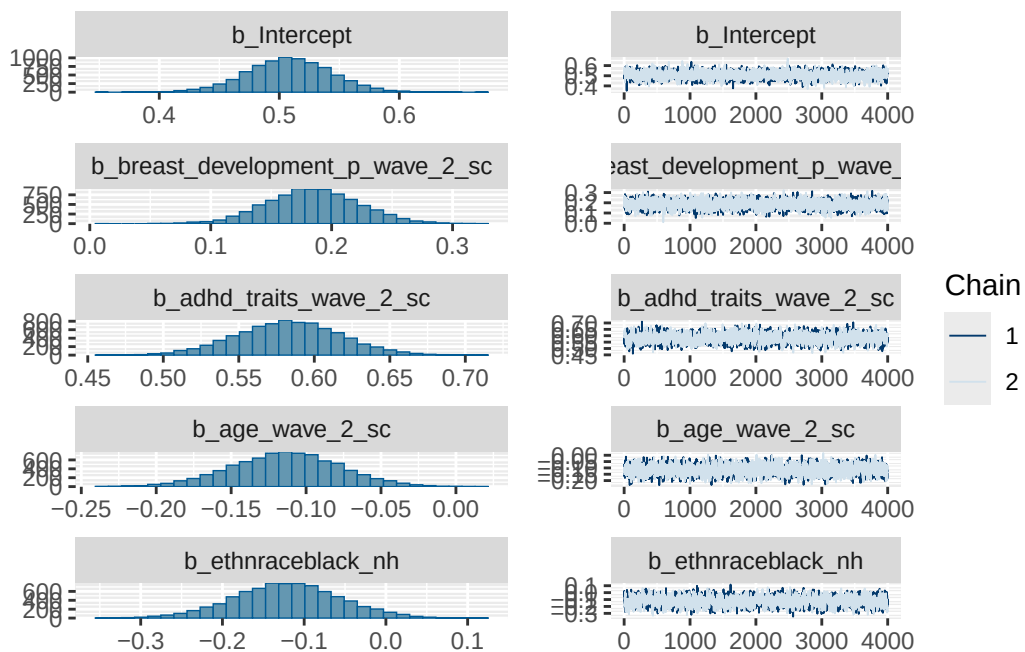


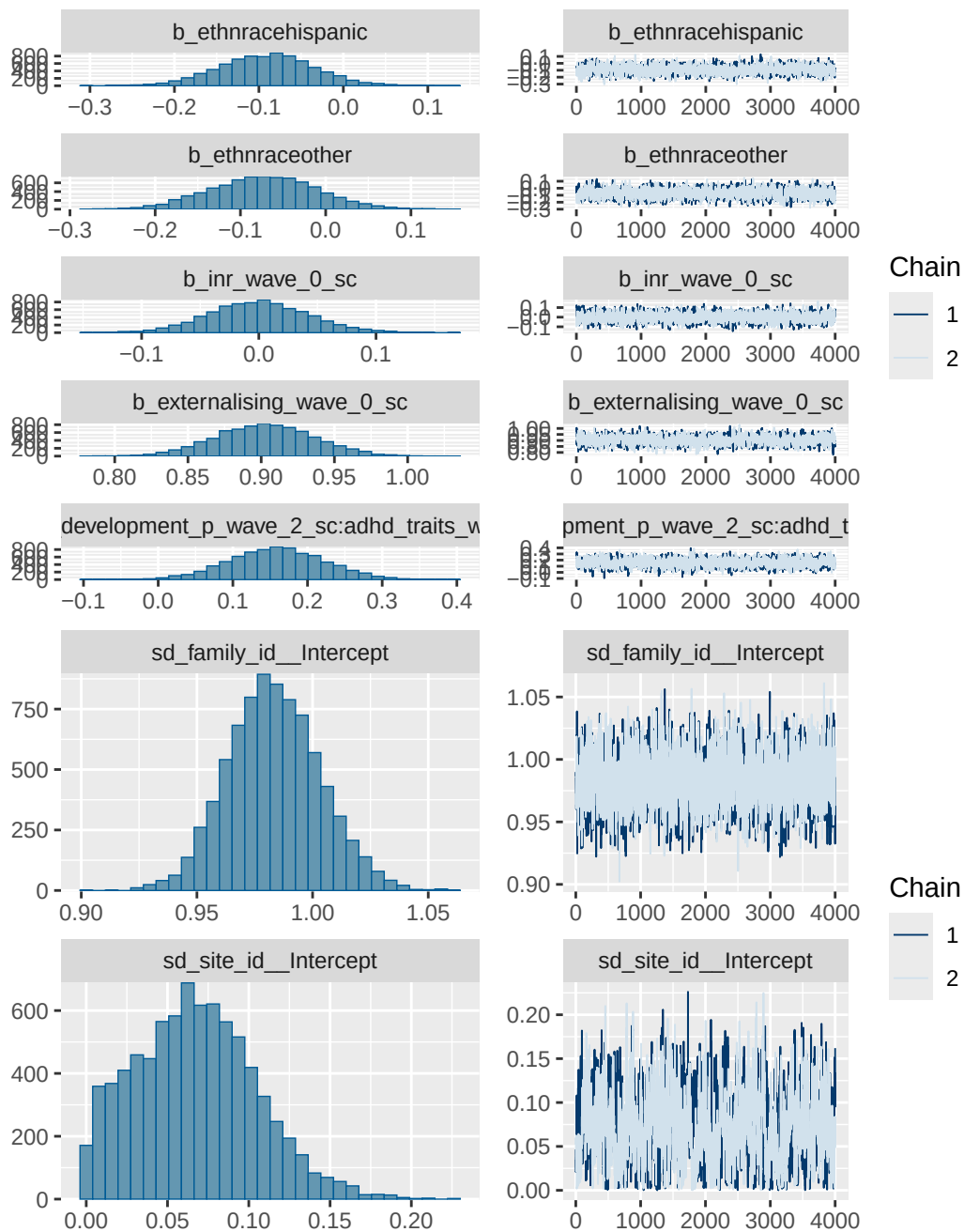


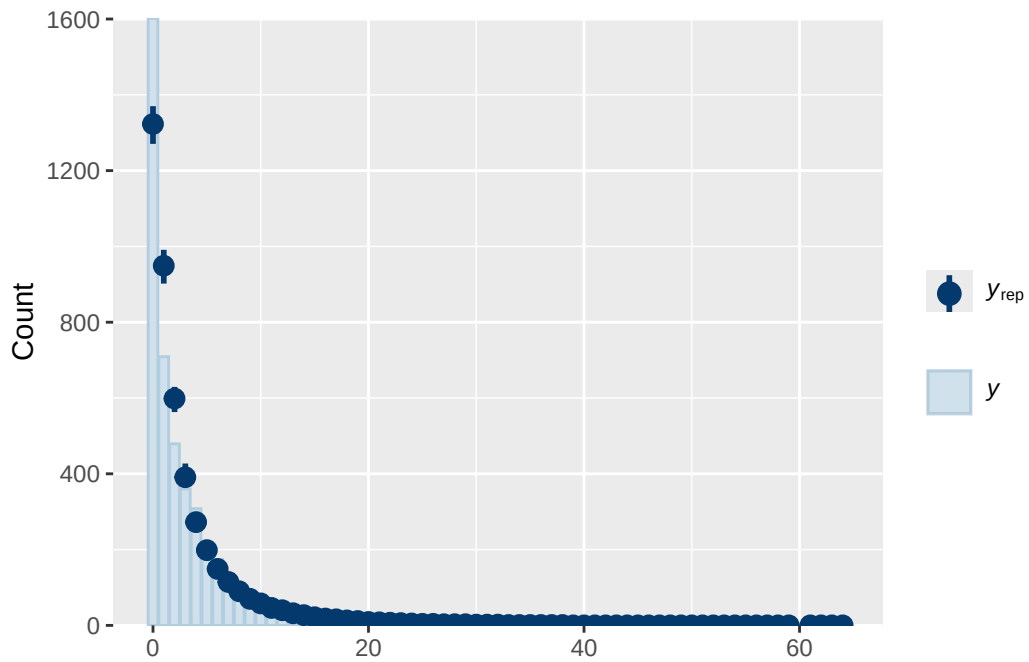
```

      default
0      default
0      (vectorized)
0      (vectorized)
0      (vectorized)
0      (vectorized)

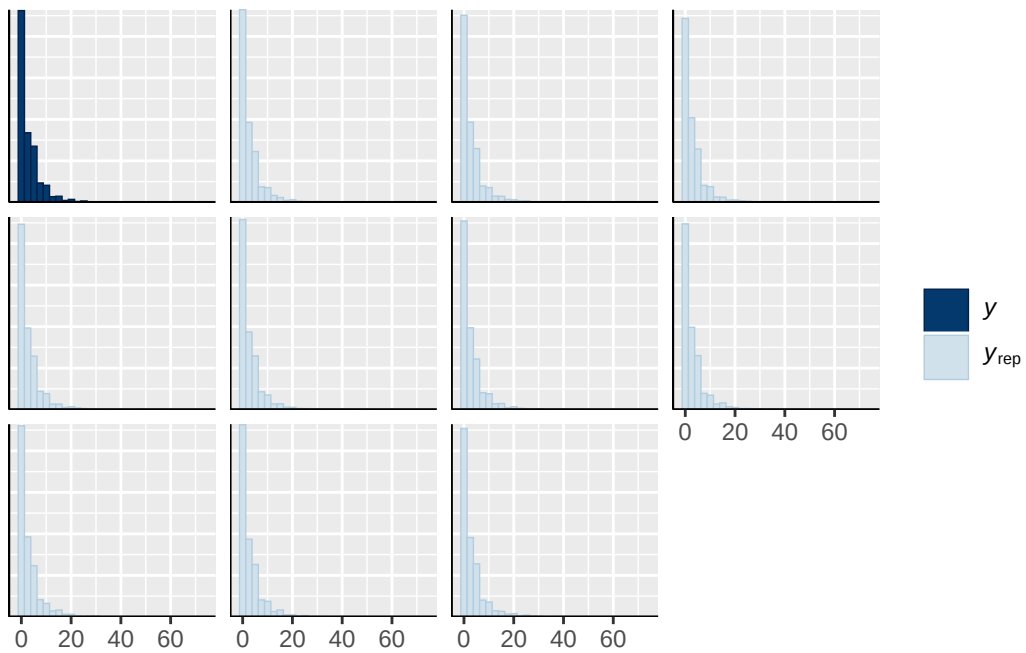
```



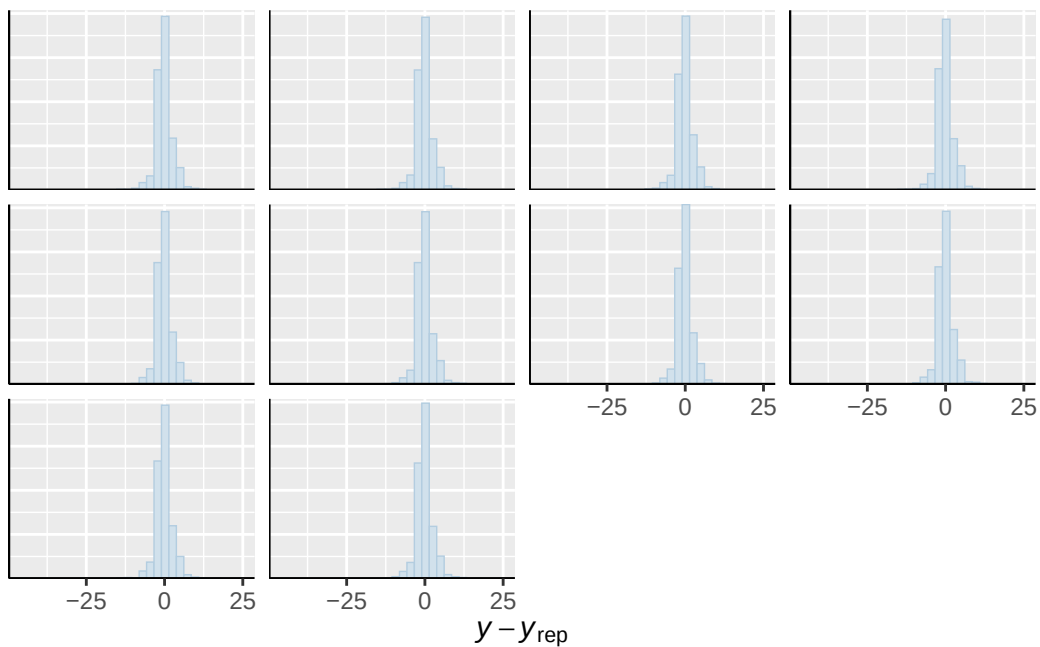




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

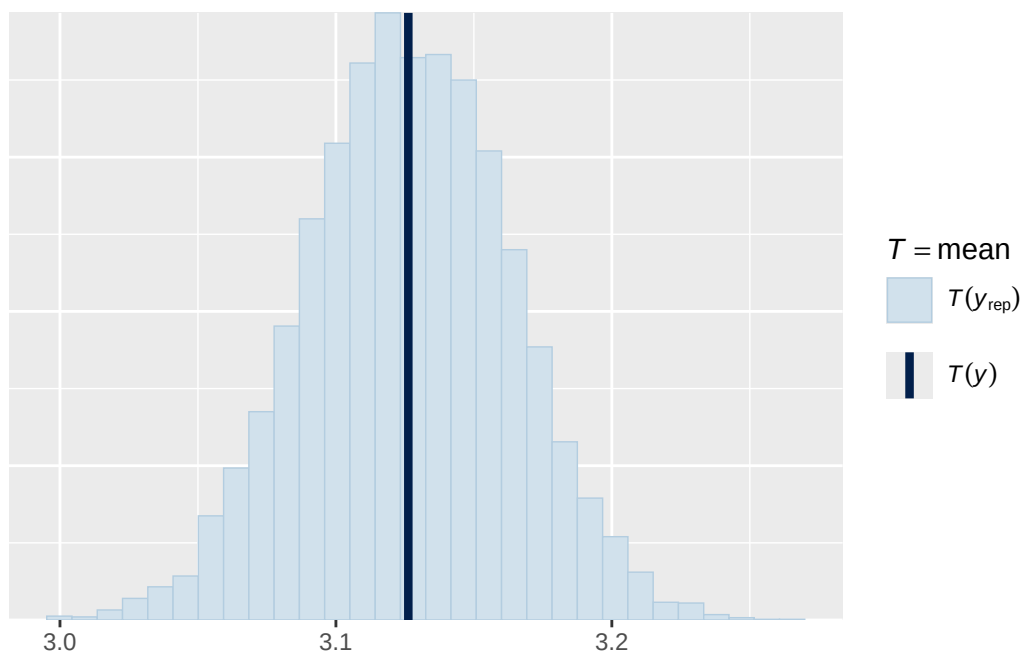


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

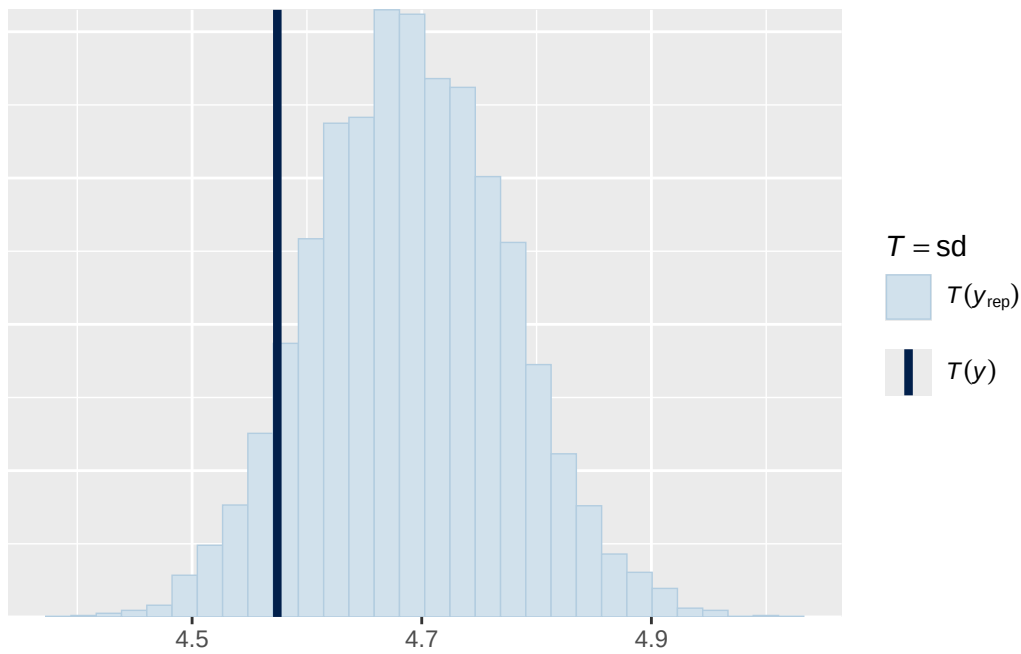


Using all posterior draws for ppc type 'stat' by default.

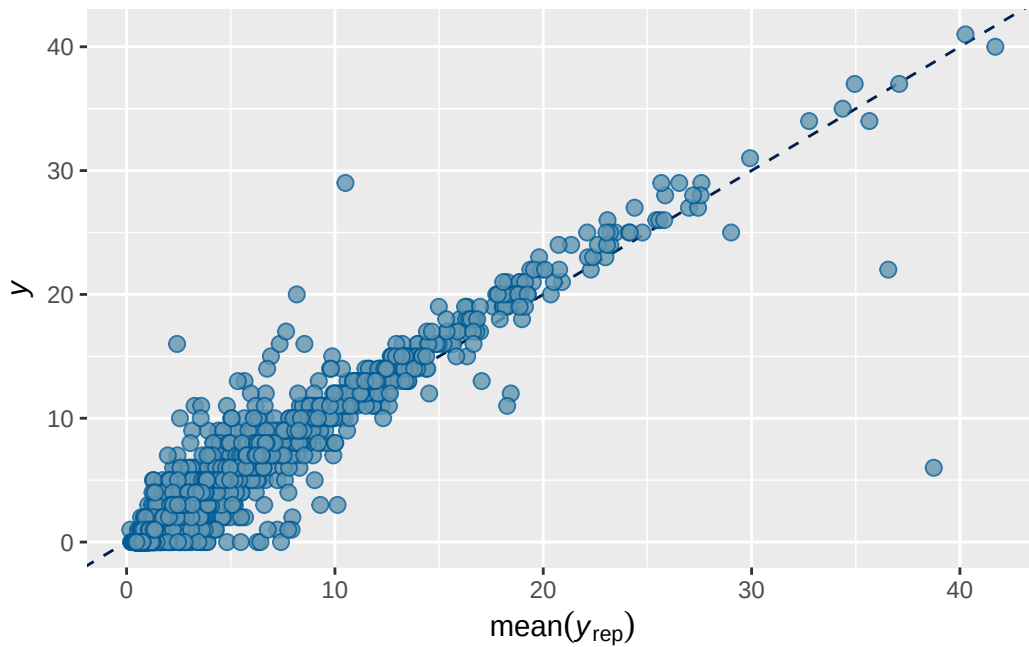
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1 longitudinal prior 2 output

```

      Estimate   Est.Error      Q2.5    Q97.5
R2 0.7393702 0.007767695 0.7238316 0.754201

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4494)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3913)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.899    0.021   0.857   0.940 1.000    1230    2009

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.090    0.036   0.021   0.163 1.006     556     535

```

### Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS
Intercept          0.148    0.040   0.070   0.227 1.001    1838
menarche_status_p_wave_2Y 0.101    0.042   0.019   0.182 1.001    1819
age_wave_2_sc       -0.111    0.041  -0.189  -0.032 1.003    1935
ethnraceblack_nh    -0.093    0.066  -0.224   0.039 1.000    1639
ethnracehispanic    -0.047    0.058  -0.158   0.068 1.000    1431
ethnraceother       -0.006    0.062  -0.128   0.118 1.000    1613
inr_wave_0_sc        0.062    0.043  -0.020   0.149 1.000    1648
adhd_traits_wave_0_sc  1.315    0.033   1.253   1.381 1.002    1270
Tail_ESS
Intercept          2837
menarche_status_p_wave_2Y 2607
age_wave_2_sc       2664
ethnraceblack_nh    2775
ethnracehispanic    2236
ethnraceother       2739
inr_wave_0_sc        2050
adhd_traits_wave_0_sc 2183

```

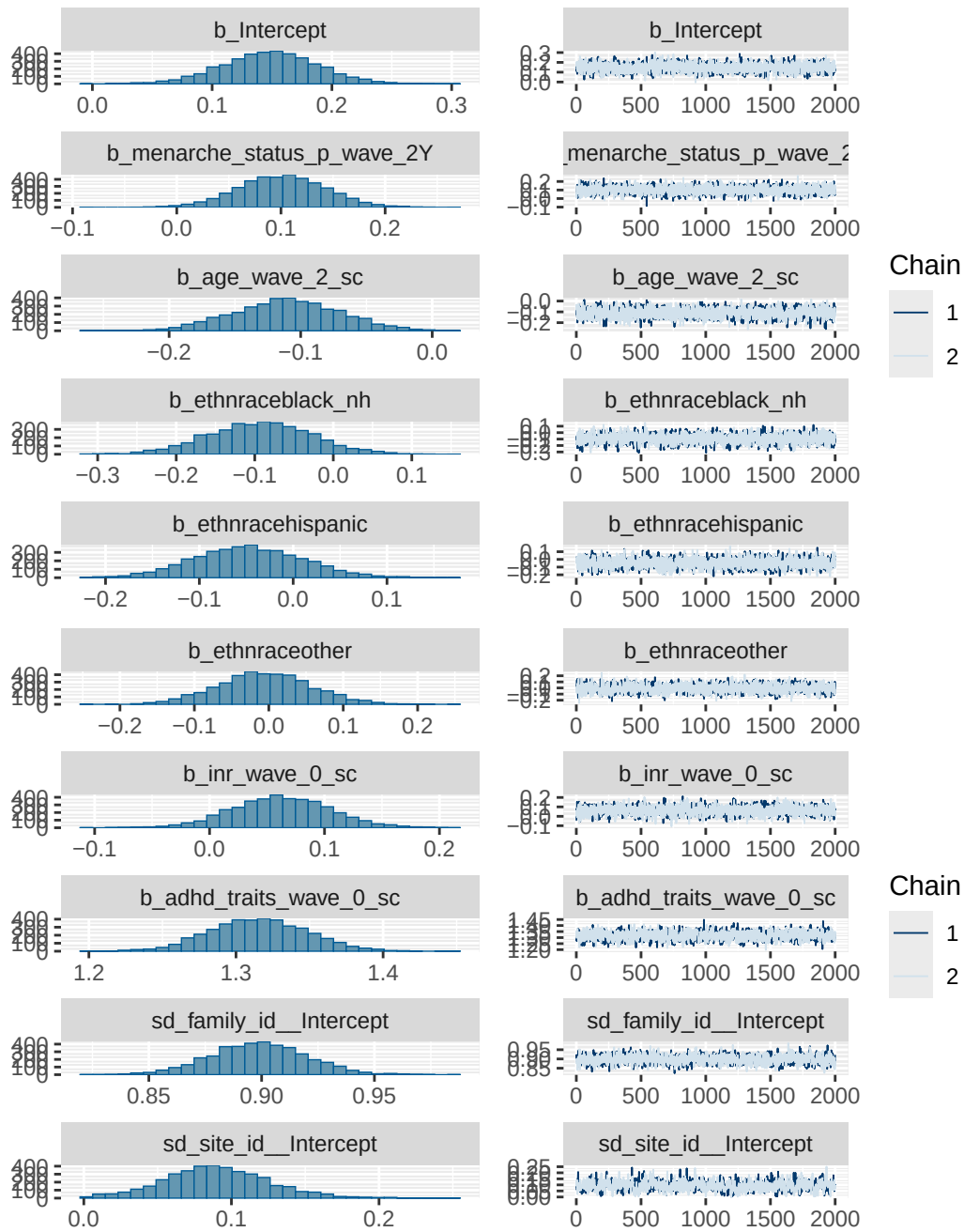
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS

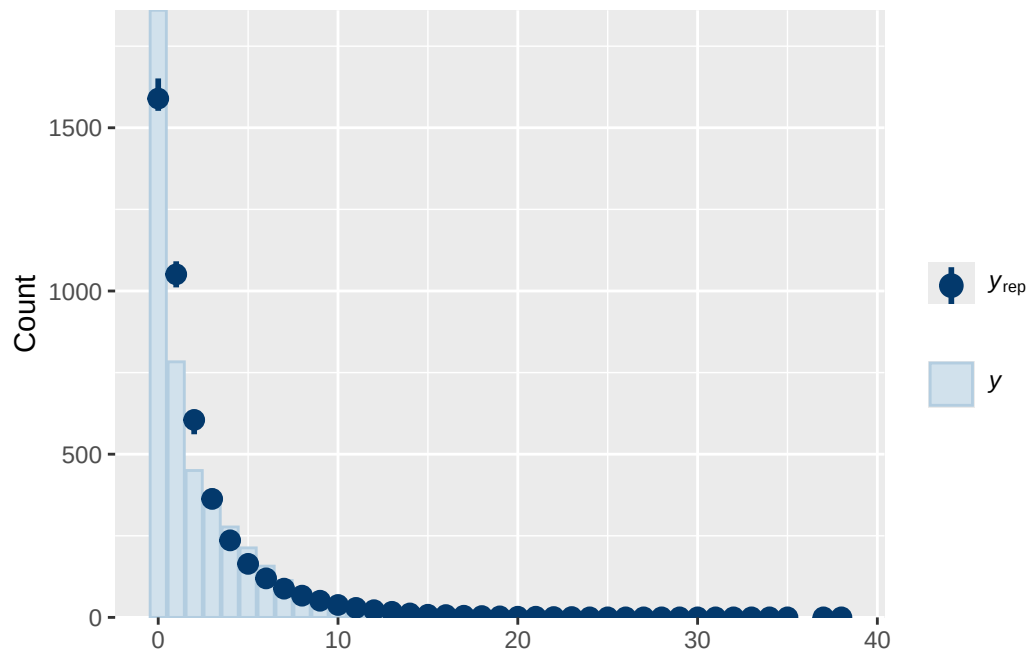
and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

## Model 1 longitudinal prior 2 diagnostics

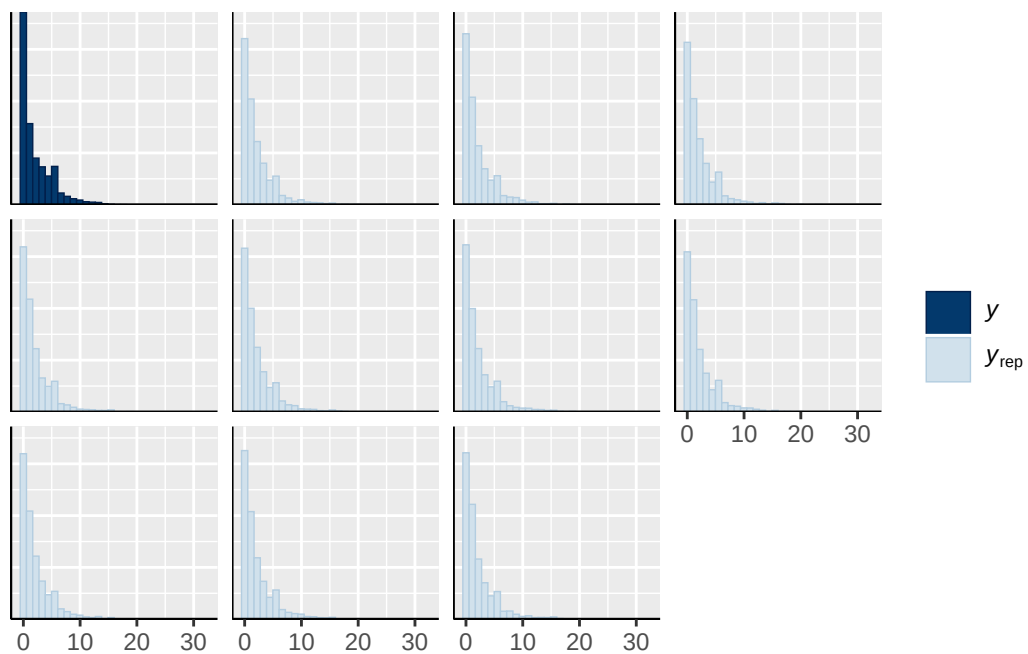
|                      | prior         | class        | coef                      | group     | resp | dpar |
|----------------------|---------------|--------------|---------------------------|-----------|------|------|
|                      | normal(0, 10) | b            |                           |           |      |      |
|                      | normal(0, 10) | b            | adhd_traits_wave_0_sc     |           |      |      |
|                      | normal(0, 10) | b            | age_wave_2_sc             |           |      |      |
|                      | normal(0, 10) | b            | ethnraceblack_nh          |           |      |      |
|                      | normal(0, 10) | b            | ethnracehispanic          |           |      |      |
|                      | normal(0, 10) | b            | ethnraceother             |           |      |      |
|                      | normal(0, 10) | b            | inr_wave_0_sc             |           |      |      |
|                      | normal(0, 10) | b            | menarche_status_p_wave_2Y |           |      |      |
| student_t(3, 0, 2.9) | Intercept     |              |                           |           |      |      |
| student_t(3, 0, 2.9) | sd            |              |                           |           |      |      |
| student_t(3, 0, 2.9) | sd            |              |                           | family_id |      |      |
| student_t(3, 0, 2.9) | sd            |              | Intercept                 | family_id |      |      |
| student_t(3, 0, 2.9) | sd            |              |                           | site_id   |      |      |
| student_t(3, 0, 2.9) | sd            |              | Intercept                 | site_id   |      |      |
| nlpar lb ub tag      |               | source       |                           |           |      |      |
|                      |               | user         |                           |           |      |      |
|                      |               | (vectorized) |                           |           |      |      |
|                      |               | (vectorized) |                           |           |      |      |
|                      |               | (vectorized) |                           |           |      |      |
|                      |               | (vectorized) |                           |           |      |      |
|                      |               | (vectorized) |                           |           |      |      |
|                      |               | (vectorized) |                           |           |      |      |
|                      |               | (vectorized) |                           |           |      |      |
|                      |               | default      |                           |           |      |      |
| 0                    |               | default      |                           |           |      |      |
| 0                    |               | (vectorized) |                           |           |      |      |
| 0                    |               | (vectorized) |                           |           |      |      |
| 0                    |               | (vectorized) |                           |           |      |      |
| 0                    |               | (vectorized) |                           |           |      |      |



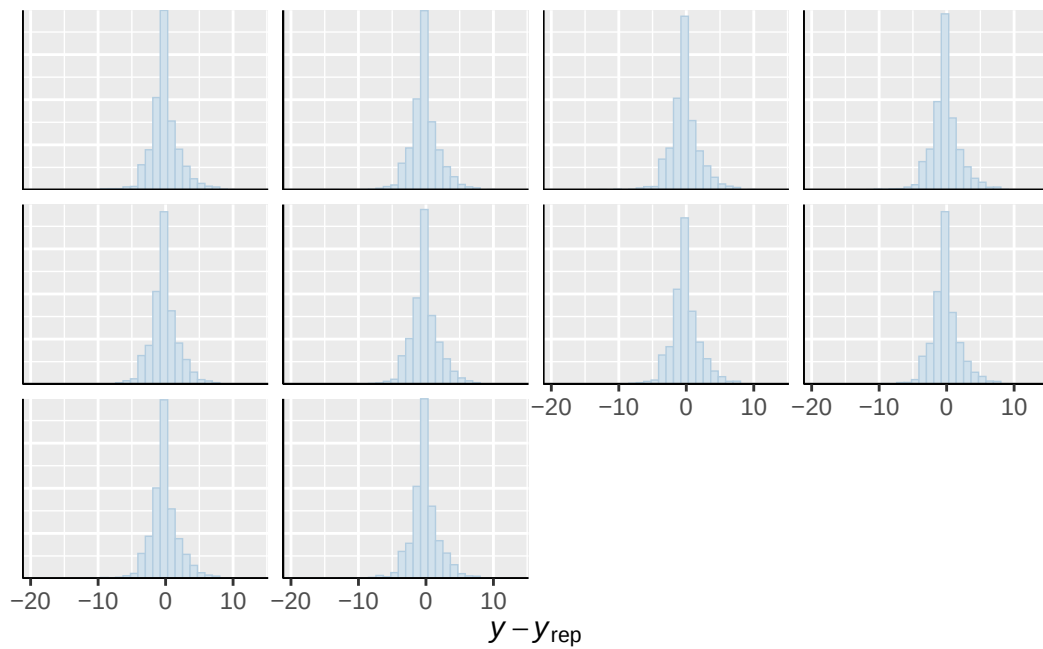




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

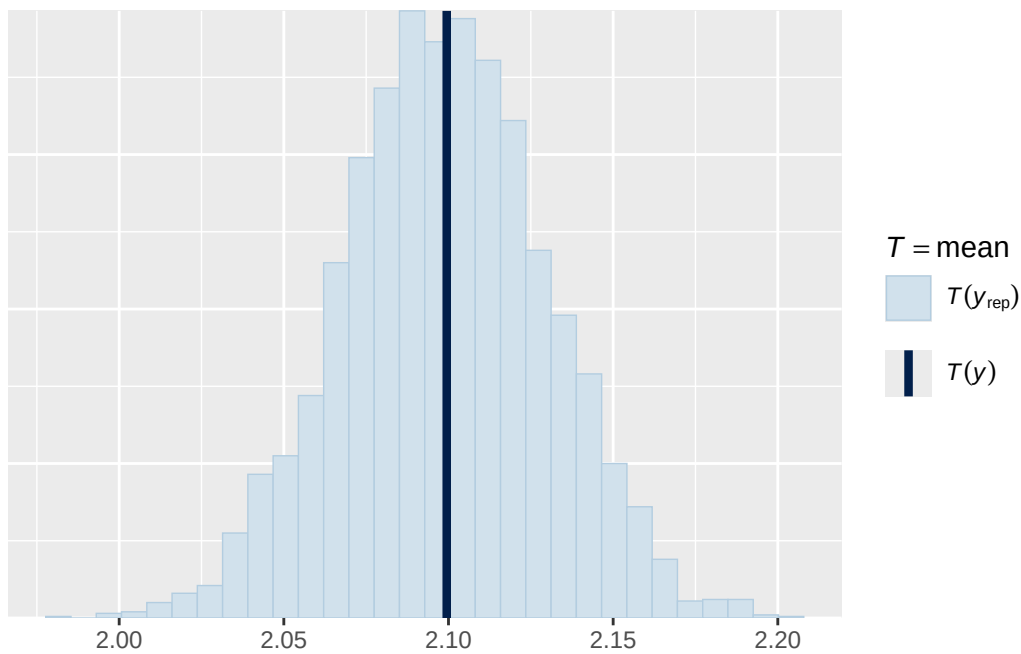


Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

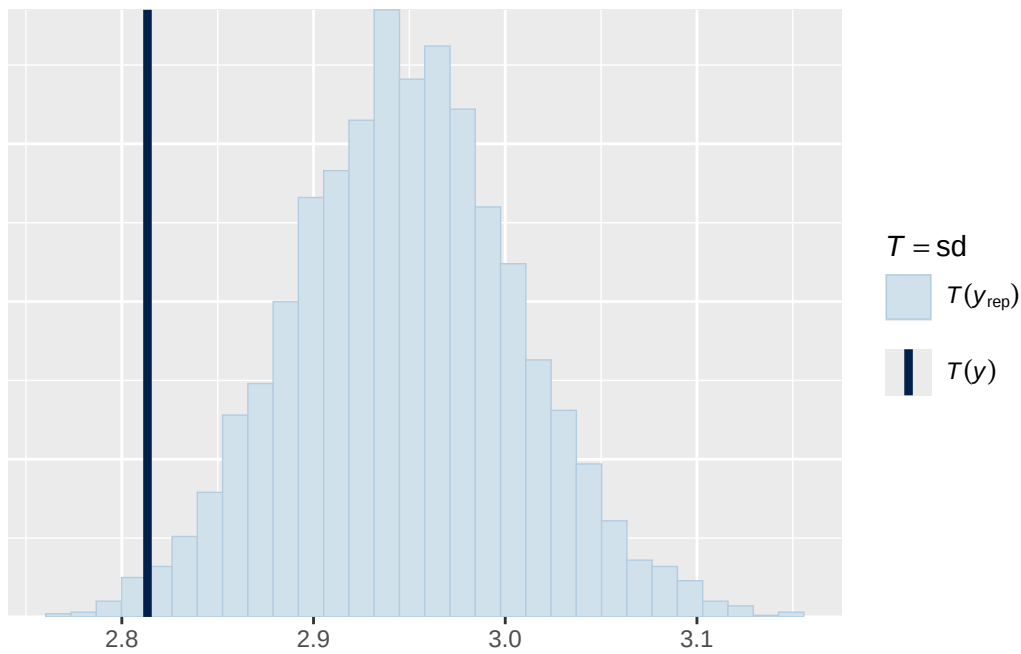


Using all posterior draws for ppc type 'stat' by default.

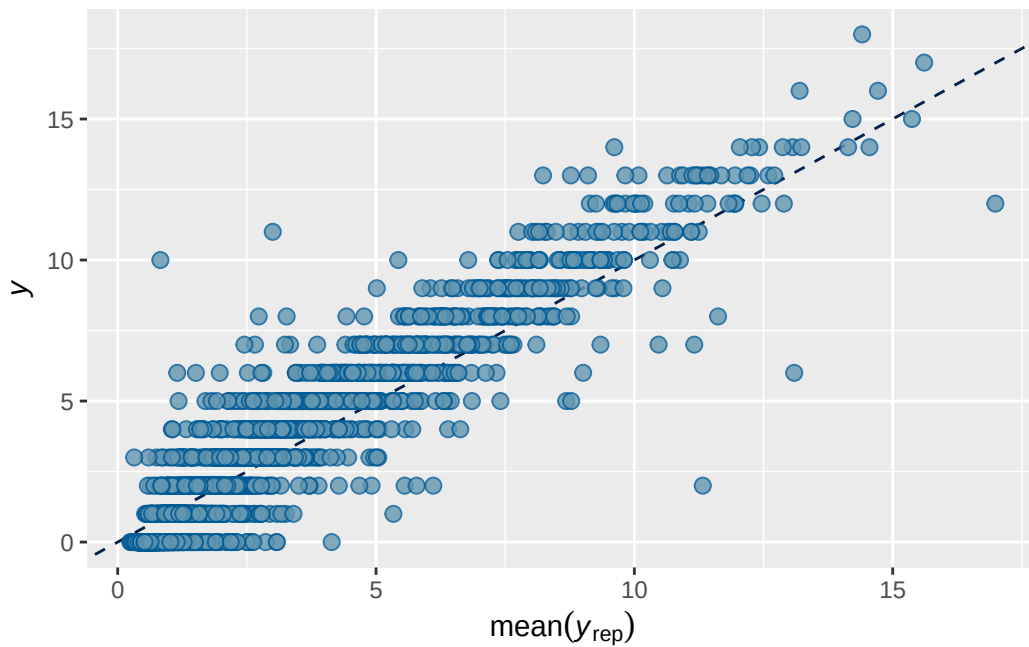
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 1B longitudinal prior 2 output

```

      Estimate   Est.Error      Q2.5    Q97.5
R2 0.7357083 0.007529665 0.7207626 0.750273

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + inr_wave_0_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.886    0.021   0.846   0.927 1.004    2360    4681

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.093    0.036   0.022   0.168 1.001    1068    1112

```

### Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat
Intercept          0.191    0.037   0.116   0.264 1.001
breast_development_p_wave_2_sc 0.125    0.040   0.047   0.204 1.000
age_wave_2_sc      -0.104    0.040  -0.182  -0.026 1.000
ethnraceblack_nh   -0.091    0.065  -0.216   0.037 1.001
ethnracehispanic   -0.035    0.057  -0.145   0.076 1.001
ethnraceother      -0.007    0.062  -0.129   0.118 1.001
inr_wave_0_sc       0.087    0.043   0.003   0.170 1.000
adhd_traits_wave_0_sc 1.322    0.034   1.256   1.388 1.001

      Bulk_ESS Tail_ESS
Intercept          3537    4081
breast_development_p_wave_2_sc 3892    4613
age_wave_2_sc      3724    5234
ethnraceblack_nh   2931    4747
ethnracehispanic   3005    4569
ethnraceother      3139    4952
inr_wave_0_sc       3087    4724
adhd_traits_wave_0_sc 2630    4155

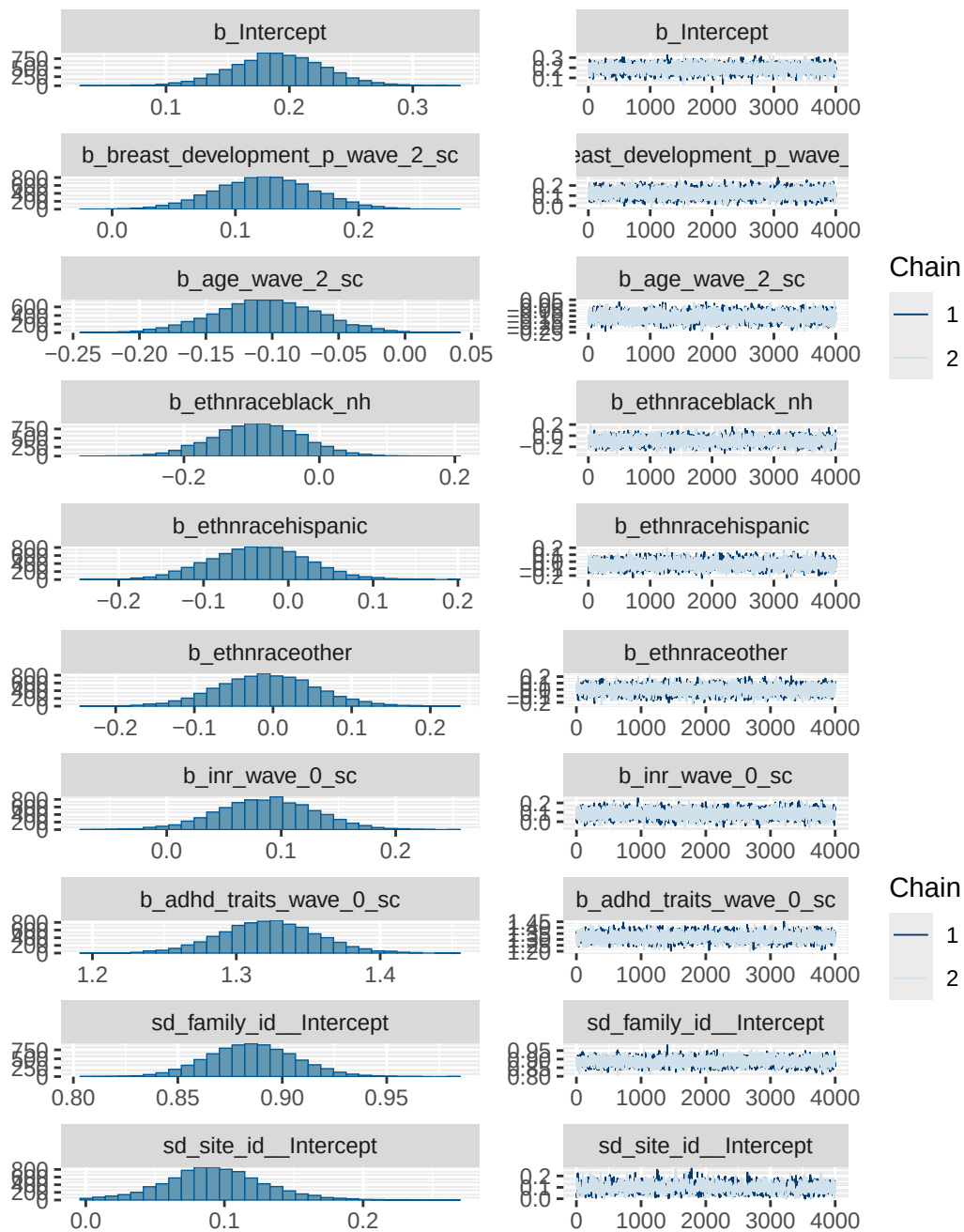
```

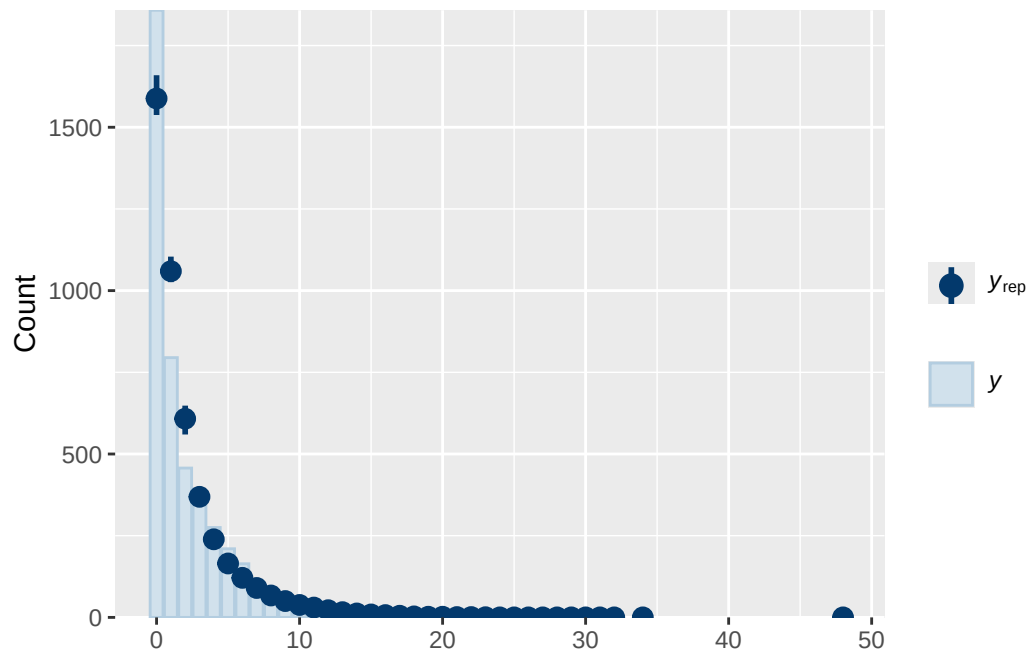
Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS

and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

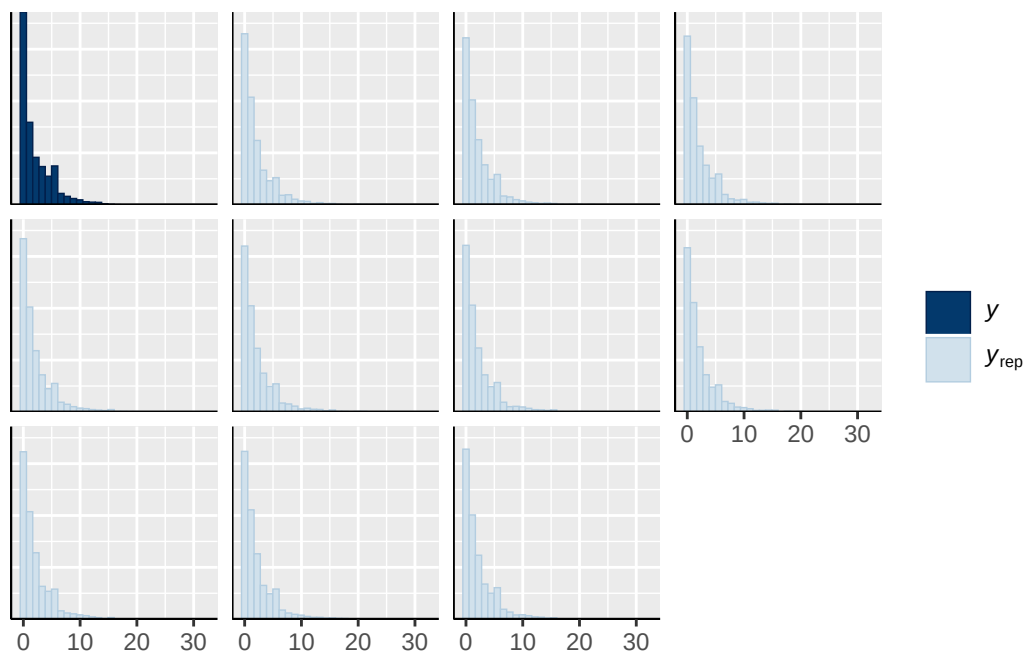
## Model 1B longitudinal prior 2 diagnostics

|                      | prior         | class        | coef                           | group     | resp |
|----------------------|---------------|--------------|--------------------------------|-----------|------|
|                      | normal(0, 10) | b            |                                |           |      |
|                      | normal(0, 10) | b            | adhd_traits_wave_0_sc          |           |      |
|                      | normal(0, 10) | b            | age_wave_2_sc                  |           |      |
|                      | normal(0, 10) | b            | breast_development_p_wave_2_sc |           |      |
|                      | normal(0, 10) | b            | ethnraceblack_nh               |           |      |
|                      | normal(0, 10) | b            | ethnracehispanic               |           |      |
|                      | normal(0, 10) | b            | ethnraceother                  |           |      |
|                      | normal(0, 10) | b            | inr_wave_0_sc                  |           |      |
| student_t(3, 0, 2.7) | Intercept     |              |                                |           |      |
| student_t(3, 0, 2.7) | sd            |              |                                |           |      |
| student_t(3, 0, 2.7) | sd            |              |                                | family_id |      |
| student_t(3, 0, 2.7) | sd            |              | Intercept                      | family_id |      |
| student_t(3, 0, 2.7) | sd            |              |                                | site_id   |      |
| student_t(3, 0, 2.7) | sd            |              | Intercept                      | site_id   |      |
| dpar nlpar lb ub tag |               | source       |                                |           |      |
|                      |               | user         |                                |           |      |
|                      |               | (vectorized) |                                |           |      |
|                      |               | (vectorized) |                                |           |      |
|                      |               | (vectorized) |                                |           |      |
|                      |               | (vectorized) |                                |           |      |
|                      |               | (vectorized) |                                |           |      |
|                      |               | (vectorized) |                                |           |      |
|                      |               | (vectorized) |                                |           |      |
|                      |               | default      |                                |           |      |
| 0                    |               | default      |                                |           |      |
| 0                    |               | (vectorized) |                                |           |      |
| 0                    |               | (vectorized) |                                |           |      |
| 0                    |               | (vectorized) |                                |           |      |
| 0                    |               | (vectorized) |                                |           |      |



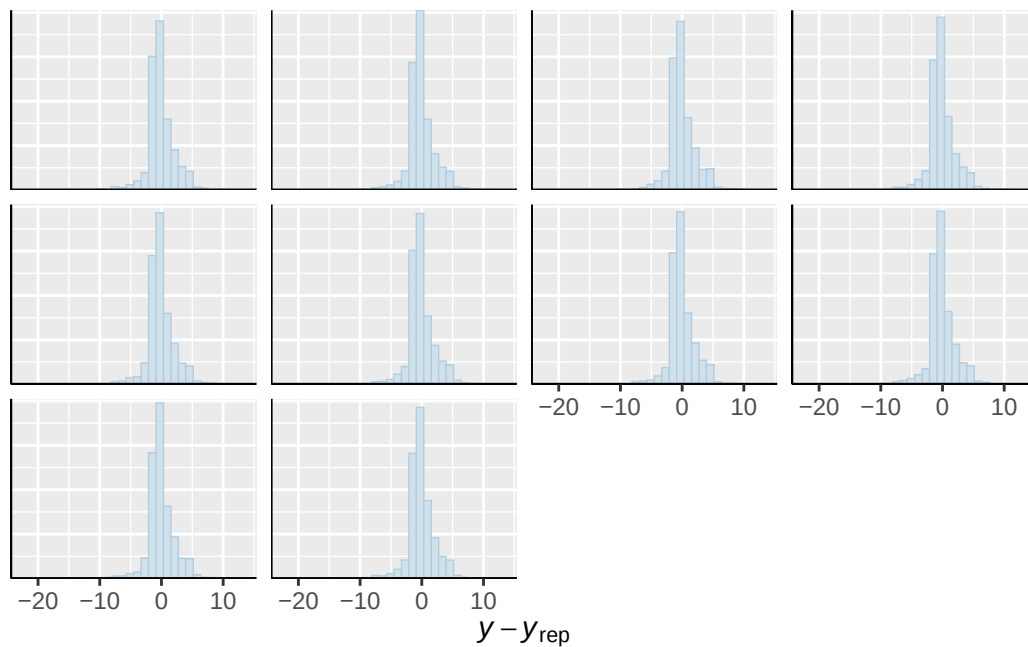


``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



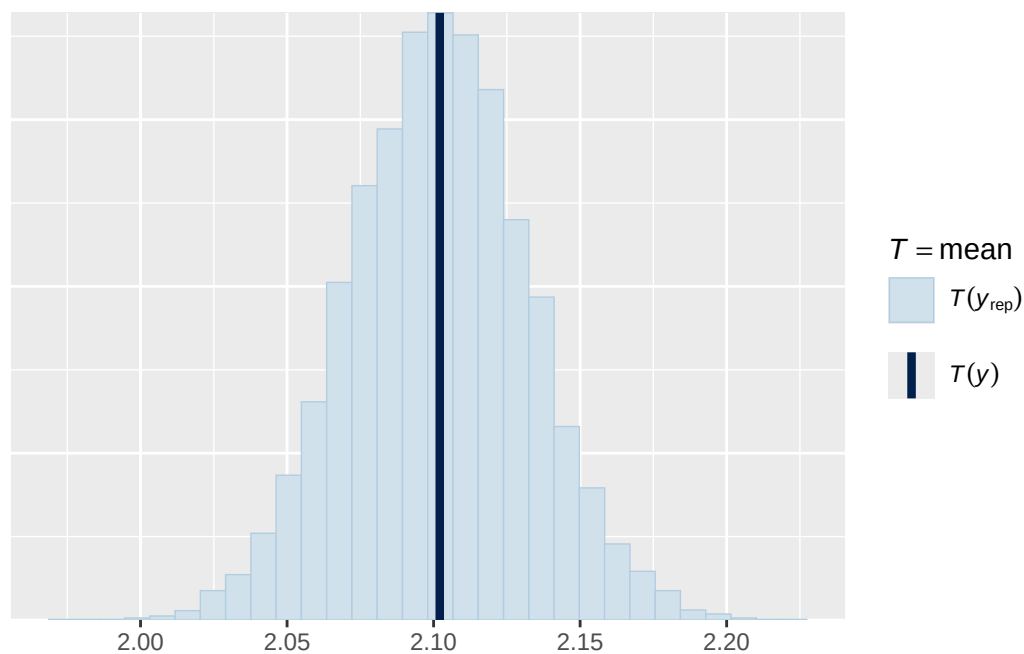
Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



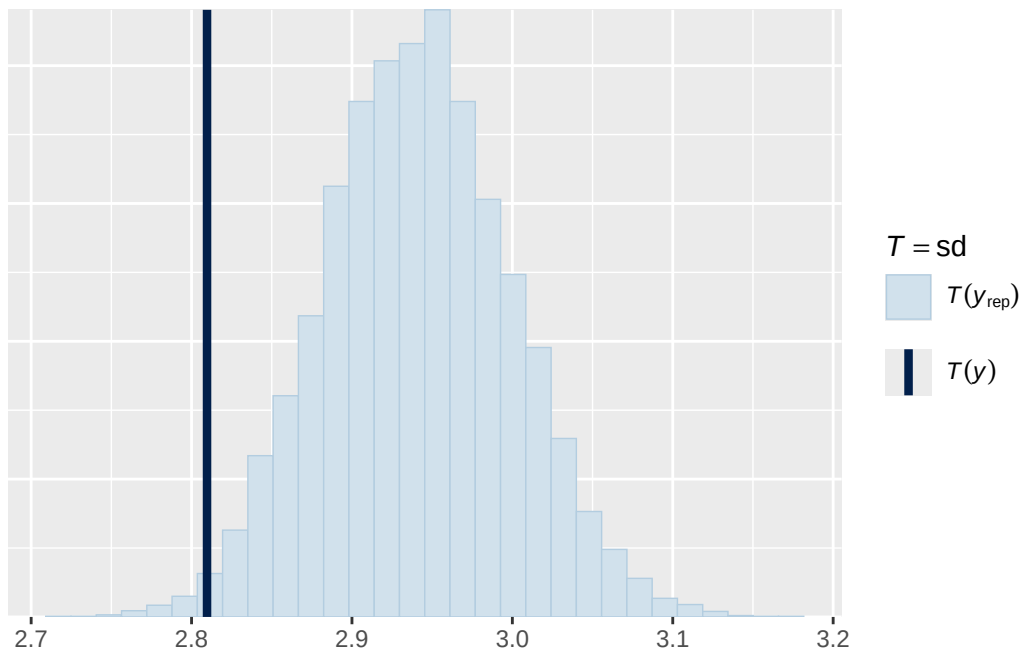


Using all posterior draws for ppc type 'stat' by default.

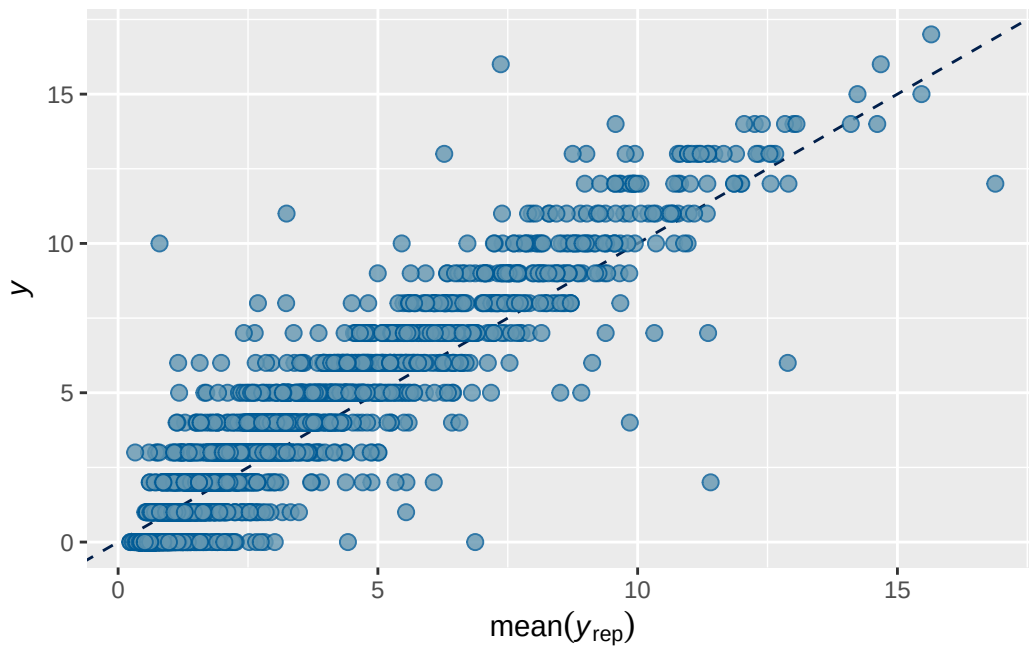
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.



## Model 5B longitudinal prior 2 output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8232708 0.005211247 0.8127758 0.8330009

```

```

Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

### Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.985    0.019   0.947   1.023 1.000    2396    4408

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.066    0.038   0.004   0.150 1.003     747    1960

```

### Regression Coefficients:

|                                                      | Estimate | Est.Error |
|------------------------------------------------------|----------|-----------|
| Intercept                                            | 0.510    | 0.035     |
| breast_development_p_wave_2_sc                       | 0.189    | 0.038     |
| adhd_traits_wave_2_sc                                | 0.588    | 0.034     |
| age_wave_2_sc                                        | -0.119   | 0.037     |
| ethnraceblack_nh                                     | -0.134   | 0.066     |
| ethnracehispanic                                     | -0.099   | 0.058     |
| ethnraceother                                        | -0.082   | 0.064     |
| inr_wave_0_sc                                        | -0.002   | 0.043     |
| externalising_wave_0_sc                              | 0.912    | 0.035     |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.173    | 0.064     |

|                                | 1-95% CI | u-95% CI | Rhat  |
|--------------------------------|----------|----------|-------|
| Intercept                      | 0.440    | 0.577    | 1.000 |
| breast_development_p_wave_2_sc | 0.115    | 0.264    | 1.001 |
| adhd_traits_wave_2_sc          | 0.520    | 0.653    | 1.001 |
| age_wave_2_sc                  | -0.193   | -0.046   | 1.000 |
| ethnraceblack_nh               | -0.261   | -0.006   | 1.000 |
| ethnracehispanic               | -0.212   | 0.015    | 1.000 |
| ethnraceother                  | -0.209   | 0.044    | 1.000 |
| inr_wave_0_sc                  | -0.087   | 0.082    | 1.001 |

|                                                      |          |          |       |
|------------------------------------------------------|----------|----------|-------|
| externalising_wave_0_sc                              | 0.845    | 0.980    | 1.000 |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 0.046    | 0.298    | 1.001 |
|                                                      | Bulk_ESS | Tail_ESS |       |
| Intercept                                            | 3957     | 5113     |       |
| breast_development_p_wave_2_sc                       | 4506     | 5827     |       |
| adhd_traits_wave_2_sc                                | 3055     | 4147     |       |
| age_wave_2_sc                                        | 4175     | 5517     |       |
| ethnraceblack_nh                                     | 2842     | 4278     |       |
| ethnracehispanic                                     | 3424     | 5003     |       |
| ethnraceother                                        | 3134     | 4414     |       |
| inr_wave_0_sc                                        | 3475     | 5000     |       |
| externalising_wave_0_sc                              | 2660     | 4781     |       |
| breast_development_p_wave_2_sc:adhd_traits_wave_2_sc | 3070     | 4827     |       |

Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

### Model 5B longitudinal prior 2 diagnostics

```
prior      class
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
coef      group resp dpar nlpar
adhd_traits_wave_2_sc
age_wave_2_sc
breast_development_p_wave_2_sc
```

```

breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
      ethnraceblack_nh
      ethnracehispanic
      ethnraceother
externalising_wave_0_sc
      inr_wave_0_sc

```

```

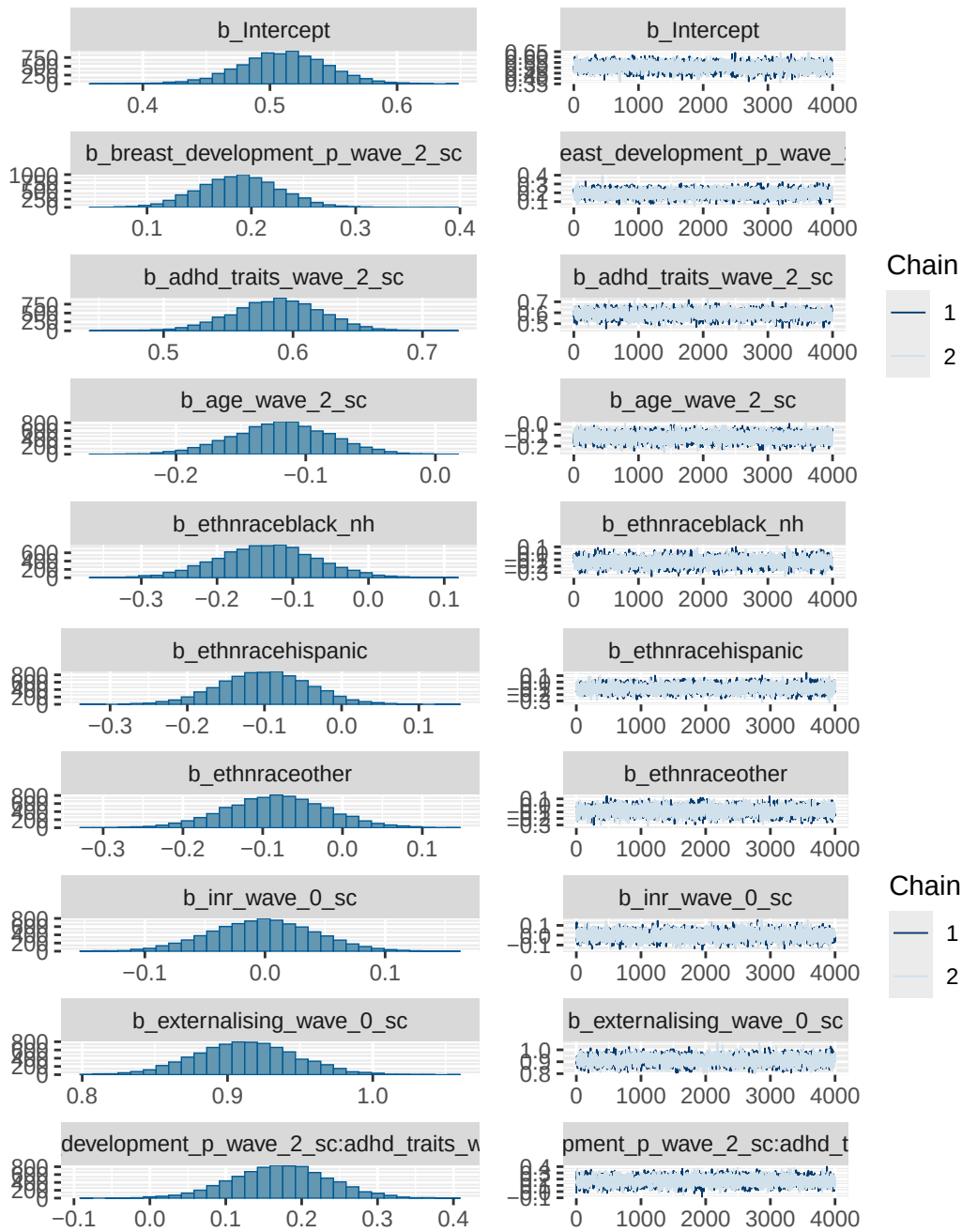
family_id
Intercept family_id
      site_id
Intercept site_id

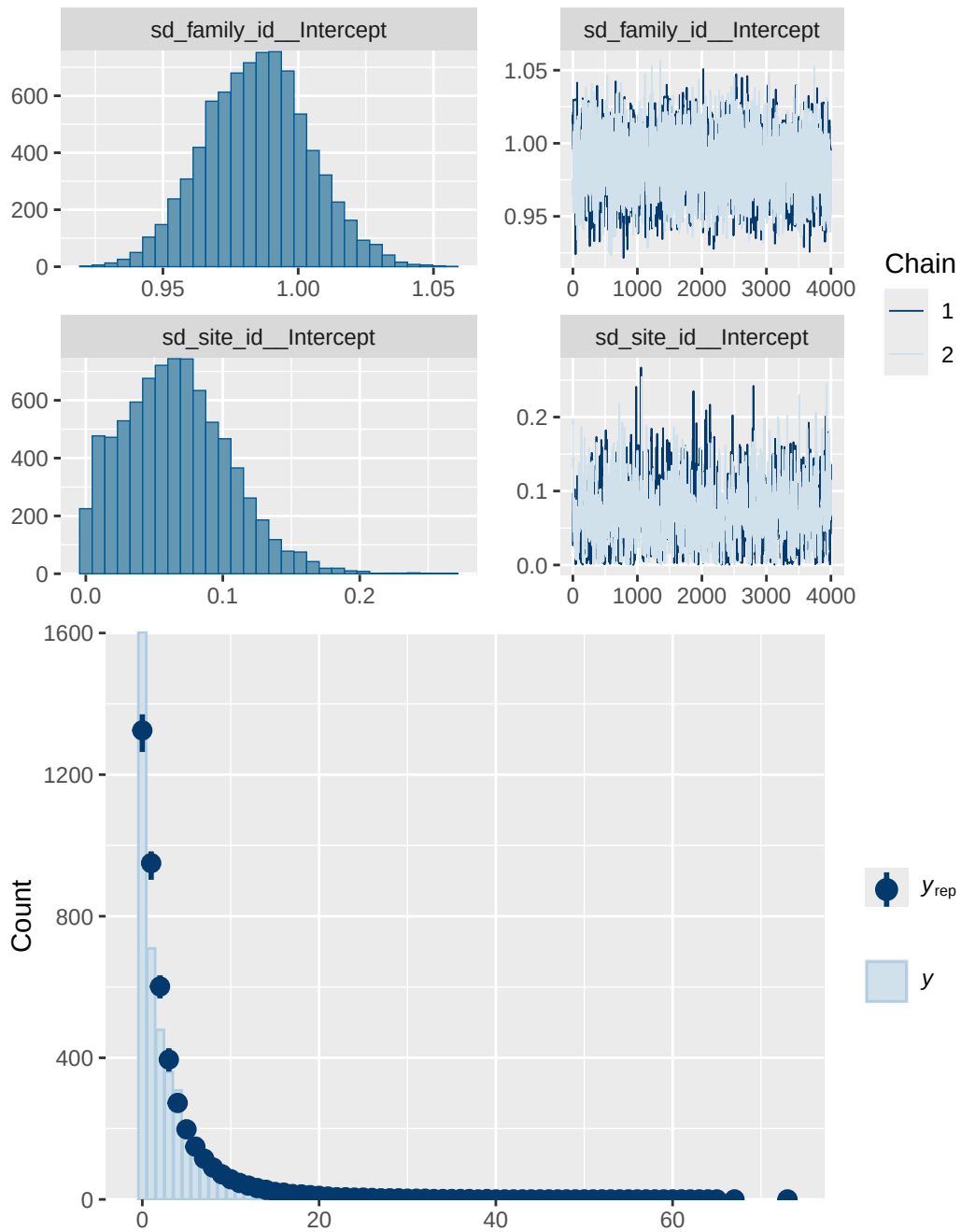
```

```

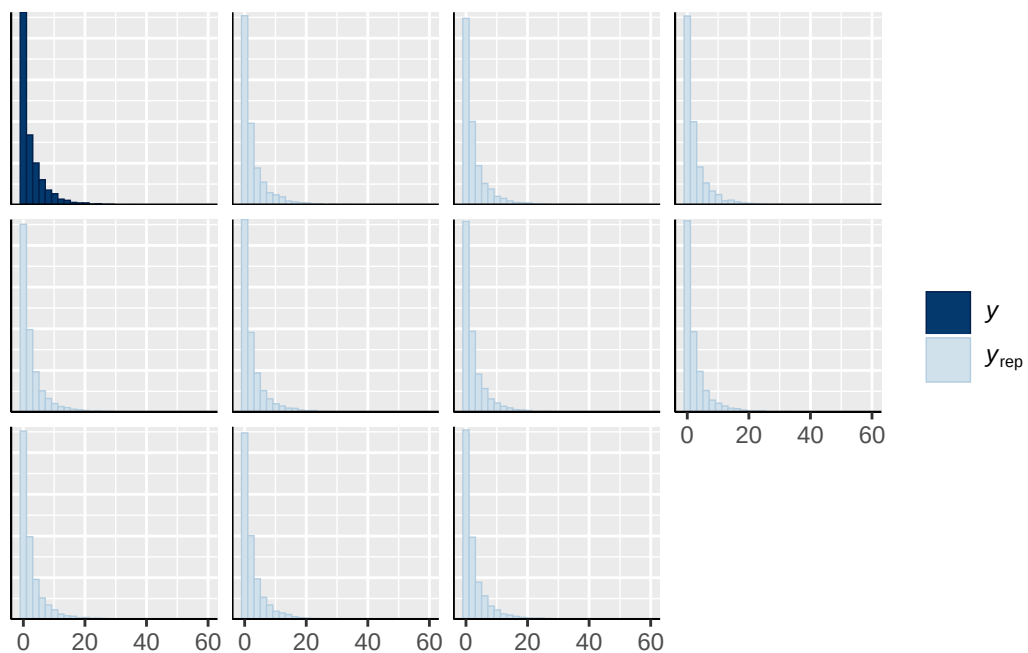
lb ub tag      source
      user
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      default
0      default
0      (vectorized)
0      (vectorized)
0      (vectorized)
0      (vectorized)

```

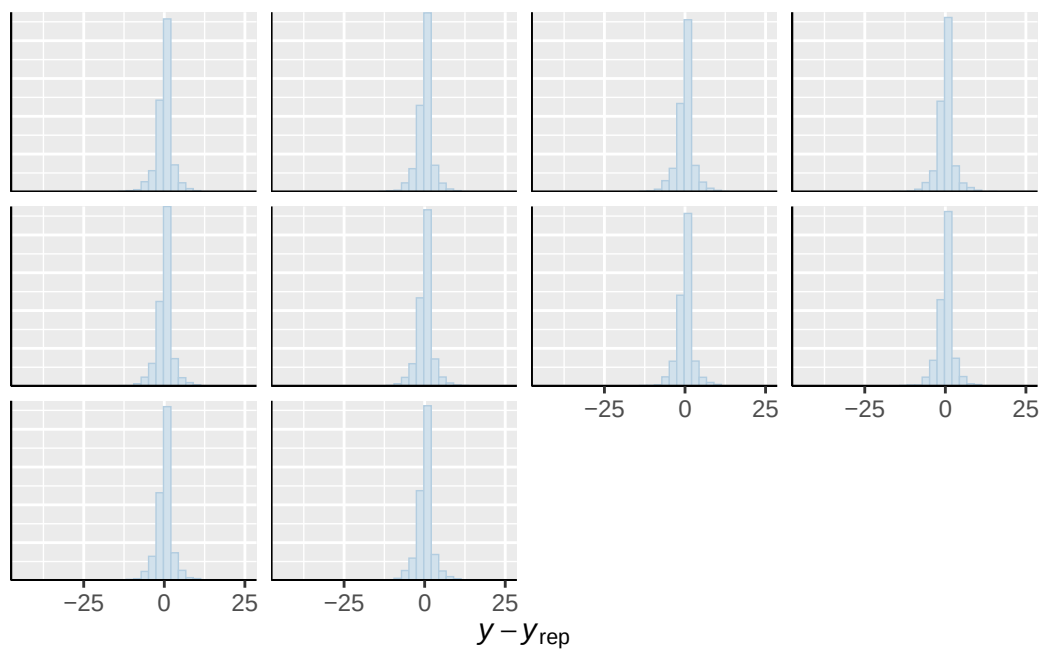




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



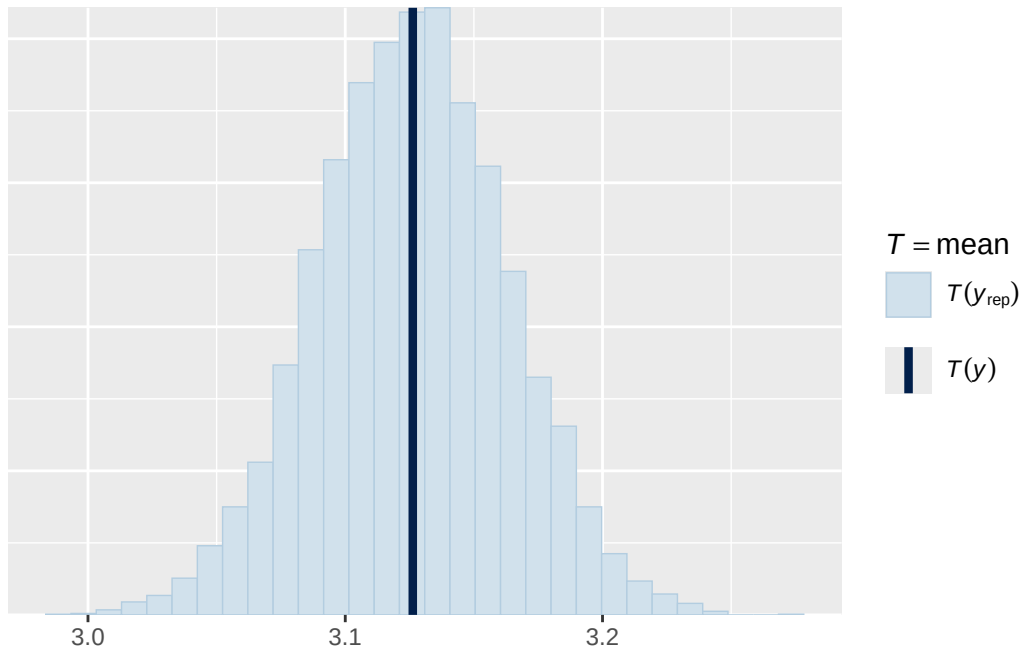
Using 10 posterior draws for ppc type 'error\_hist' by default.  
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



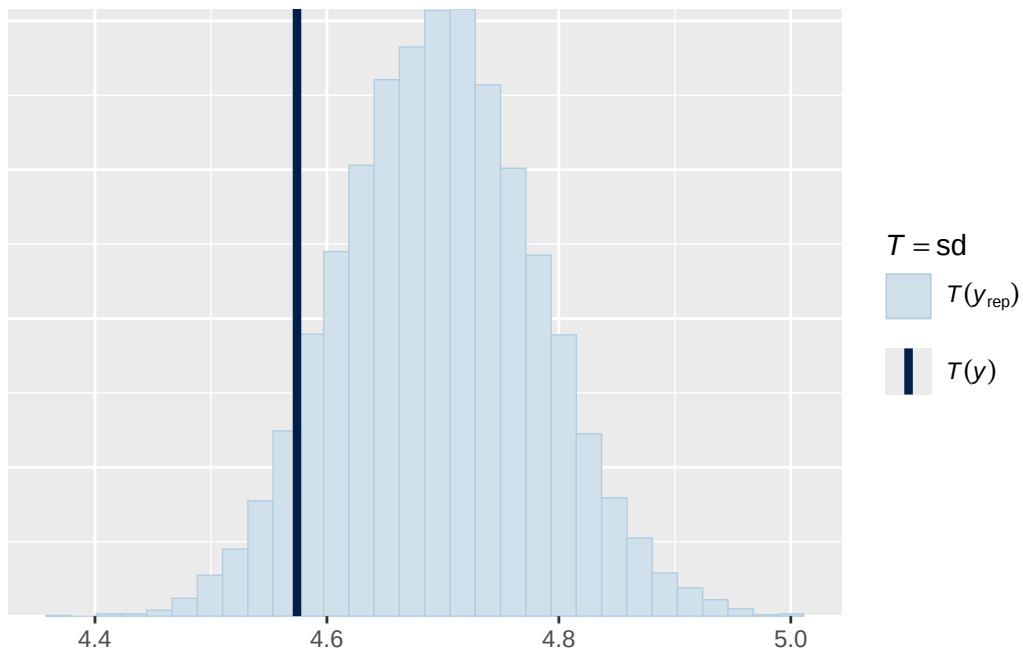
Using all posterior draws for ppc type 'stat' by default.



Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter\_avg' by default.

